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Republic of Lebanon

Ministry of Education and Higher Education

■
CHEMISTRY
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Secondary Education

Third Year

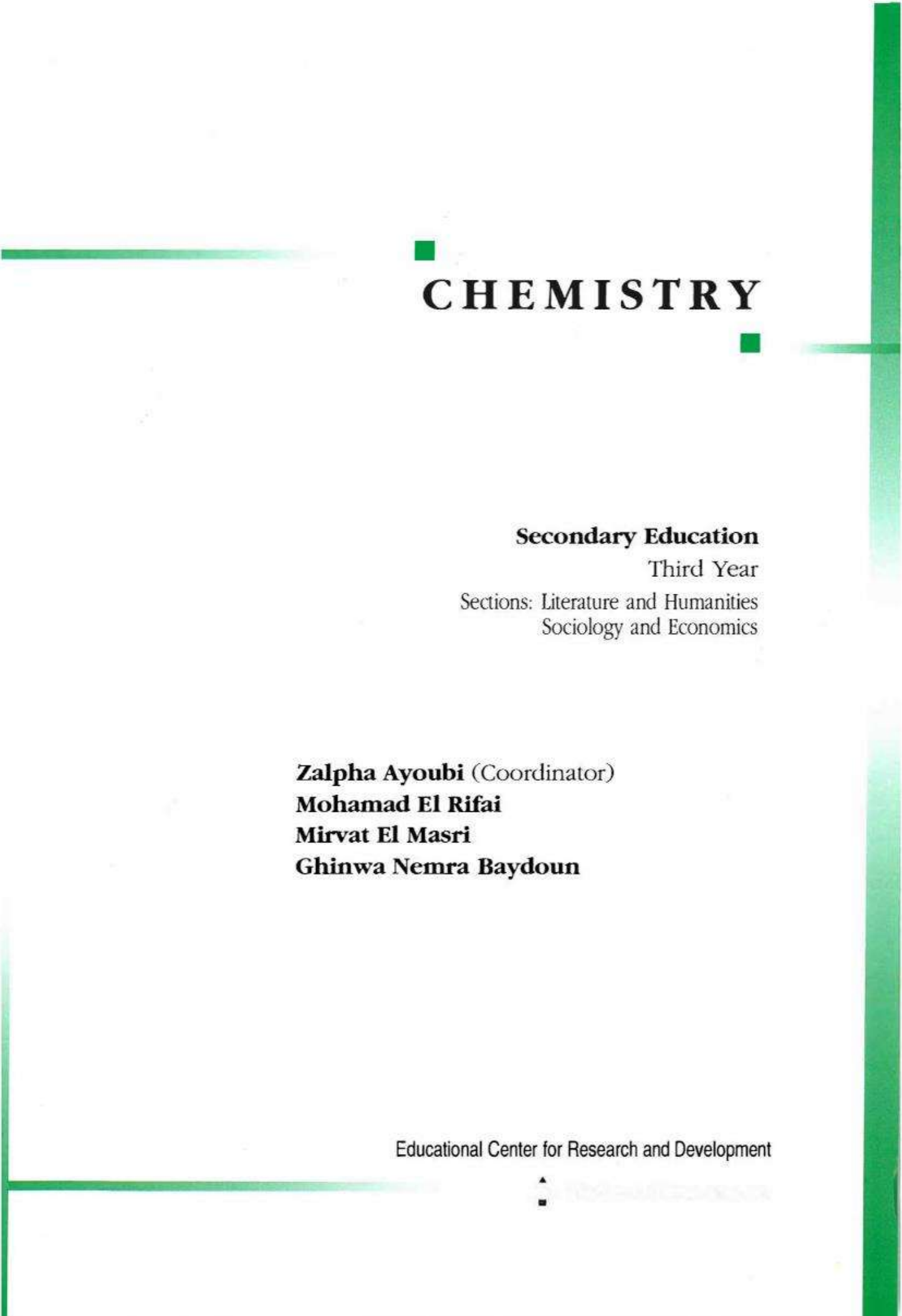
Sections: Literature and Humanities
Sociology and Economics

Educational Center for Research and Development



National
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New Curricula



CHEMISTRY

Secondary Education

Third Year

Sections: Literature and Humanities
Sociology and Economics

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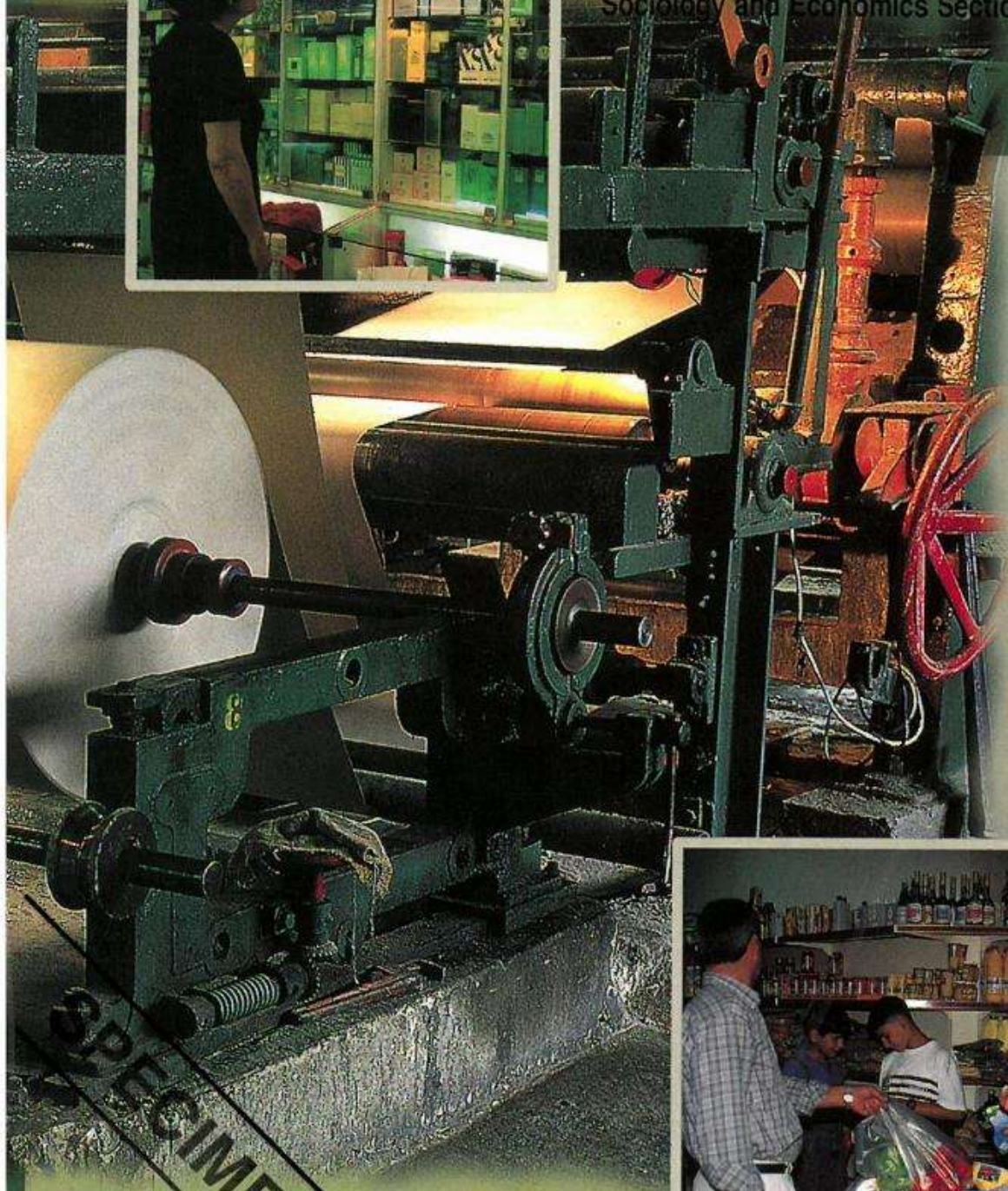
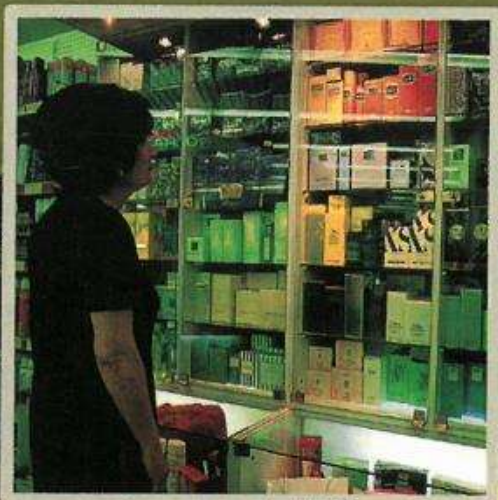
Educational Center for Research and Development



Chemistry

Secondary Education
Third Year
Literature and Humanities
Sociology and Economics Sections

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National
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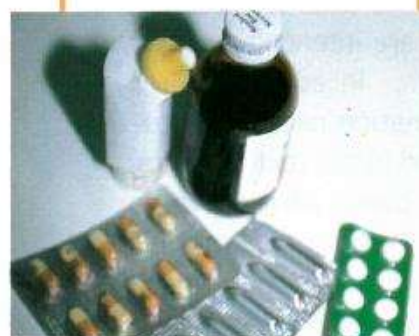
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INTRODUCTION

The chemistry textbook for the third year secondary Literature and Humanities and Sociology and Economics sections attempts to achieve the goals of the scientific culture chemistry curriculum. The main goal is to prepare students to, based on scientific data and information, take efficient decisions in situations related to everyday activities, especially to health and environment.

The curriculum of the Literature and Humanities section consists of three units, food chemistry, perfumes and cosmetics, and current medicinal drugs. The Sociology and Economics curriculum contains two additional units, treatment of wastes and chemistry and economy.

Each unit is composed of one or more chapters. The content and the experimental activities of each chapter are intended to cover the stated learning objectives of the curriculum. In addition, three types of extra information are presented. Information needed by students to better understand the content is included in the text in *Italics*. Information found to be important to widen students' perspectives is included at the margins. Information on special, usually recent, topic found to be important and of interest is included at the end of each chapter as a document. Each chapter also concludes with review questions and research activities. At the end of each unit, a variety of unit questions are proposed to evaluate students' understanding of the whole unit.

It is our hope that this chemistry book, together with the other science books for the Literature and Humanities and Sociology and Economics sections, serve to the attainment of the general scientific culture goals. Comments and suggestions from all who are involved and interested in the educational process are always welcomed.

Authors

The National Textbook Project

The Center for Educational Research and Development (CERD) has embarked on an extensive workshop for assessing and developing the educational framework and curricula which have been placed into effect more than three years ago. With full realization of the fact that the educational cycle must continue normally through its components, and until the development process attains its aspired objectives, we are placing in the hands of students, teachers and directors of public schools, this corrected version of textbooks issued by CERD as part of the National Textbook Series.

This version is an interim stage incorporating the corrected typographical and linguistic errors discovered by CERD specialists as well as teachers and students through their daily dealings with the books. The process of assessment and development of the framework and curricula will take into consideration all the comments that have been made, or will be made, in this regard.

It is expected that once the curricula are developed and aligned with the general and specific objectives set for them, the textbooks will be realigned with the new curricular and framework requirements, including tying the content of a course to the number of teaching hours set for it during the school year, taking into consideration vertical alignment within the same course as well as the horizontal alignment with the rest of the courses.

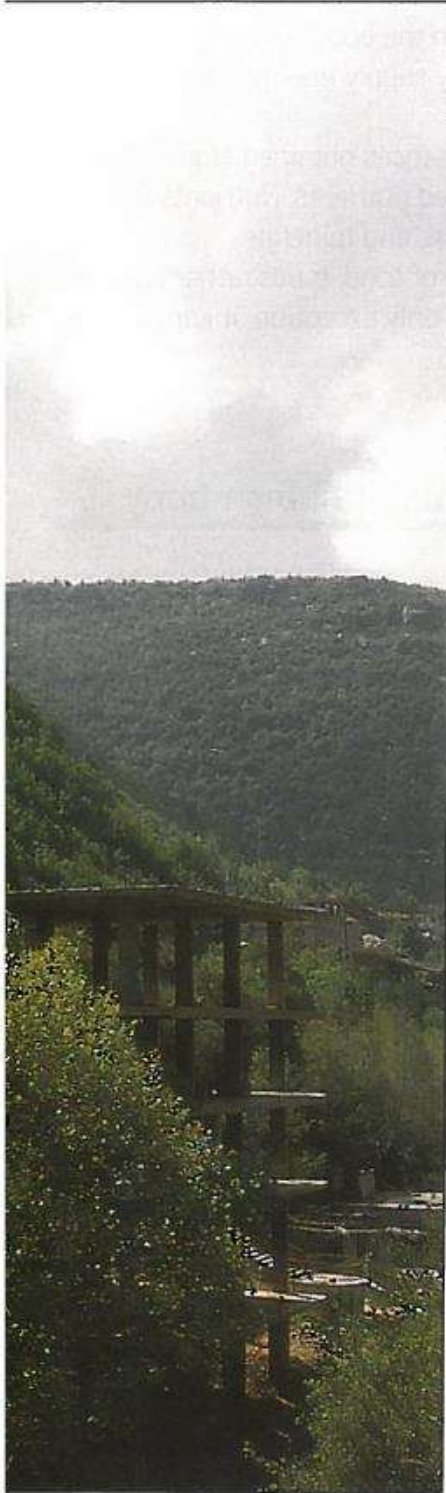
I take this opportunity to invite all school administrators, teachers and students and all officials concerned in public and private schools alike, to promptly send their comments on these curricula and books as their contribution to enrichment of this momentous national process.

This workshop, which was launched under the kind sponsorship of His Excellency the Minister of Education and Higher Education in implementation of Decree No. 10227 embodying the educational curricula and their objectives, fits in with CERD's proclaimed new motto "Together We Build Through Education".

It is our earnest desire to see this national, all-inclusive workshop attracting the greatest amount of interest and participation to define the safest and soundest educational options that directly affect our children, as we vow to continually modernize education and develop its ways and means to keep abreast of modern developments and progress in science and technology.

Dr. Leila MALEEHA
President CERD

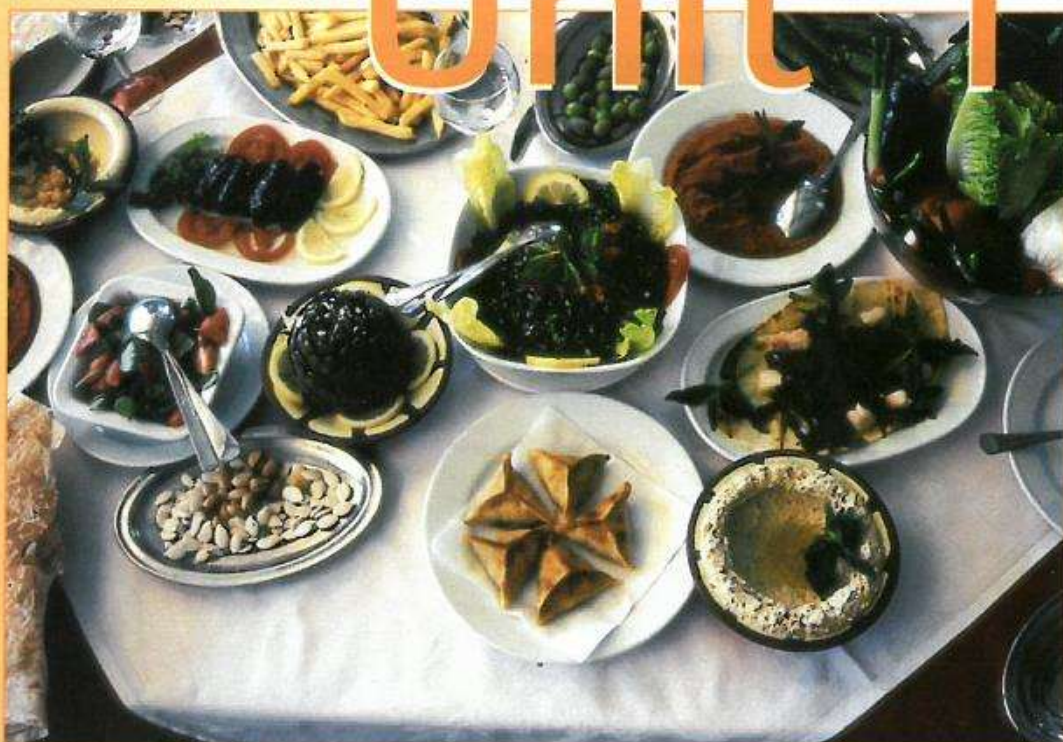
Carbohydrates 1



Objectives:

- Define food and nutritional requirements of human body.
- List the functions of carbohydrates in the human body.
- Define and classify carbohydrates.
- Identify the most important monosaccharides, disaccharides, and polysaccharides.

Unit 1



FOOD CHEMISTRY

- Chapter 1: Carbohydrates
- Chapter 2: Lipids
- Chapter 3: Proteins
- Chapter 4: Minerals and Vitamins
- Chapter 5: Nutritional Requirements
- Chapter 6: Principal Foods
- Chapter 7: Food Diet



3- Galactose

Galactose is an aldohexose like glucose with the same general formula. The two compounds differ only in the position of the hydroxyl group on C₄. (Figure 1.6).

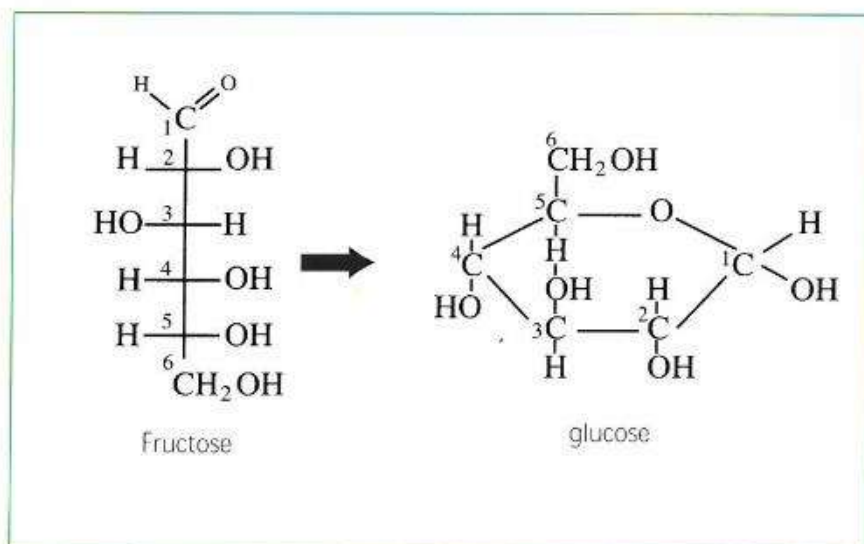
Galactose has a less sweet taste than sucrose. It is one of the components of algae and linseed. It is the most abundant sugar after glucose but it is seldom encountered as a free monosaccharide.

Fructose	Fru
Galactose	Gal
Glucose	Glu
Ribose	Rib

Table 1.1 Abbreviations of common monosaccharides

Cyclic structure of monosaccharides

The linear structural formulas are good but incomplete representation of monosaccharides. Certain properties cannot be explained by using the linear formula so it is necessary to refer to the cyclic form. In the cyclic form, the carbonyl group (on C1 or C2) reacts intramolecularly with the hydroxyl group on C5 (in general). The result is a cyclic form of 5- or 6- membered ring.



In the crystalline solid state, monosaccharides are found in the cyclic form. In solutions, the two forms, linear and cyclic, coexist.



Fig. 1.5: Fructose is found in honey in equal amount as glucose.

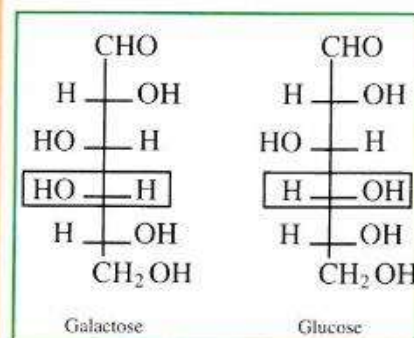


Fig. 1.6: Structural formulas of glucose and galactose.

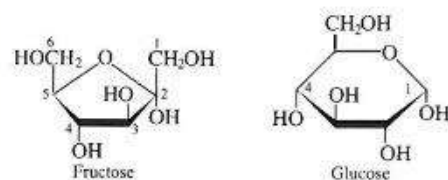


Fig. 1.7: Cyclic forms of fructose (5-membered ring) and glucose (6-membered ring).



Fig. 1.2: Serum contains the monosaccharide glucose.

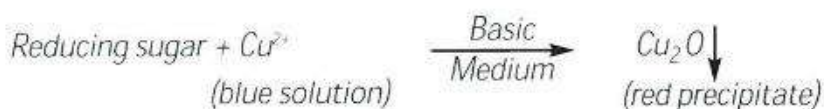
Blood sugar is glucose with the concentration of 1g/l.



Fig 1.4: Glucose is grape sugar.

In this chapter, the carbon chain of monosaccharides is written vertically.

mine qualitatively and quantitatively reducing sugars. For example:



Glucose, fructose and galactose are three monosaccharides. They have the same molecular formulas, $\text{C}_6\text{H}_{12}\text{O}_6$, but different structural formulas.

1- Glucose (Dextrose)

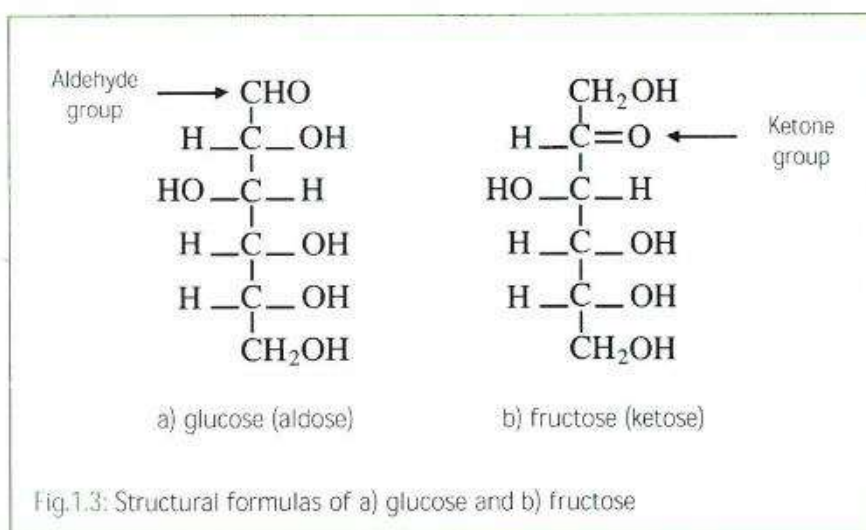
Glucose is often referred to as grape sugar and is found naturally in fruit juice, honey (35%), and vegetables such as onions and potatoes. It has a less sweet taste than sucrose (table sugar).

Glucose is an aldohexose (Figure 1.3 a) with the carbonyl group (aldehyde) on C_1 and the hydroxyl groups on C_2 , C_3 , C_4 , C_5 , and C_6 . It is a reducing sugar.

2- Fructose (Levulose)

Fructose is found in sweet fruit juice. In honey, it is found in equal amounts with glucose. Fructose is a little bit sweeter than sucrose. It is difficult to crystallize.

Fructose is a keto-hexose with the same general formula as glucose ($\text{C}_6\text{H}_{12}\text{O}_6$). Its structural formula (Figure 1.3 b) shows the carbonyl group (ketone) on C_2 . Like glucose, fructose is a reducing sugar.



1.1 What is food?

Foods are substances taken into the body by mouth and they maintain life and growth. That is, they supply energy and build and replace tissues.

Most foods are complex substances obtained from animals and plants. Food components are called nutrients. Nutrients include carbohydrates, lipids, proteins, vitamins, and minerals.

There is no one complete kind of food, foods rather complement each other. Mother's milk is the only exception, it can be considered a complete food.

1.2 Nutritional requirements of human body.

A balanced diet must cover the nutritional requirements of the human body in such a way that an individual is in good physical and mental conditions. Some of these requirements are to renew cellular constituents and to build body tissues of young people. These are called structural requirements and can be gotten from water, some minerals (calcium and phosphorus), and proteins. Other requirements are for energy and they can be gotten mainly from carbohydrates and lipids and partly from proteins. Finally, life would not be possible without some substances needed for protection and control of body functioning. These are vitamins and some minerals such as sodium and potassium.

The idea of nutritional balance is complex because this balance is a function of individual needs, activities, physiological state, and environment.

1.3 What are carbohydrates?

Carbohydrates are the most abundant molecules on earth. Every year, photosynthesis, from plants and algae, transforms more than 100 billion tons of carbon dioxide and water to carbohydrates such as cellulose and starch. Certain carbohydrates, such as sugar and starch, are the essential constituents of food for humans in many regions of the world. Oxidation of carbohydrates is considered the main source of energy for many non-photosynthetic cells and thus is considered the fuel for the human body.

Simple carbohydrates are substances composed of carbon,

B- Oligosaccharides:

Oligosaccharides are carbohydrates that contain two to ten monosaccharide units. Disaccharides are those in which two monosaccharides are linked together by a glycosidic linkage.

The glycosidic linkage is the carbon-oxygen-carbon bond formed between the hydroxyl group of one sugar and the carbonyl group of another or the carbonyl groups of two sugars, with the elimination of water molecule.

In the first case, the resulting disaccharide acts as a reducing agent since the reducing carbonyl group of one of the monosaccharides is free, i.e. is not engaged in the formation of the glycosidic linkage. In the second case, the resulting disaccharide is not a reducing sugar since the two carbonyl groups are engaged in the formation of the glycosidic linkage.

The condensation of two monosaccharides gives a disaccharide with the elimination of water. The inverse reaction, hydrolysis of disaccharides by acids or enzymes, produces the two monosaccharides.

The most important oligosaccharides are disaccharides. Sucrose, lactose, and maltose are three disaccharides.



Fig. 1.8: Cubes of sugar.



Fig. 1.9: Lactose is milk sugar.

1- Sucrose

Sucrose is common table sugar that is produced commercially from beets and sugar cane. It results from the condensation of the two monosaccharides glucose and fructose. Sucrose is a very important carbohydrate food. It is the only sugar that is crystallized and purified to be used by humans. It can be easily assimilated in the human body.

2- Lactose

Lactose is the major sugar found in milk. It is synthesized in the mammary glands and is found in milk in variable rates ranging from 10 to 80 g/L. Lactose is formed by the condensation of galactose and glucose. It can be hydrolyzed either by an acid or by the enzyme lactase forming an equimolar mixture of glucose and galactose. Lactose is slightly soluble in water at room temperature (10 times less than sucrose). It has less sweet taste than sucrose (1/6 that of sucrose).

hydrogen, and oxygen with the general formula $C_n(H_2O)_m$. The most abundant carbohydrates contain six or multiple of six carbon atoms per molecule. Examples are glucose with the general formula $C_6(H_2O)_6$ and sucrose with the general formula $C_{12}(H_2O)_{11}$. Furthermore, carbohydrates contain a carbonyl group (aldehyde or ketone) and hydroxyl groups.

1.4 Role of carbohydrates in the human body

Carbohydrates are the most important organic substances that enter in the chemical composition of plants and animals. They have the following main functions in the human body:

- Carbohydrate polymers linked to groups such as proteins and lipids play an important role in the functioning of the organism.
- Carbohydrate storage, in the form of glycogen, provides a short-term energy reserve.

Carbohydrate oxidation provides energy for the body.

1.5 Classification of carbohydrates.

Carbohydrates are classified on the basis of their molecular size into monosaccharides, oligosaccharides, and polysaccharides.

A Monosaccharides

Monosaccharides are white crystalline solids that are soluble in water and insoluble in non-polar solvents. Most of them have sweet taste. They are composed of non-branched chain in which all carbon atoms are linked by simple bonds.

Monosaccharides are the simplest units of carbohydrates that can not be broken down by hydrolysis. According to the nature of the carbonyl group, they can be classified into aldoses (aldehyde group) and ketoses (ketone group).

Monosaccharides can act as reducing agents due to the presence of the carbonyl group which can be oxidized by oxidizing agents in basic medium (Fehling solution).

Salts of heavy metals (Cu^{2+} , Ag^+ , Fe^{2+} , etc...) are reduced by heat in basic medium forming a precipitate. This reaction is used to deter-

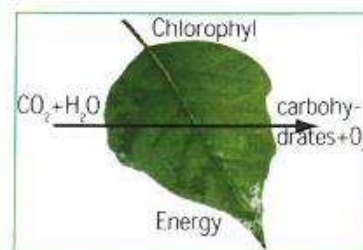


Fig.1.1: Photosynthesis of carbohydrates.

The term "saccharide" comes from the Latin word "saccharum" meaning sugar.

Monosaccharides can also be classified according to the number of carbon atoms per molecule:

- Trioses (3 carbon atoms)
- Tetrose (4 carbon atoms)
- Pentose (5 carbon atoms)
- Hexose (6 carbon atoms)
- Heptose (7 carbon atoms)

Activity

Activity 1.1: Detecting reducing sugars

Objective:

To identify reducing sugars using fehling test

Materials:

- Fehling solution A
- Fehling solution B
- 5 g/L solutions of : glucose, fructose, lactose, and sucrose
- Test tubes and test-tube rack
- Bunsen burner
- Test-tube holder
- Balance
- Spatula
- Graduated cylinders (10 mL)

Procedure:

- In a test tube place 2 mL of each of solution A, solution B, and glucose solution.
- Heat to boiling
- Record observations
- Repeat for fructose, lactose, and sucrose solutions.

Questions

1-What is a positive fehling test?

2-Which of the above sugars is a reducing sugar?

3- Maltose

Maltose is a disaccharide that is made up of two glucose units (figure 1.10).



It is produced whenever starch is hydrolyzed as when seeds germinate in plants and during starch digestion in human beings.

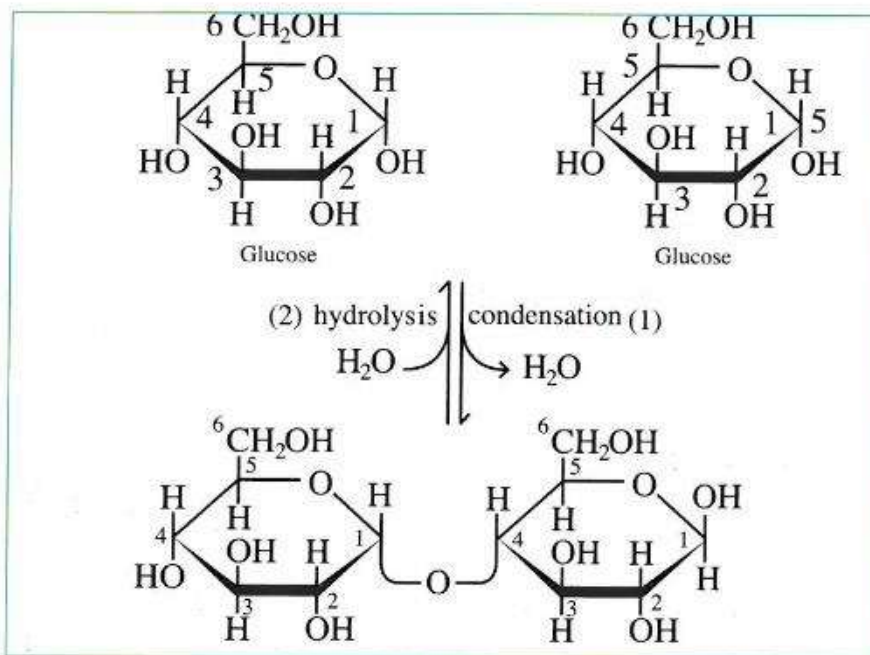


Fig. 1.10: (1) Formation of maltose and (2) hydrolysis of maltose.

2- Glycogen

Glycogen was isolated from the liver by Claude Bernard in 1856. It is referred to as animal starch for it is the essential storage polysaccharide in animals. Liver and muscle cells are the storage sites for glycogen in animals. Glycogen also exists in bacteria, algae and yeast.

Glycogen is a highly branched polymer of glucose units that can be represented by a tree structure. Like starch, glycogen gives glucose upon hydrolysis in acidic medium. Glycogen is soluble in water at room temperature and gives brown color with iodine.

3- Cellulose

Cellulose is the structural component of the cell walls of plants. It is a fibrous substance that is insoluble in water. It is a linear unbranched polymer of glucose units joined by glycosidic linkages. Cellulose is highly resistant to hydrolysis and necessitates the action of strong acid. It cannot be hydrolyzed in the human's digestive tract because humans lack the enzyme (cellobiase) capable of catalyzing the hydrolysis of cellulose. So, cellulose cannot be considered a source of nutrition for human beings. Ruminant animals can digest cellulose because the enzyme (cellobiase) is secreted by microorganisms found in the intestinal tract of those animals. Cellulose (fiber) has favorable action on the digestive transit. The indigestible "bulk" helps peristalsis.

The long cellulose chain contains large number of hydroxyl groups, which can associate together by hydrogen bonding to form a fibrous, compact, and nonsoluble structure, that is why they are used in paper industry.

C- Polysaccharides

A polysaccharide is a polymer that contains hundreds or thousands of monosaccharide units bonded to each other by glycosidic linkages. In some polysaccharides, as cellulose, the monosaccharides are bonded together in a linear chain. In others, like glycogen, there is extensive branching of the chains. Polysaccharides differ in the physical and chemical properties from the monosaccharides they are formed of. Polysaccharides differ from one another because of the nature of monosaccharide units, the type of linkage between them, the length of the chain and the degree of branching.

Starch, cellulose and glycogen are polysaccharides that result from the condensation of large number of the same monosaccharide, glucose. Starch and glycogen are the most important storage polysaccharides in nature (starch in plants and glycogen in animals).

1- Starch

Starch is a polysaccharide which only contains glucose units. It is found in plant tissues in a dense granular form. It is formed in large amounts in tubercles like potatoes and cereal grains like corn and rice. It is also found in high quantities in fruits like bananas and several legumes.

Two different polysaccharides can be isolated from most starches:

amylose and amylopectin. Amylose, an unbranched glucose polymer, usually accounts for 20 to 30% of the starch. It has a helical structure that forms a blue color complex with iodine. Amylopectin, a highly branched glucose polymer, accounts for the remaining 70 to 80% of the starch. It absorbs water during cooking causing swelling of the starch granules. Starch rich in amylopectin is easier to dissolve in water at 95 °C than starch rich in amylose.

Grains rich in starch :

Wheat:60 to 70 %

Rice :70 to 80 %

Corn :65 to 75 %

Percentage of amylose and amylopectin in natural starches.

	Amylose%	amylopectin%
Potatoes	23	77
Wheat	20	88
Corn	25	75
Sweet peas	40	60
Bariana	17	83

a) Amylose molecule consists of 50 to 500 glucose units joined in a linear chain.



b) Amylopectin molecule consists of up to 100,000 glucose units joined in a branched chain structure.

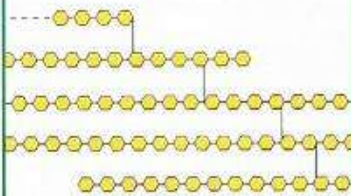


Fig.1.11: (a) Amylose and (b) Amylopectin are polysaccharides of starch.

Reversible transformation between starch and glucose that occurs when fruits matures (after harvesting) has a great influence on the quality of the fruit (texture).

• Hydrolysis of starch:

Starch can be hydrolyzed by the action of an acid or an enzyme. Heating starch in acidic medium breaks down starch molecules into shorter chains of glucose units known as dextrines. Dextrines will further break down into the disaccharide maltose and finally into glucose.



Hydrolysis of starch may also be brought about by enzymatic action. During digestion, the enzyme amylase, found in saliva and in pancreas, breaks down starch into maltose.

• Effect of heat

Starch is insoluble in cold water. If a suspension of starch in water is heated to about 60 to 80°C, starch granules will absorb water and may swell to about five times their volume. At 80°C, the starch granules break up and the content becomes dispersed in water forming a water-starch viscous solution. If the solution is enough concentrated a gel will be produced. This gelatinous material is important in cooking, it thickens sauces and soups.

The strength of a starch gel depends on the following factors:

- the proportions of starch and water present. The more starch is the stronger the gel will be.
- the proportion of amylose in the starch. Gelling increases with the increase of amylose in the starch.
- the presence of sugar. Sugar reduces gel strength.
- the presence of an acid. Acids hydrolyze starch and reduce gel strength forming a paste.

Lipids

2



Objectives

- Define and classify lipids.
- List the functions of lipids in the human body.
- Identify triglycerides, phospholipids, and cholesterol.



Activity

Activity 1-2 :Testing foods for starch

Objective:

Detect starch in different foods using iodine solution.

Materials

- Tincture of iodine solution
- Starch, sucrose, bread, potato, apple

Procedure:

- Add one or two drops of iodine solution to the starch and sucrose samples.
- Record your observations
- Add one or two drops of iodine solution to the different samples of foods.
- Record your observations.

Questions

- 1-What is a positive test for starch?
- 2-Which of the tested foods contain starch?

Review Questions

1. Define food.
2. Identify the human nutritional requirements.
3. Define carbohydrates.
4. List the functions of carbohydrates in the human body.
5. Name the three classes of carbohydrates.
6. Define monosaccharides, oligosaccharides and polysaccharides.
7. Identify the three monosaccharides: glucose, fructose, and galactose.
8. Identify the three disaccharides: sucrose, lactose, and maltose.
9. Identify the three polysaccharides: starch, cellulose, and glycogen.
10. Write the molecular and the linear structural formulas of glucose.
11. Write the word equation of the formation of sucrose, lactose, and maltose.
12. Describe the solubility and hydrolysis of starch.
13. Describe the effect of heat on starch.

Document 1

The secret of fiber

The fiber, bulk, or roughage of our foods is largely cellulose, a carbohydrate we find indigestible because of our lack of the enzyme cellobiase.

Foods rich in fiber include fruits, vegetable, bran, and nuts. For maintaining good health it is recommended that these be part of our daily diets.

Dietary fiber seems to reduce the risk of cancer, especially cancer of colon. How and why dietary fiber might have this beneficial effect is uncertain. One bit of speculation holds that the secret of fiber may lie in its mechanical effect on our large intestine. As fiber absorbs a large amount of water when it passes through us, it may stimulate the intestine and promote a rapid transit of the fibrous bulk through out our bodies. This rapid passage of fiber also speeds the elimination of other substances, cancer-causing substances, so their contact time with intestinal tissue is shortened, and their opportunity for acting on intestinal tissue and generating cancer is diminished.

Another possibility may simply be that as we increase the proportion of fiber-containing foods in our diets, we necessarily decrease the proportion of meats and other fatty foods. Since diets high in fats seem to be associated with cancer, this hypothesis could easily account for the protective effect of fiber.

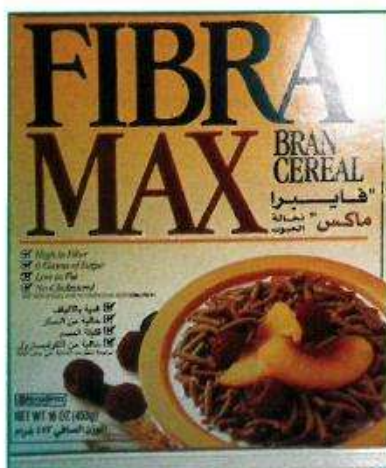


Figure 1.12: Fiber, including the fiber of grains, forms an important part of our diets.

The term "lipid" comes from the Greek "lipos", which means "fat".

A wax is a monoester formed from the reaction of a long chain alcohol with a fatty acid molecule. Waxes are usually used as protective coatings. They are also widely used in the cosmetics industry.

Essential fatty acids are fatty acids needed by the human body. They must be obtained from dietary sources because they can not be synthesized within the body from other substances. There are two essential fatty acids: linoleic acid and linolenic acid. Both are 18- carbon polyunsaturated acids.

2.1 what are lipids?

Lipids are a class of natural compounds that include triglycerides (fats and oils), waxes, phospholipids, and steroids. Most of these compounds are esters of fatty acids and alcohol. Their common property is insolubility in water and solubility in organic solvents such as ether, chloroform, and ethanol. Lipids in food can be identified by the formation of a translucent grease spot on paper.

2.2 Role of lipids in the human body

Lipids have various functions. Fats and oils are the principal form of energy storage. They are an important source of energy in the human body. Other lipids such as steroids and phospholipids constitute the main components of cell membranes. Cholesterol plays a vital biological role in chemical synthesis within the human body. It is the starting material for the synthesis of numerous steroid hormones and vitamin D.

2.3 Fatty acids

Most lipids are natural derivatives of fatty acids. Fatty acids are carboxylic acids that contain long, unbranched hydrocarbon chain generally 4 to 36 carbon atoms in length.

They are classified as saturated (the chain contains no double bonds) and unsaturated (the chain contains one or more double bonds).

The physical properties of fatty acids and their derivatives are mainly determined by the length of their hydrocarbon chains and their degree of unsaturation (number of double bonds per molecule). For example, saturated fatty acids containing 12 to 24 carbon atoms in their chain are solids at room temperature while unsaturated fatty acids of the same length are liquids.

The length of the chain and the number of double bonds of fatty acids can be identified by two numbers: the number of carbon atoms and the number of double bonds. For example palmitic acid, saturated fatty acid composed of 16 carbons and no double bonds, is represented by 16 : 0, and oleic acid, an unsaturated fatty acid composed of 18 carbons and one double bond, is represented by 18: 1.

Hydrogenation affects the nutritional value of lipids. For example, hydrogenation of linoleic acid decreases the amount of essential fatty acids. Hydrogenation also decrease the value of fat soluble vitamins.

The degree of unsaturation (number of double bonds) of fatty acids or glycerides can be represented by the iodine value. Iodine value can be determined experimentally by the fixation of iodine molecules on the double bonds according to the following reaction:



As unsaturation of fats increases the iodine value increases.



Fig. 2.5: Margarine is obtained from liquid oil.

$$\text{Iodine value} = 0.2 \times 127 (V_2 - V_1) / 10 \times m$$

where 0.2 = normality of the thiosulfate solution

and 127 = atomic weight of iodine

-Repeat the same procedure for butter.

Questions

- 1-How do the iodine values of the two fatty substances compare?
- 2-What can you conclude?

2.6 Cholesterol

Steroids constitute a third category of lipids. They are lipids with structures based on the steroid nucleus. (Figure 2.6)

Cholesterol is the most abundant steroid in the human body. It is not found in plants. Cholesterol is an alcohol with an hydroxyl group on carbon 3 of the steroid nucleus. In addition, cholesterol has methyl groups bonded to carbon atoms 10 and 13 and a small branched hydrocarbon chain on carbon 17 (Figure 2.7).

Review Questions

- 1- Define lipids.
- 2- List the functions of lipids in the body.
- 3- Name the two main groups of lipids.
- 4- Define triglycerides, phospholipids, and cholesterol.
- 5- Distinguish between fats and oils.
- 6- Define saturated and unsaturated fatty acids.
- 7- Describe the hydrogenation of fats.

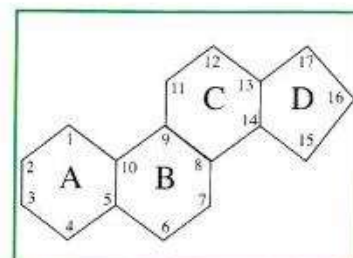


Fig. 2.6: Steroid nucleus formed from 4 fused rings: three are 6- membered rings and the fourth is a 5- membered ring.

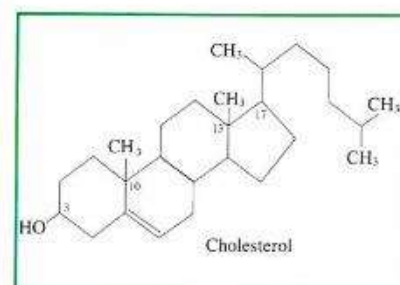


Fig. 2.7: Cholesterol is found in cell membranes and nerve tissues.

Activity

Activity 2.1: Determining the iodine value of fats.

Objective: Determine the iodine value of soybean oil and butter

Materials:

- Wijs reagent
- Potassium iodide solution (100 g/L)
- Sodium thiosulfate solution (0.2 N)
- Carbon tetrachloride
- Distilled water
- Starch solution
- Soybean oil and butter
- Two 250mL Erlenmeyer flasks with stoppers
- Pyropipette
- Graduated cylinders (100, 50, and 20 mL)
- Burette and stand
- Balance
- Small weighing tube

Procedure:

- In a weighing tube weigh an exact mass of about 0.2 g soybean oil (mass= m).
- Introduce this tube into a dry erlenmeyer flask.
- Add 20 mL carbon tetrachloride.
- Stir regularly till the oil is dissolved. Make sure that the carbon tetrachloride does not splash on the stopper.
- Add, using pyropipette, 20 mL of wijs reagent.
- Leave the closed flask in the dark for about 15–20 minutes. Stir once or twice.
- Add, using graduated cylinders, 100 mL of distilled water and 20 mL of potassium iodide solution.
- Titrate using the 0.2 N sodium thiosulfate solution till the brown color changes to colorless. Starch solution is used as an indicator. Let V_1 be the volume of the sodium thiosulfate used to titrate the soybean oil solution.
- Repeat the above experiment, except for the use of soybean oil, in another erlenmeyer flask containing 20 mL of wijs reagent and 20 mL of carbon tetrachloride. Let V_2 be the volume of the sodium thiosulfate solution used to titrate this solution.
- Calculate the iodine value using the following equation:

B- Complex lipids

They have the same structure as simple lipids except that one of the fatty acid residues is replaced with other groups such as groups containing N, P, or S, and sugar (either one or more). Phospholipids are complex lipids that contain in their structure a phosphoric acid residue. They play an important role in the functioning of the human body. These substances are widely found in small quantities except in egg yolk and nervous tissues where they are found in large quantities.

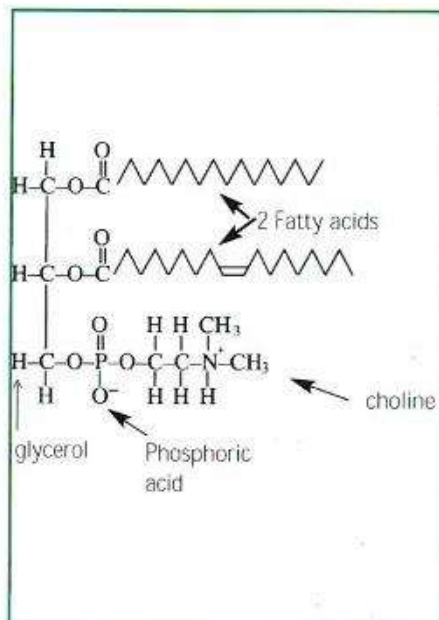


Fig. 2.4 (b): Structural formula of a phosphatidyl choline molecule containing stearic acid (18:0) and oleic acid (18:1).

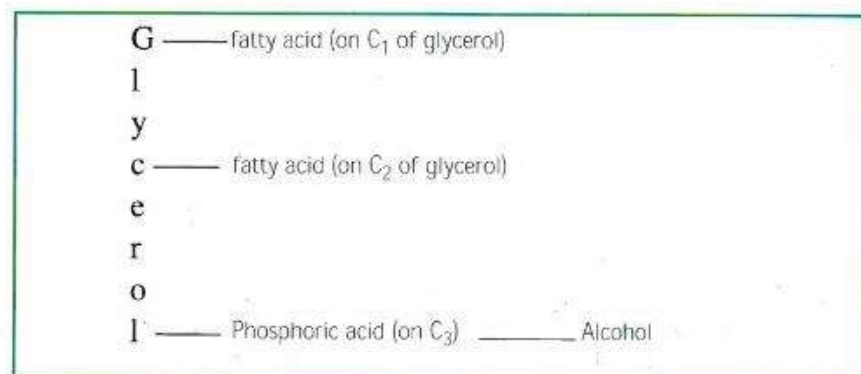
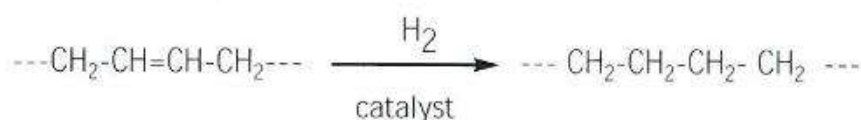


Fig. 2.4 (a): General structure of phospholipids: R₁ and R₂ are two long chain fatty acids linked to glycerol through C₁ and C₂. C₃ of the glycerol is linked by a phosphoester bond to an amino alcohol.

2.5 Hydrogenation of fats

Hydrogenation is a reaction where hydrogen is added across the carbon – carbon double bonds of the unsaturated fatty acids of triglycerides, thus decreasing the degree of unsaturation as some double bonds are converted into single bonds. With this change, there is a corresponding change in the physical and chemical properties of triglycerides. Hydrogenation increases the melting point of the hydrogenated fat and thus changes the liquid fat to solid fat at room temperature. Hydrogenation is used to produce solid shortenings or margarines from liquid vegetable oils.

Hydrogenation is accomplished by passing hydrogen gas over the oil in the presence of Ni or Cu as a catalyst. The oil is heated to more than 100 °C for a short time at high pressure (more than 760 torrs). After hydrogenation, oil is centrifuged and filtered to recover the catalyst.



2.4 Classification of lipids

Lipids can be classified as simple and complex lipids.

A— Simple lipids

They are esters of fatty acids and alcohols which contain C, H and O in their molecules. For example, glycerides are esters of the alcohol, glycerol, and fatty acids (Figure 2.1). Triglycerides are the most abundant natural simple lipids.

The three polar hydroxyl groups of glycerol and the polar carboxyl groups of fatty acids are attached by an ester bond to give triglycerides, non-polar molecules (neutral fats). They are hydrophobic: strongly insoluble in water. This explains why a mixture of water and oil forms two phases. Oil floats on water because its density is less than that of water.

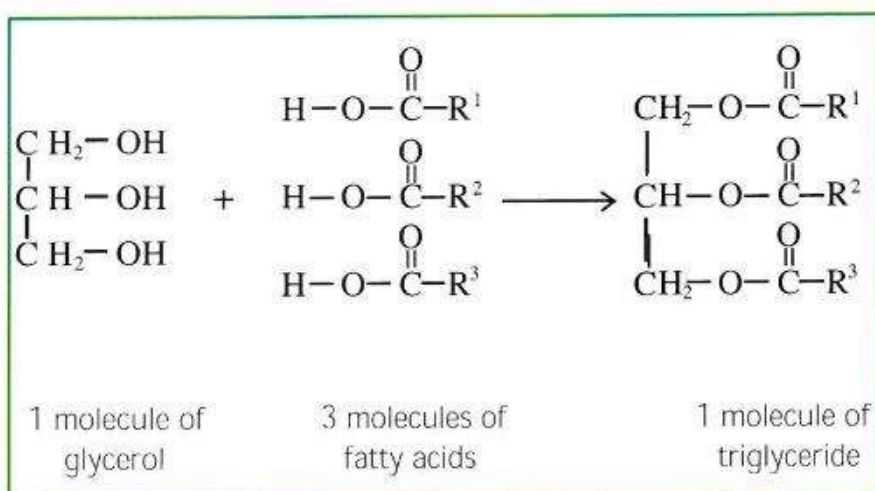


Fig.2.2: Formation of triglycerides from glycerol and fatty acids. (R_1 , R_2 , and R_3 may be identical or different).

In general, at room temperature, fats are solids and oils are liquids. Fats are obtained from animal sources, and oils are obtained from plant sources.

The melting points of triglycerides depend on both the length of their hydrocarbon chains and their degree of unsaturation. The melting point decreases with the increase in the degree of unsaturation and in the length of the hydrocarbon chain.

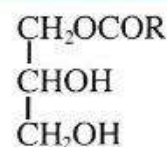


Fig. 2.1 a): Monoglycerides result from the esterification of one hydroxyl group of glycerol with one fatty acid.

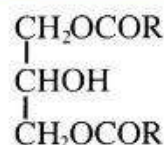


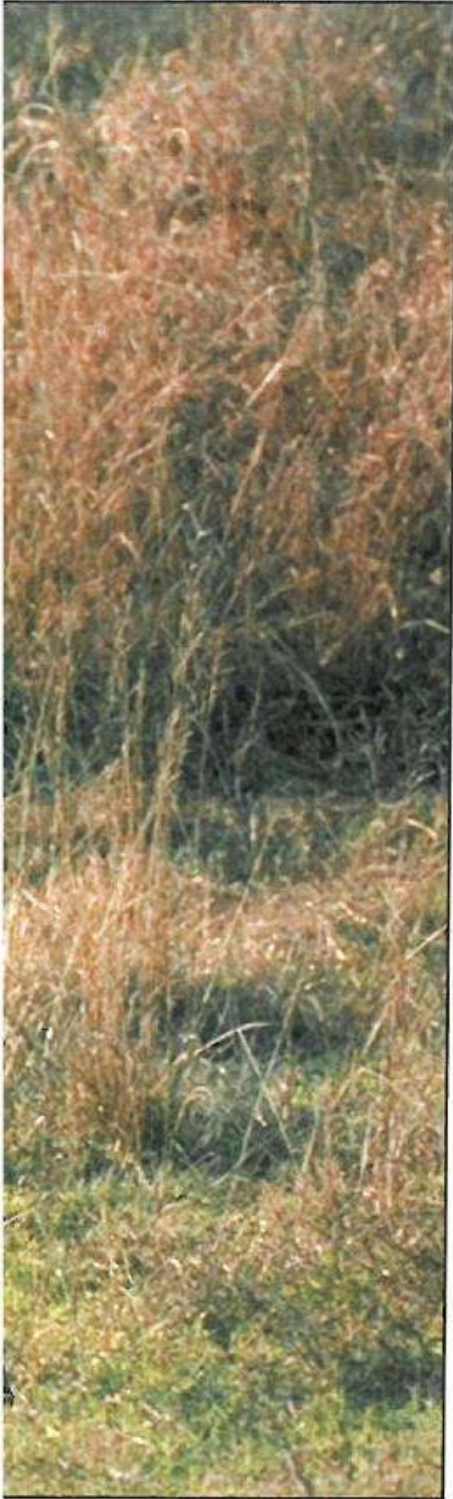
Fig. 2.1 b): Diglycerides result from the esterification of two hydroxyl groups of glycerol with two fatty acids.



Fig 2.3: Olive oil and sunflower oil are liquids at room temperature.

Proteins

3



Objectives

- Define proteins.
- Describe the structures of proteins.
- Explain denaturation of proteins.
- List the functions of proteins in the human body.

The word "protein" comes from the Greek "proteios" which means "of prime importance"



Fig.3-1: Keratin is the protein of hair, nails, horns ...

3.1 What are proteins?

Proteins are macromolecules or polymers in which the monomer units are amino acids. They are the most abundant macromolecules in living cells. Millions of different proteins can be found in one cell. Cells can produce proteins of different properties and activities by joining the same twenty amino acids in different combinations and sequences.

Proteins constitute 80 to 90% of the dry weight of the total organic material in animals and 15 to 20% in plants. Therefore, proteins are essential for life.

Proteins are polymers formed by joining monomers (twenty amino acids) by a special type of covalent bonding called peptide bond.

All proteins contain carbon, hydrogen, oxygen, and nitrogen. Some proteins also contain sulfur, phosphorus, copper, or iron. Proteins differ from carbohydrates in that they contain nitrogen.

3.2 What is an amino acid?

An amino acid is an organic compound that contains at least one carboxylic acid group ($-\text{COOH}$) and one basic amino group ($-\text{NH}_2$). The amino acids found in proteins are α -amino acids in which the amino group is attached to the α -carbon atom of the carboxylic acid chain. The general structural formula of an amino acid is shown in Figure 3.2. In an α -amino acid, the carboxyl group and the amino group are attached to the same carbon atom.

The nature of the side chain (R group) distinguishes α -amino acids from each other.

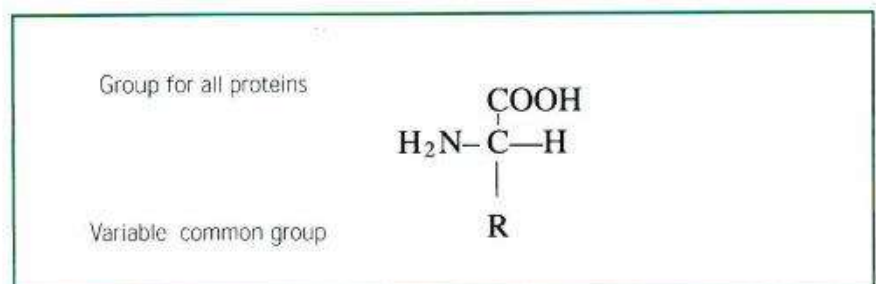


Fig.3-2: General structural formula of an α -amino acid



Artificial fat substitutes

Artificial sweeteners (sugar substitutes) have been an accepted part of the diet of most people for many years. New in the 1990s are artificial fats, substances that create the sensations of "richness" of taste and "creaminess" of texture in food without the negative effects associated with dietary fats (heart disease and obesity). It is dietary fats and oils that give food its rich flavor and creamy texture.

Food scientists have been trying to develop fat substitutes since 1960s. Now available for consumer use are two types of fat substitutes: calorie-reduced fat substitutes and calorie-free substitutes. They differ in their chemical structures and consequently in how the body handles them.

Simplese, the best-known calorie-reduced fat substitute, received marketing approval in 1990. It is made from the protein of fresh egg whites and milk by a procedure called microparticulation. This procedure produces tiny, round protein particles so fine that the tongue perceives them as a fluid rather than as being solid. Their fineness creates a sensation of smoothness, richness, and creaminess on the tongue, characteristics normally associated with dietary fat.

In the body, Simplese is digested and absorbed, contributing to energy intake. But 1 gram of Simplese provides 1 1/3 cal, compared with the 9 cal provided by 1 g of fat. Simplese is used only to replace fats in formulated foods such as salad dressings, cheeses, sour creams, and other dairy products.

Simplese is unsuitable for frying or baking, because it turns rubbery or rigid (gels) when heated. Consequently, it is not available for home use.

Olestra, a sucrose polyester and the best-known calorie-free fat substitute, received marketing approval in 1996. It is produced by heating soybean oil with sucrose in the presence of methyl alcohol. The resulting product has a structure somewhat similar to that of a triacylglycerol; sucrose takes the place of the glycerol molecule, and six to eight fatty acids are attached to it rather than the three in a triacylglycerol. Unlike triacylglycerols, however, Olestra cannot be hydrolyzed by the body enzymes and therefore passes through the digestive tract undigested.

Olestra looks, feels, and tastes like dietary fat and can be a substitute for fats and oils in foods such as shortenings, oils, margarines, snacks, ice creams, and other desserts. It has the same cooking properties as fats and oils.

In the digestive tract, Olestra interferes with the absorption of both dietary and body-produced cholesterol; thus it may lower total cholesterol levels. A concern about its use revolves around its interference with vitamin E absorption. All products containing Olestra must carry the following label: "Olestra may cause abdominal cramping and loose stool. Olestra inhibits the absorption of some vitamins and other nutrients. Vitamins A, D, E, and K have been added".

Adapted from : Stoker, H.S. (1998). Organic and biological chemistry. Boston: Houghton Mifflin Company. P.258

A c t i v i t y

Activity 3-1 :Denaturation of proteins

Objective: Carry out the denaturation of egg white by heat, acids, and salts.

Materials:

- 5 test tubes and tube rack
- 10 ml graduated cylinders
- Medicine dropper
- Spoon
- Albumin solution
- 5% acetic acid solution
- Table salt
- Bunsen burner

Procedure:

- Place 10 ml of the albumin solution into each of the 5 test tubes.
- Leave tube 1 as control.
- Add 3 drops of acetic acid solution to tube 3.
- Add one spoon of salt to tube 4.
- Add 3 drops of acetic acid and one spoon of salt to tube 5.
- Heat tubes 2 and 5 to boiling.
- Observe.

Questions

- 1-What happened in each tube?
- 2-How would you explain the above results?

The 20 amino acids constitute the alphabet of proteins. As words are made of different number of letters joined in different sequences, each protein is made of a number of amino acids joined in a proper sequence.

3.3 What's a peptide bond?

$$\begin{array}{c} \text{R}_1 \\ | \\ \text{H}_2\text{N}-\text{CH}-\text{C}(=\text{O})-\text{OH} + \text{H}-\text{N}-\text{CH}(\text{R}^2)-\text{COOH} \\ \downarrow \text{H}_2\text{O} \\ \begin{array}{c} \text{R}_1 \quad \text{H} \quad \text{R} \\ | \quad | \quad | \\ \text{H N}-\text{CH}-\text{C}(=\text{O})-\text{N}-\text{CH}-\text{COOH} \end{array} \end{array}$$

A chain of three amino acids joined by two peptide bonds constitutes a tripeptide. Four amino acids constitute a tetrapeptide. In general, chains of two to ten amino acids joined by peptide bonds are oligopeptides (molecular weights range from 200 to 1000 amu). Ten to 100 amino acids constitute polypeptides. Proteins are

3.7 Role of proteins in the human body

Proteins have important roles in the functioning of the human body. They serve structural as well as functional roles. Examples are:

- Enzymatic activity: enzymes are, partly or totally, made up of proteins. All cellular chemical reactions are catalyzed by enzymes.
- Transport: proteins of blood plasma transport molecules or ions from one organ to another. For example hemoglobin carries oxygen from the lungs to peripheral tissues.
- Nutritive proteins: examples are casein (protein of milk) and albumin (protein of eggs).
- Storage proteins: examples are proteins stored in some grains that are necessary for their growth during germination.
- Proteins of contraction: examples are actin and myosin in meat.
- Proteins of defense: examples are antibodies and immunoglobulins.
- Regulatory proteins: regulate some physiological activities. An example is insulin, which regulates carbohydrate metabolism.
- Structural proteins: are the major components of tissues like hair, nails, skin, bird feathers...

3.8 Biological value of proteins

The nutritional value of proteins varies with their source. Proteins can be obtained from both animal and plant sources. The biological value of a protein is used as a measure of protein quality. Biological Value (BV) is the percentage of absorbed protein which is converted to body protein.

$$BV = \frac{\text{Retained proteins}}{\text{Absorbed protein}} \times 100$$

In general, foods from animal sources have higher BV values than foods from plant sources.

Protein source	BV %
Eggs	97
Meat	82
Fish	79
Milk	77
Soya beans	73
Rice	67
Peas	64
Corn	50
Wheat	49

Table 3.2: Biological Value of proteins of different sources.

polypeptides with more than 100 amino acids and molecular weights greater than 10 000 amu.

Acidic, basic, or enzymatic hydrolysis of proteins produces the simple amino acids the protein is formed of.

3.4 Simple proteins and conjugated proteins

Some proteins like albumin of egg white, globulin, and glutelline are made up entirely of amino acid residues. These are called simple proteins. Conjugated proteins contain other chemical components in addition to the amino acids. These additional components are called prosthetic groups. Conjugated proteins may be further classified according to the nature of the prosthetic group to lipoproteins, glycoproteins, and phosphoproteins. Examples of conjugated proteins are shown in the table below:

conjugated protein	Prosthetic group	Example
Lipoproteins	Lipid	Lipoproteins of blood
Glycoproteins	Carbohydrates	Immunoglobulin
Phosphoproteins	Phosphate group	Casein of milk

3.5 Protein structure

The arrangement of atoms of one protein in space is called conformation. This is a well-defined structure that characterizes natural biological proteins.

Protein structure should be considered at four levels of organization. The combination of all four levels of structure controls the protein function.

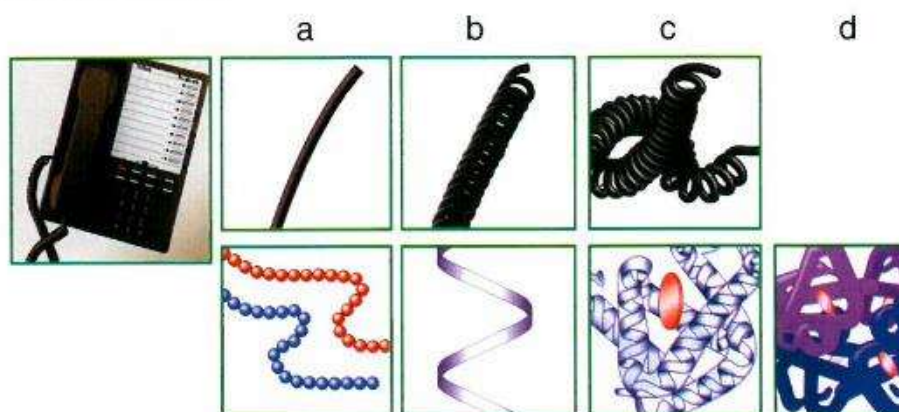


Fig 3.4 :

- a- Primary structure of proteins
- b- Secondary structure of proteins
- c- Tertiary structure of proteins
- d- Quaternary structure of proteins

1-Primary structure

The primary structure of a protein is the sequence of amino acids in its peptide chain (s). That is, the order in which the amino acids are connected to each other.

2-Secondary structure

Peptide chains are not necessarily straight but they usually have a helical or elliptical structure. This structure is referred to as the secondary structure of proteins.

3-Tertiary structure

Different types of chemical bonds contribute to the tertiary structure of proteins. They cause the same polypeptide chain or different chains to fold, twist, and bend resulting in the tertiary structure of proteins.

4-Quaternary structure

It is found only in proteins that have structures involving two or more polypeptide chains that are not covalently bonded to each other. The quaternary structure involves the associations among the separate chains in a protein.

3.6 Protein denaturation

Proteins of living animal and plant cells are called natural or native proteins. These proteins have definite organized structures that can be changed under special conditions, i.e denatured. Protein denaturation is the partial or complete disorganization of protein's characteristic three dimensional shape as a result of disruption of quaternary, tertiary, and secondary structural interactions. Protein denaturation does not affect the primary structure of proteins. Denatured proteins are generally less soluble in water than the initial proteins and they often precipitate from solution. Denaturation is, in general, irreversible and affects the physical properties of proteins such as viscosity and the biological properties such as the activity of enzymes and hormones .

Denaturing agents can be:

1-Chemical agents: such as acids, bases, ethanol and organic solvents, and ions such as chlorides, bromides, and iodides. Synthetic detergents are also active denaturing agents even at low concentrations.

2-Physical and mechanical agents: such as heat, radioactive rays, UV-rays, high pressure, and ultrasonic vibrations

Cooking food denatures the protein present but does not alter its nutritional value. The primary structure of proteins remains intact.

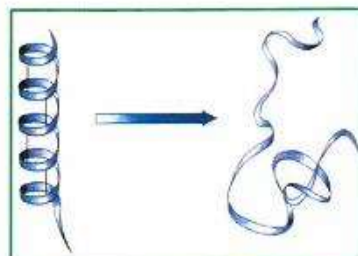


Fig.3.5 a): Denaturation of proteins.



Fig. 3.5 (b) Coagulation of egg proteins under the action of heat.

Review Questions

- 1- Name the main elements that constitute proteins.
- 2- Write the general formula of α -amino acids.
- 3- Define peptide bond.
- 4- Define protein.
- 5- Name the two classes of proteins.
- 6- Describe protein structure.
- 7- Define denaturation of proteins.
- 8- Name the physical and chemical factors that cause denaturation of proteins.
- 9- Define biological value of proteins.
- 10- Describe the functions of proteins in the human body.

Is and Vitamins

4



Objectives

- Define and classify minerals.
- Define and classify vitamins..
- List the functions of minerals and vitamins.

Document 3

Kwashiorkor

Kwashiorkor (Kwash-e-OR-kor), a potentially fatal condition, develops from a protein deficiency that results from the virtually complete absence of essential amino acids from the diet. The word kwashiorkor comes from the language of Ghana, a country on the west-coast of Africa. There, nursing infants normally receive sufficient protein from their mother milk. With the coming of a second child, the first is commonly weaned and placed on a diet consisting almost exclusively of readily available fresh fruits and plants rich in carbohydrates but poor in proteins such as potatoes. This new diet contains only small quantities of protein, and low-quality protein at that. The first born soon suffers from a severe protein deficiency and develops a variety of symptoms, including a swollen abdomen, loss of color from the hair, patchy and cracked skin, loss of appetite, and general restlessness. Death often follows. The word kwashiorkor originated as the name of an evil spirit that, tradition held, afflicted the first child after the second was born. Though kwashiorkor occurs principally in the tropics, its effects can be found worldwide.



Fig. 3.6: The loss of hair coloring and a swollen abdomen are typical symptoms of the protein deficiency known as kwashiorkor.

Mine





Fig. 4.1: Iodine is added to table salt to increase the iodine intake.

4.1 Minerals

Minerals are chemical elements other than carbon, hydrogen, oxygen, and nitrogen needed for the proper functioning of the human body. They are present in food, mostly in the form of inorganic salts, e.g. sodium chloride.

Minerals, such as calcium, play a structural role in the formation of bones and teeth and in the composition of secreted milk. Minerals also play important functional roles by activating the enzyme systems (trace elements) and controlling the pH of body fluids and their electrical neutrality. In addition, minerals are constituents of compounds that have special physiological functions such as hydrochloric acid in the stomach, iodine of the thyroid hormones, and iron of hemoglobin.

Mineral nutrients differ from organic nutrients (carbohydrates, lipids, and proteins) in that the body uses them, in general, in the form of mono-atomic ions (Ca^{2+} , K^+ , Na^+ , Mg^{2+} , Cl^-) or poly-atomic ions (HPO_4^{2-} , H_2PO_4^-).

Based on the quantity needed by the body, minerals are divided into two classes:

1-Macro minerals which are needed by the body in amounts greater than 100 mg/ day. There are six major macro minerals, calcium, phosphorus, magnesium, sodium, potassium, and chlorine.

2-Trace minerals are needed by the body in amounts less than 20 mg/ day. Fluorine, iron, copper, zinc, and iodine are important trace minerals.

4.2 Role of minerals in the human body

Calcium:

- forms and maintains both bones and teeth.
- regulates nerve transmission and muscle contraction
- is involved in blood clotting

Phosphorus:

- is part of bone structure
- is important in energy transfer
- is important to muscle contraction and nerve transmission

Potassium:

- is essential in nerve, muscular, and cardiac activities
- is involved in carbohydrate and protein metabolism

Chlorine:

- is involved in food digestion (HCl)
- keeps body fluids electrically neutral

Sodium:

- contributes to nerve transmission

Review Questions

- 1- Define minerals.
- 2- List the functions of minerals in the human body.
- 3- Name the two classes of minerals and give examples.
- 4- Define vitamins.
- 5- Name the two classes of vitamins and give examples.
- 6- List daily needs, sources, and functions of vitamins A, B series, D, E, folic acid, and niacin.

Research Activity

Conduct research on different ways of cooking and find out which is the best for retaining minerals and vitamins.

Document 4

Document 4-A Average requirements of minerals

Mineral	Unit*	Children** (per day)	Adult (per day)
Ca	mg	500-1200	600-1000
p	mg	500-1200	600-1000
NaCl	g	1-8	8-10
K	mg	-	3000-4000
Mg	mg	100-400	300-400
Fe	mg	10-20	10-20
Cu	mg	0.5-2	2.5-5
Zn	mg	5-10	10-20
I	µg	50-150	100-150

Document 4-B Average requirements of vitamins

Vitamin	Unit*	Children** (per day)	Adult (per day)
A	IU	2000-5000	3000-6000
D	IU	300-500	300-500
E	mg	5-10	10-20
B ₁ (thiamine)	mg	0.5-1.5	1-2
B ₂ (Riboflavin)	mg	1-2	1.5-2
B ₃ (Niacin)	mg	5-15	15-20
B ₁₂	mg	0.5-2	2-4
C	mg	30-50	40-60

*For vitamin A: IU = 0.3 µg retinol = 0.6 µg carotene = 0.35 µg retinol acetate

*For vitamin D: IU = 0.025 µg vitamin D3

**Depends on age and weight

water-soluble vitamins are present in fruits and vegetables and fat-soluble-vitamins are present in butter and liver. Table 4.1 summarizes the daily needs, sources, and functions of some vitamins.

Vitamin	Daily Needs	Sources	Functions
B ₁ (Thiamine)	1.5 mg	Yeast, cereals, beans, peas, dried fruits, liver, nuts, meat, eggs.	Involved in the metabolism of carbohydrates. Supports nervous system function.
B ₂ (Riboflavin)	1.8 mg	Yeast, cereals, liver, milk, eggs, meat.	Involved in the metabolism of carbohydrates, lipids, and proteins. Maintains mucous membranes.
B ₃ (Niacin)	18 mg	Yeast, wheat, nuts, fish, liver, meat.	Necessary for growth. Vital to release of energy from glucose
B ₉ (Folic acid)	300 µg	Green vegetables, liver, kidney, eggs, meat.	Necessary for the formation and maturation of red and white blood cells. Used in new cells synthesis.
B ₁₂ (Cobalamin)	3 µg	Yeast, meat, liver, kidney, fat fish.	Involved in the formation of red blood cells and nervous cells.
C (Ascorbic acid)	80 mg	Fresh fruits and vegetables.	Involved in the formation of connective tissues and red blood cells. Promotes absorption of iron Protects other vitamins from oxidation.
A (Retinol)	800-1000 µg	Eggs, butter, cheese, liver, milk, orange vegetables (carrots).	Develops and maintains cell membranes. Important for proper bone growth Important for vision.
D (Calciferol)	10 µg	Liver, fish-liver oils, milk, butter.	Promotes phosphorus and calcium absorption in the intestine and their fixation in bones
E (Tocopherol)	12 mg	Cereals, leafy green vegetables (spinach, lettuce, cabbage), liver, milk, butter, vegetable oils, egg	Prevents the oxidation of vitamins A and C and fatty acids. Favors the synthesis of good cholesterol (HDL cholesterol).

Table 4.1: Daily needs, sources, and functions of some vitamins

Magnesium:

- is essential to normal muscular contraction
- contributes to protein synthesis
- is essential for energy metabolism

Fluorine:

- strengthens bones and diminishes dental decay

Iron:

- is a component of oxygen-carrying proteins (hemoglobin and myoglobin)

Copper:

- participates in the formation of hemoglobin
- is part of 11 enzymes

Zinc:

- is part of insulin and some 154 enzymes

Iodine:

- is essential for the synthesis of thyroid hormones
- regulates rate of energy use

4.3 Vitamins

Vitamins are organic compounds that cannot be synthesized by the human body but are essential, in small amounts, for the proper functioning of the human body. They must be obtained from the diet.

Unlike other groups of nutrients, vitamins are not chemically similar to each other. Each vitamin has a specific function or set of functions in the body. The lack of these vitamins reduces the growth rate and causes specific deficiency diseases.

There are 13 known vitamins. Solubility characteristics divide these vitamins into two classes: the water-soluble vitamins (hydro-soluble) and the fat-soluble vitamins (lipo-soluble).

Nine vitamins (vitamin C and eight B vitamins) are water-soluble. These vitamins are rapidly eliminated from the body in the urine and are unlikely to be toxic except when taken in unusually large doses.

Four fat-soluble vitamins denoted by the letters A, D, E, and K are stored in fat tissues. These vitamins are more likely to be toxic when consumed in excess of need.

4.4 Daily needs, sources, and functions of some vitamins

The distinction between water-soluble and fat-soluble vitamins is useful to determine where the vitamins are present. For example:

Iron is absorbed by the body as Fe^{2+} . Iron in foods is present as Fe^{3+} . To increase absorption of iron, the body must reduce the Fe^{3+} to Fe^{2+} . Vitamin C aids in this reduction.

Seafood is a good natural source of iodine. Iodine is also found in cereals, vegetables, and milk.

The term "vitamin" is derived from Latin "vita", and from the fact that these substances were all thought to contain the "amine" functional group



Fig.4.2: Although many people think that citrus fruits (50 mg per 100 g) are the best source of vitamin C, peppers (128 mg per 100g) are richer in vitamin C.

Requirements 5



Objectives

- Describe the nutritional requirements of the human body.
- Define metabolism, catabolism, and anabolism.
- Describe the catabolism of carbohydrates, lipids, and proteins.

Blood glucose can either be utilized directly by the body for energy or can be stored in small quantities as glycogen. When glucose exceeds the body requirements, it will be converted to reserved lipids.

B-Energy from lipids

Lipids constitute an important source of energy that produces upon decomposition 9 kcal per gram. This energy is produced by the degradation of fats to fatty acids and glycerol which upon oxidation releases energy.



Fig. 5.3: Digestion of fatty substances (catabolism)

C-Energy from proteins

Decomposition of food proteins provides amino acids that are mainly used to synthesize proteins (e.g. enzymes) needed by the human body. Excess amino acids that are not utilized to synthesize proteins are oxidized to produce energy.



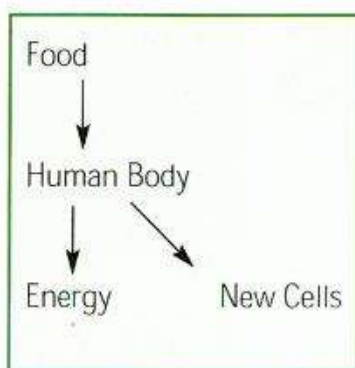
Fig. 5.4: Digestion of food proteins.

Review Questions

- 1- Describe the nutritional requirements of the human body.
- 2- Define metabolism, catabolism, and anabolism.
- 3- Describe the production of energy by the decomposition of carbohydrates and lipids.
- 4- Describe the transformation of proteins in the human body.

Research Activity

Calculate the calorie content of a typical Lebanese breakfast



$$1 \text{ kcal} = 4.18 \text{ kJ}$$

5.1 Nutritional requirements of the human body

Food is transformed in the human body to produce energy and to synthesize new molecules.

So, nutritional requirements of the human body to accomplish these two essential functions can be divided into "energy" requirements and "growth and maintenance" requirements.

A Energy requirements

These requirements are provided by the energy released from the oxidation of the nutrients in the cell. Some of this energy is converted into mechanical energy, some into heat, and some is used to maintain metabolic processes.

The energy requirements are covered mainly from carbohydrates and lipids and partly from proteins. These three groups of nutrients provide about 4, 9, and 4 kilocalories per gram respectively.

1 g of carbohydrates provides 16 kJ (3.75 kcal)

1 g of lipids provides 38 kJ (9 kcal)

1 g of proteins provides 17 kJ (4 kcal)

The energy value of food depends on the quantities of carbohydrates, lipids and protein in the food. It can be calculated if the composition of food is known as shown in the following example:

If 100 g of milk contains

4.7 g carbohydrates

3.8 g lipids

3.3 g proteins

Calculate the energy value of 100 g of milk.

Carbohydrates provide $4.7 \times 16 \text{ kJ} = 75.2 \text{ kJ}$

Lipids provide $3.8 \times 38 \text{ kJ} = 144.4 \text{ kJ}$

Proteins provide $3.3 \times 17 \text{ kJ} = 56.1 \text{ kJ}$

Therefore, 100 g of milk provide 275.7 kJ of energy

The following table shows the energy value of some foods

Food	Kcal per 100 g
Butter	740
Cheddar cheese	406
Table sugar	394
White flour	356
Beef	282
Eggs	147
Potatoes	87
Bananas	79
Milk	65
Apples	46

Table 5.1: Energy value of some foods

B Growth and maintenance requirements

These requirements are provided by proteins which, when hydrolyzed, provide the body with the essential amino acids needed to build and replace cells and tissues. They include:

1-Growth requirements: associated with gaining weight. They are particularly very high at young age and diminish as a person gets older.

2-Gestation requirements: correspond to the fetus growth requirements. They are very important during the last three months of pregnancy.

3-Requirements of women during lactation periods. They depend on the quality and quantity of the secreted milk.

5.2 Metabolism, anabolism, and catabolism

Thousands of biochemical reactions, catalyzed by enzymes, occur in cells in special sequences called pathways.

In a pathway, the products of one reaction are the reactants of another.

Certain pathways include reactions in which large molecules are broken down into smaller ones. The sum of such reactions is called catabolism and is associated with the release of energy.

The reactions involved in the transformation of starch to maltose and those involved in the oxidation of glucose are examples of catabolic reactions.

Other pathways include reactions in which small molecules are joined together to form larger molecules. Such reactions usually consume energy in order to proceed and their sum is called anabolism. Reactions involved in the synthesis of proteins, such as enzymes, are examples of anabolic reactions.

In any one cell of the body catabolic and anabolic reaction pathways occur simultaneously and are called metabolism.

5.3 Energy production

A-Energy from carbohydrates

Decomposition of carbohydrates produces energy. Digestion of polysaccharides (starch, glycogen, dextrines) and disaccharides produces mainly glucose which upon oxidation produces energy at the rate of 4 kcal per gram. Glucose constitutes a source of energy for rapid use.

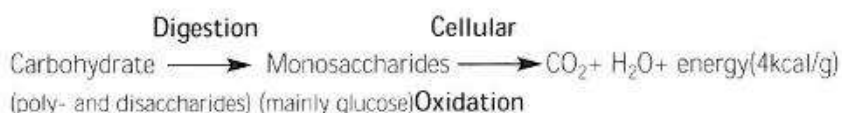


Fig 5.2: Digestion of poly- and disaccharides (catabolism).

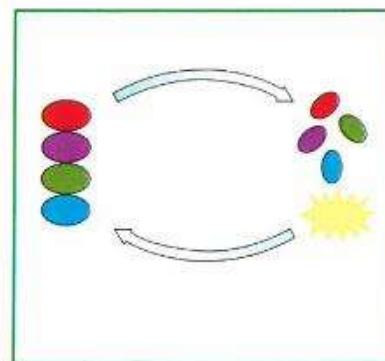


Fig. 5.1: The processes of catabolism and anabolism are opposite in nature. The first usually produces energy, and the second usually consumes energy.

Glucose is the only metabolite that provides energy to the brain and the nervous cells. Fructose is utilized directly by muscles and liver cells.

Nutritiona



Principal Foods 6



Objectives

- List the average composition of some principal foods.
- Define and classify additives.

Document 5

The human body's response to feasting, to fasting, and to starvation

	Component to be broken down	End products of digestion or catabolism	Fate
Feasting (overeating) Ingestion of food in excess of energy needs causes the body to store a limited amount as glycogen and the rest as fat	Dietary carbohydrates	Glucose	Stored glycogen in liver and muscle
	Dietary fat	Fatty acids	Stored as body fat
	Dietary protein	Amino acids	Nitrogen lost in urine (urea) Body proteins
Fasting (no food ingestion) When no food is ingested, the body uses its stored glycogen and fat for energy	Glycogen in liver and muscle	Glucose	Brain energy
	Body fat	Fatty acids	Energy
Starvation (continued fasting beyond glycogen depletion) When glycogen stores are depleted, body protein is broken down to amino acids which are used to synthesize glucose. The glucose serves as an energy source for the brain and nervous system. Also, in the liver, fats are converted to ketone bodies, which are another energy source for the brain.	Body protein	Amino acids → Glucose	Nitrogen lost in urine (urea) Brain energy
	Body fat	Fatty acids → Ketone bodies	Energy

Adapted from: Stoker, H.S. (1998), Organic and biological chemistry. Boston: Houghton Mifflin Company. P.476

2- Cheese:

Cheese is prepared from milk by precipitating the milk protein casein, by adding rennet, which lowers the pH of milk and coagulates the protein, and filtering the liquid out. Table 6.3 shows the average percent composition of cheddar cheese.

Nutrient	%
Proteins	26
Lipids	33
Carbohydrates	0
Water	37
Vitamins and minerals	2.5

Table 6.3: Average composition of cheese (cheddar).

3- Yogurt

Yogurt is the product of fermentation of milk by using pure culture bacteria. Lactose is converted into lactic acid, which together with a variety of minor products is responsible for the sour taste of yogurt. Since it is made from milk, the composition of yogurt is similar to milk.

4- Lactose intolerance

Lactose is found only in milk. It is hydrolyzed in the presence of the enzyme "lactase" to give glucose and galactose. The enzyme is secreted in the small intestine.

For many people, the digestion and absorption of lactose is a problem. This problem, called "lactose intolerance", is a condition in which people lack the enzyme. Deficiency of lactase can be caused by a genetic defect, by physiological decline with age, or by injuries to the mucosa lining the intestines. When lactose molecules remain in the intestine undigested, they attract water causing fullness, nausea, and diarrhea. Bacterial fermentation of lactose further along the intestinal tract produces lactic acid and gas adding to the discomfort.

C- Meat

The composition of meat is very variable. It contains about 20% proteins, 10-30% lipids, 1% vitamins and minerals and the rest is water. Meat is a valuable and important source of vitamin B and iron. Table 6.4 shows the average percent composition of beef and lamb.

Nutrient	Beef %	Lamb%
Protein	16	15
Lipid	24	30
Carbohydrates	0	0
Water	59	54
Vitamins and minerals	1	1

Table 6.4 Composition of beef and lamb meats

6.1 The principal foods

Foods provide the essential nutrients for the body. Certain foods are rich in proteins such as meat and milk, others are rich in carbohydrates such as fruits and vegetables, and others are rich in lipids such as oils and fats. Composition and some properties of some principal foods are discussed in this chapter.

A - Cereals

Cereal grains are the seeds of a group of plants (cultivated grasses). They are the most important source of food for humans and animals as they are rich in carbohydrates. Table 6.1 shows the average percent composition of three of the most important cereals: wheat, rice, and corn.

Grain	Protein%	lipids%	Carbohydrates%	Fiber % (1)	Vitamins and minerals	water
Wheat	7-18	1.5-2	60-69	2-2.5	5-12	13
Rice	7-12	2	63	9	6	12
Corn	7-12	4-8	67-72	2	1.5-2	11

Table 6.1: Average percent composition of wheat, rice and corn.

Average composition of wheat flour – Role of gluten

The major component of the wheat grain is the endosperm. It is composed of granules containing mainly starch and reserve proteins. The two reserve proteins in the grain of wheat are glutenin and gliadin.

Gluten is a lipoprotein. It is a cohesive, viscous, and elastic substance. It contains gliadin which is responsible for the extensibility of the dough and glutenin which is responsible for its elasticity.

White flour is mainly the endosperm grounded into fine powder. The average composition of white wheat flour is: 12% proteins, 2% lipids, 70% starch, 2% other carbohydrates, 0.5% minerals and vitamins, and 12% water.

B- Milk and dairy products

1- Milk

Milk is composed of a variety of nutrients either dissolved in water or dispersed in the form of colloid. Milk is a valuable source of proteins especially the phosphoprotein, casein. Table 6.2 shows the average percent composition of milk.

Nutrient	%
Proteins	3.3
Lipids	3.8
Carbohydrates	4.7
Water	87.6
Vitamins & minerals	0.7

Table 6.2: Average composition of milk

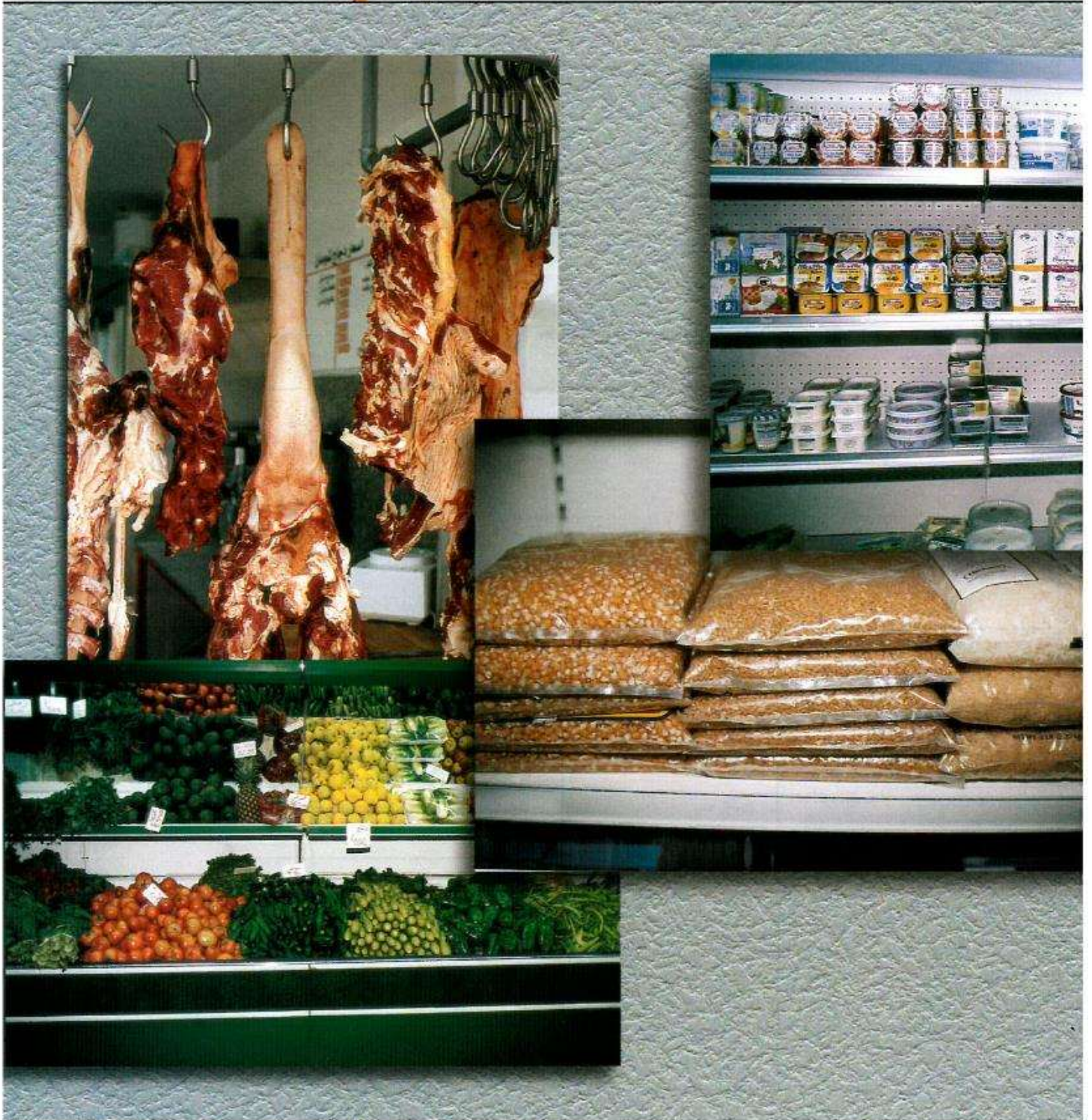


Fig. 6.1: Cereals are rich in carbohydrates



Fig.6.2: Milk and dairy products

The



Research Activities

- 1- Write a research report on famines and food supplies in developing countries. Emphasize the need for cereals.
- 2- Write a report on the laws dealing with food safety in Lebanon.

Soya

There is no doubt that soya is considered to be the most important cultivated plant in the world. Soya beans are very rich in proteins and, to a lesser degree, in lipids. This makes soya a good and wide field for scientific and technological research studies.

Soya plants were originated in Asia. Fifty years ago, it was industrialized on large basis in the U.S.A and Brazil. The seeds were adapted to be suitable to grow under different climatic and pedo-logical conditions so they can be planted in Europe.

Defatted soya-bean flour contains about 50% protein and is used for feeding animals. It cannot be used as human food unless it is subjected to special treatments to eliminate the toxic constituents and for protein enrichment.



Fig. 6.4: Soya beans are rich in proteins

Serial numbers of antioxidants are in the range E 300 to E 399.

Serial numbers of colorants are in the range E100 to E199.

Serial numbers of emulsifiers are in the range E400 to E499

C- Sweeteners

Sweeteners may be natural such as aspartame (dipeptide) or synthetic such as saccharin and cyclamates. Toxicity of sweeteners require more complex research studies. Most of them are new molecules with no long history as human food and not enough research has been done on them.

D- Colorants

Colorants are added to food to make them more attractive or to replace color lost during processing. Many colors are of natural origin such as riboflavin (vitamin B2), chlorophyll, and caramel. Others are synthetic such as tartazine (an azo dye).

In general, most synthetic colors contain an azo group ($-N=N-$) which may cause cancer if not metabolized. However, they do not present high risk if they are taken within the acceptable daily intake (ADI). Colorants may have some allergic effects.

E- Emulsifiers, stabilizers, and thickeners

Emulsifiers and stabilizers are used to stabilize food emulsions. Thickeners are utilized to reinforce food texture.

F- Miscellaneous food additives

In addition to the above five categories, there are many other types of additives such as flavors, flavor enhancers, substances added to food in order to adjust or control its pH

Review Questions

- 1- List the most important cereals.
- 2- Give the average composition of wheat, milk, yogurt, cheese, meat, and eggs.
- 3- Explain lactose intolerance.
- 4- Name the proteins of an egg.
- 5- Explain rancidity of fats.
- 6- Describe vegetal proteins and proteins of microorganisms.
- 7- Define and classify additives.
- 8- Describe the effects of food additives on health.

Minerals and vitamins in an egg are distributed as:

Minerals

- P: 0.2% in the yolk
- Ca: 0.055% mainly in the yolk
- K: 0.14% distributed equally between egg white and the yolk
- Na: 0.13% mainly in the egg white
- Fe: 2.5% major part in the yolk

Vitamins 0.2%

- A, D, and thiamine in the yolk
- Riboflavin distributed equally between egg white and yolk

D- Eggs

The average mass of an egg is 40 to 60 g. About two thirds of the edible part (which represents 90% of an egg) is the egg white and one third is the yolk. Table 6.5 shows the average percent composition of an egg.

Nutrient	Edible part %	Egg white%	Yolk %
Proteins	12	9	16
Lipids	11	trace	31
Carbohydrates	0	0	0
Water	75	88	51
Vitamins and minerals	1	1	1

Table 6.5: Average composition of an egg

The egg white is basically a colloidal solution of proteins, mainly albumin, in water, together with small quantities of vitamins and mineral salts. The yolk is composed of dispersed lipids and proteins in water. The main proteins of the yolk are vitellin and vitellinin. Egg yolk is insoluble in water.

E- Fatty substances

In a previous chapter, the properties and composition of fatty substances (fats and oils) were discussed.

Rancidity of lipids

Rancidity or oxidation of lipids is due to the presence of double bonds in fatty acids. Polyunsaturated fatty acids are oxidized faster than mono-saturated fatty acids.

Oxidation of the unsaturated components of lipids results in the formation of a variety of low molecular mass volatile compounds such as aldehydes, ketones, and carboxylic acids which cause the unpleasant taste and objectionable odors.

F- Vegetal proteins

Cereals (wheat), some leguminous plants (soya beans, fava beans, and peanuts), and potatoes are important sources of low-cost proteins. Soya beans are particularly rich in proteins. They are also rich in lipids.

G- Proteins of microorganisms (Single-cell proteins)

To compensate for the world deficit in proteins, microorganisms are considered a new non-traditional and low-cost source of proteins to be consumed by humans and animals. These microorganisms (bacteria, yeast, and fungi) offer a great possibility that some are being produced industrially. These have the following common properties:

- they are rich in proteins: 50 to 70% of dry weight.
- they are well preserved in the form of powder.
- they have variety of culture mediums. Such as: hydrocarbons (natural gas, alkanes), alcohols (ethanol, methanol), organic waste of several industries such as paper, cheese-making and plastic industries.

Yeast is considered the most important cultivated microorganism. From a nutritional point of view, yeast is characterized by a balance of essential amino acids, except methionine, and richness in vitamin B. Molds are sources of proteins rich in sulphur-containing amino acids.

6.2 Food additives

Food additives are substances that are not consumed as food by themselves, but are rather added to food for the following purposes.

- a- To preserve and maintain the nutritional quality of food.
- b- To make food more attractive.

Additives may be natural or synthetic and can be classified as preservatives, antioxidants, sweeteners, colorants, emulsifiers and stabilizers, and miscellaneous additives.

A- Preservatives:

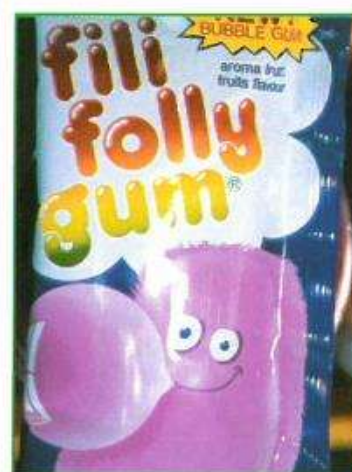
They prevent or reduce the growth of microorganisms in food. Examples are sorbic acid and its salts, benzoic acid and its derivatives, and sulphur dioxide and sulphites.

Benzoic acid and its derivatives may be toxic at high levels of consumption on long term.

B- Antioxidants

They are used to prevent the oxidative rancidity of fats and oils. They are natural antioxidants such as vitamin E and vitamin C and artificial antioxidants such as BHA (butylated hydroxyanisole) and BHT (butylated hydroxytoluene).

Toxicological data reveals that some artificial antioxidants may affect the metabolism of lipids in the liver and reproduction. They may also cause allergy and develop tumors.



BUBBLE GUM AROMA FRUTTI	BUBBLE GUM AROMA FRUTTI
INGREDIENTI: Zucchero, Sciroppo di glucosio, Gomma Base, Lattosio, Aromi, Colorante: E 120, Antiossidante: E 320.	ZUTATEN: gummi-Basis, Zucker, Glukose-Sirup, Gummi, Laktose, Aromen, Farbstoff: E 120, Antioxidant: E 320.
BUBBLE GUM FRUTTS FLAVOUR	BUBBLE GUM FRUTTS FLAVOUR
INGREDIENTS: Sugar, Glucose Syrup, Gum Base, Lactose, Flavours, Colour: E 120, Antioxidant: E 320.	INGREDIENTS: Gummi, Zucker, Glukose, Gummi, Laktose, Aromen, Farbstoff: E 120, Antioxidant: E 320.
CHICLE SABOR FRUTAS	CHICLE SABOR FRUTAS
INGREDIENTES: Azúcar, Jarabe de Glucosa, Goma Base, Lactosa, Aromas, Colorante: E 120, Antioxidante: E 320.	INGREDIENTES: Azúcar, Jarabe de Glucosa, Goma Base, Lactosa, Aromas, Colorante: E 120, Antioxidante: E 320.

Fig 6.3 Additives in foods

An "E" in front of the number of an additive means that it has been approved by the European Community.

Serial numbers of preservatives fall in the range E 200 to E 299.

Food Diet

7

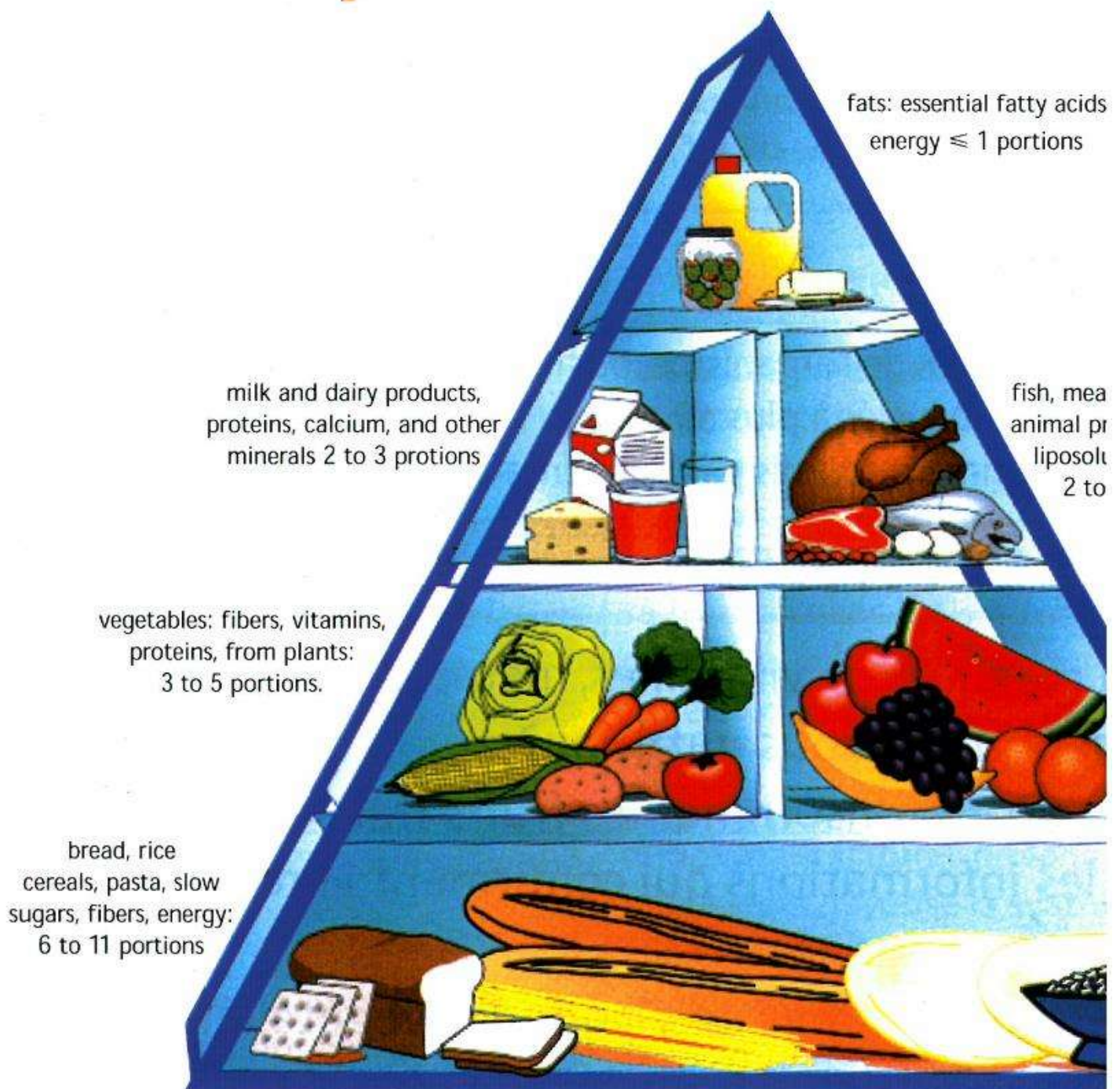
poultry,
minerals,
vitamins.
S.

vitamins, minerals
sugars, fibers:
at 4 portions.



Objectives

- Identify the energy requirements of humans.
- Define a balanced diet.
- Describe the relationship between diet and health.
- Describe the effect of cooking on the nutritive value of food.



7.1 Energy requirements of humans

The rate at which energy is used up in maintaining basal metabolism is called basal metabolic rate (BMR).

The intake of food stimulates metabolism. After a meal, metabolic rate increases. The extra energy appears in the form of heat. This is known as thermogenic effect of food.

Activity	Kcal / minute
Sitting	1.4
Writing	1.7
Standing	1.7
Typing	1.9
Domestic work	2.9
Walking	3.3 – 5.0
Running	6.0 – 15.6
Swimming	4.8 – 12.0
Tennis	4.8 – 8.4
Football	4.8 – 8.4
Agricultural Work	3.6 – 7.2

Table 7.1: Energy consumption of various activities

Energy is required by the human body for basal metabolism and physical activity. This energy is the chemical energy which comes from food that can be converted into mechanical energy and heat.

A- Basal Metabolism

Basal metabolism is the term used to describe the basic metabolic processes which keep the body alive: to keep the heart beating and the lungs functioning, to maintain body temperature, and for the numerous chemical reactions that take place in the body cells. Energy for basal metabolism of an individual varies with body size, age, activity of the thyroid gland, and the thermogenic effect of food.

B- Physical Activity

In addition to basal metabolism, the human body needs energy for physical (muscular) activity. The energy requirements for various activities have been determined and the average values for some activities are shown in Table 7.1.

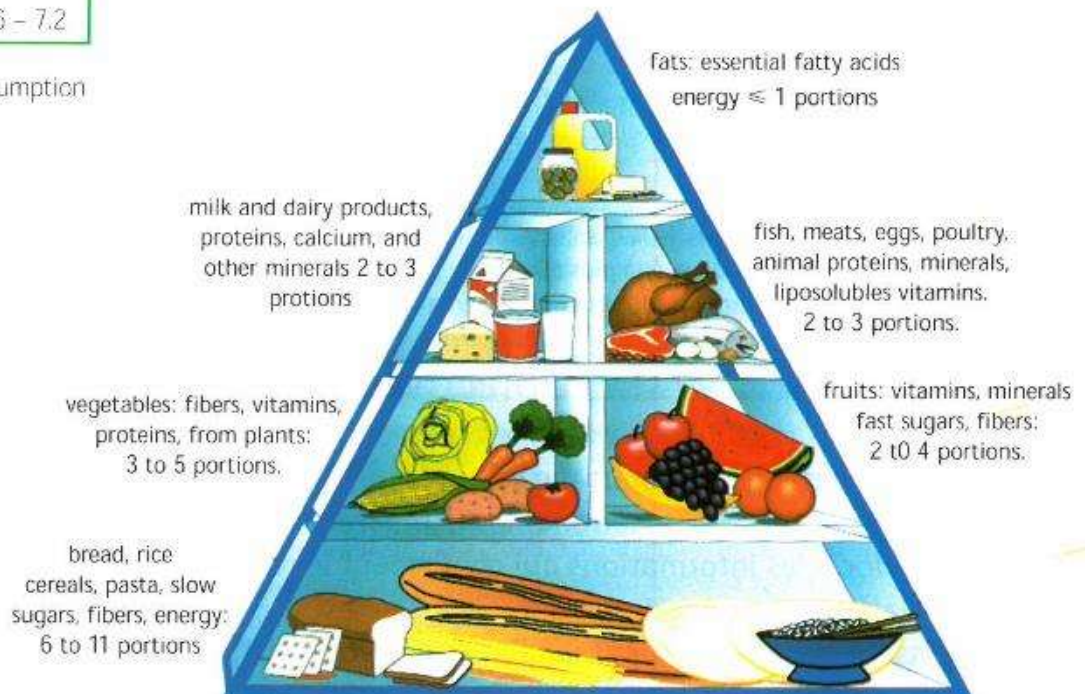


Fig. 7.1 : Pyramid of food.

7.2 What is a balanced diet?

It is impractical to calculate in detail the nutrient and energy contents of normal meals. It is important, however, that meals are nutritionally balanced, in other words, that the food consumed over the period of a day is balanced nutritionally.

One simple way to ensure a balanced diet is to include foods from the following groups each day.

- 1- Meat, fish, poultry, and eggs
- 2- Milk and dairy products
- 3- Fatty substances (butter and margarine)
- 4- Cereals and leguminous grains
- 5- Fruits and vegetables
- 6- Beverages

If daily meals contain a variety and reasonable quantities of the above six groups, the food diet can be considered balanced.

As we know, we can divide nutrients into three groups, according to their function in the organism.

- a- Materials for growth and maintenance (proteins and minerals such as calcium and phosphorus).
- b- Suppliers of energy (carbohydrates and lipids).
- c- Regulatory materials (vitamins and some minerals such as sodium and potassium).

The following illustrates how these nutrients are distributed among the six groups of food:

Group 1: meat, fish, poultry, and eggs are the main sources of proteins.

- a- Nutrients for growth and maintenance: proteins, iron, phosphorus
- b- Suppliers of energy: lipids
- c- Regulatory nutrients: vitamins

Group 2: milk and dairy products are essential to our diet because of their content of calcium.

- a- Nutrients for growth and maintenance: proteins, calcium
- b- Suppliers of energy: lipids and carbohydrates in milk
- c- regulatory nutrients: vitamins

Group 3: fatty substances (oil, butter, margarine, ...)

- a- nutrients for growth and maintenance: none
- b- suppliers of energy: lipids
- c- regulatory nutrients: fat soluble vitamins

Group 4: cereals and leguminous grains (rice, lentils, dry beans, ...)

- a- nutrients for growth and maintenance: proteins
- b- suppliers of energy: carbohydrates
- c- regulatory nutrients: some minerals

Meat is rich in iron and contains essential amino acids. The amount of protein of 100 g of meat = 100 g of fish = 2 eggs = 0.5 L of milk = 80 g of cheese



Fig. 7.2: Meat, fish, and poultry are important sources of proteins

Minimum daily requirements of calcium can be provided by 350 ml of milk + 25 g of cheese + 300 g of vegetables + 250 g of fruits



Fig. 7.3: Fruits and vegetables are important sources of minerals and vitamins

Unit Questions

I-Fill in the blanks

Foods are substances used to maintain life and growth. They are composed of _____. They include carbohydrates, _____, _____, and _____. The general chemical formula of carbohydrates is _____. The chemistry of carbohydrates is mainly due to the two functional groups, _____ group and _____ group.

Carbohydrates are usually classified into _____, _____, and _____. Glucose, the grape sugar is a _____, the sugar of milk is a disaccharide, while starch is a _____.

Proteins are _____ of α -amino acid units joined one to another by _____ bonds. The reaction of the (carboxyl) group of one amino acid and the _____ group of another amino acid leads to the formation of a _____ with the removal of one molecule of water.

Lipids are natural compounds. They are _____ in ether and chloroform, and _____ in water. They can be classified as _____ lipids such as triglycerides and _____ lipids such as phospholipids. Fats and oils are tri-esters of glycerol and are called _____. At room temperature, fats are _____, while oils are _____. Their melting points depend on the number of _____ atoms and the number of _____ in the carbon chain of the fatty acids that form the triglycerides. Phospholipids are related structurally to fats and oils, except that one of the three fatty acids is replaced by a _____ group. Cholesterol is a _____ that is present in all animal tissues and is the precursor of vitamin D.

Vitamins are organic substances that our body can not _____, we must obtain them from our diet. Vitamins such as A, D, E, and K are _____, and vitamins such as _____ and _____ are hydrosoluble.

Minerals are very important to our body, they are classified according to the needs of our body into _____ for example Ca, P, K, Na, and Mg, and _____ for example Fe, I, Cu, F, and Zn.

II-Label each of the following statements as true or false and correct the false ones.

- 1-Glucose is a monosaccharide with the general formula $C_6H_{12}O_6$.
- 2-Glucose can be hydrolyzed.
- 3-Glucose is an aldohexose.
- 4-The general formula of a disaccharide is $C_{12}H_{22}O_{11}$.
- 5-The general formula of a polysaccharide is $C_n(H_2O)_m$.
- 6-Starch is a polymer of glucose.
- 7-Glycogen is the carbohydrate reserve in animals.
- 8-Hydrogenation of a triglyceride increases its melting point.
- 9-An oligopeptide is formed from more than ten amino acids.
- 10-The primary structure of a protein describes the sequence of amino acids in its structure.
- 11-When a protein is heated it changes its primary structure.
- 12-The amount of vitamins in a balanced diet should be equal to the amount of proteins.
- 13-Proteins are hydrolyzed in the body into α -amino acids.
- 14-Minerals are provided to the body by food and drinking water.
- 15-While we are sleeping we don't need energy.
- 16-The nutritional requirements of the body in terms of energy are provided by carbohydrates, lipids and vitamins in the ratio 4, 9, 4 Kcal/g.
- 17-Human body gets most of its energy from glucose.
- 18-Saturated fatty acids are usually solids at room temperature.
- 19-Vegetables are rich in cholesterol.
- 20-A balanced diet contains the six groups of food.
- 21-Proteins are formed from various combinations of twenty amino acids.
- 22-Native proteins have the same physical properties as their denatured form.
- 23-Food additives can be used to preserve and improve the quality of food.
- 24-Proteins are the major structural component of living cells.
- 25-Maltose, lactose, and glucose are reducing sugars.

III-Choose the correct answer

- 1-Sucrose is hydrolyzed into :
- a-glucose and fructose
 - b-two glucose units
 - c-fructose and galactose
 - d-glucose and galactose

Vegetarians and vegans

Vegetarians, for a variety of reasons, do not eat meat but many do consume other animal products such as milk, cheese, and eggs. The term "lactovegetarian" is used to describe a person whose diet includes milk and cheese but no meat, poultry, fish, or eggs. A "lacto-ova-vegetarian" consumes milk, cheese and eggs but not meat, poultry or fish. Generally speaking, there are few nutritional problems with these types of diet. The nutrient most likely to be deficient is iron. This can be supplied by eggs (if eaten), cereals, pulses and green vegetables.

The term "vegan" is used to describe a person who chooses, usually for humanitarian or health reasons, a very strict diet that contains no foods of animal origin. Nutrient requirements, with the exception of vitamin B₁₂, can be met by a diet composed only of plant foods, but this requires careful planning and the inclusion of a wide variety of foods. In the average diet the nutrients provided largely by foods of animal origin include protein, riboflavin, vitamin D and iron. For vegans protein needs can be met by cereals, pulses, nuts and TVP (textured vegetable protein); riboflavin requirements can be met by nuts, cereals, potatoes and soya; vitamin D requirements by margarine and exposure to sunlight; and iron requirements by pulses, green vegetables, cereals and cocoa. There are proprietary yeast extra products available which contain vitamin B₁₂.

Sometimes vegan diets have a low energy density, particularly if they contain a lot of watery fruits and vegetables. Energy density can be kept at a satisfactory level by eating plenty of cereals, nuts and pulses. If well planned, a vegan diet is healthy and adheres to many of the current nutritional guidelines. Obesity and coronary heart disease are less common among vegans than the rest of the population.

B- Effect of cooking on proteins

Cooking proteins, if done moderately, may rarely give an unfavorable effect. Cooking in water or by steaming will lead to the dissolution of amino acids. From a nutritional point of view, the harm is not great if the bouillon is consumed. Cooking proteins (collagen of meats and egg whites) improves their digestibility. It is easier for the digestive enzymes to "work on" denatured (unraveled) proteins.

C- Effect of cooking on lipids

Cooking fatty substances leads to a number of transformations that affect their nutritional values. Excessive heating (200 °C) causes fats to oxidize because of the unsaturation of fatty acids. This makes the fats un consumable as a result of the formation of polymers and some toxic compounds.

D- Effect of cooking on minerals and vitamins

Treatment of food with large amounts of water such as washing or boiling will lead to a great loss of minerals and water-soluble vitamins. Some vitamins are also thermosensitive such as thiamine and vitamin C. Vitamin A is destroyed at high temperatures in the presence of oxygen. Pyrodixine is not stable at high temperatures while folic acid is stable to heat in acidic medium.

Review Questions

- 1- Identify the energy requirements of humans
- 2- List the factors that affect energy requirements
- 3- Define a balanced diet
- 4- Describe the relationship between diet and health
- 5- Describe the effect of cooking on the nutritive values of food.

Research Activities

- 1- Compare the nutrients in different brands of cereals.
- 2- Write a research report on one of the following:
Vegetarian diets, diabetic diet, and low – sodium diet.

Group 5: fruits and vegetables are rich in dietary fiber)

a- nutrients for growth and maintenance: minerals such as calcium

b- suppliers of energy: poor in calories

c- regulatory nutrients: minerals and vitamins

Group 6: beverages

Sixty percent of human's body weight is water and about 1.5 liters of water are eliminated daily. To compensate for this loss, our diet should contain beverages, such as water and juices, and fresh fruits and vegetables.

In addition to nutritional value, there are a number of other factors, which must be considered when planning balanced meals. These factors include:

1- Cost and availability

2- Variety and appearance (color, texture, flavor)

3- Consumer preferences

4- Religious and cultural food habits

7.3 Diet and health

Nutritional knowledge has increased and ideas about what constitutes a healthy diet have changed considerably in recent years. Scientific investigations and research have shown relations between certain diseases and various factors in the diet. They found that there are strong links between a diet high in fat and refined carbohydrate (white flour and sugar) and low in fiber and coronary heart disease, obesity, and tooth decay. It is now realised that by modifying the diet we can reduce the risk of developing these diseases. Various reports have been published over the last few years which recommend changes in our dietary pattern in order to reduce the risk of developing diseases.

Recommendations are made for reducing fat intake, particularly saturated fats, and sugar (sucrose) intake and increasing fiber intake. Other recommendations are made for reducing the intake of salt.

7.4 Effect of cooking on nutritive value of food

The nutritional quality of food may be modified if subjected to culinary or technological treatments. These modifications may be beneficial or harmful and varies according to the nature of the nutrients.

A- Effect of cooking on carbohydrates

Starch is slightly digestible. Cooking starch in water facilitates the digestibility due to gelatinisation.

Oils rich in polyunsaturated fatty acids (e.g. corn oil) are not as good for frying or baking in very hot ovens as oils less rich in unsaturated fatty acid (e.g. peanut oil) or partially hydrogenated fats.

VI-Relate the compounds in column A to their structure in column B

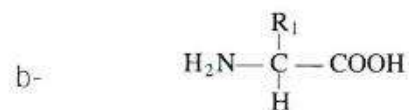
Column A

Column B

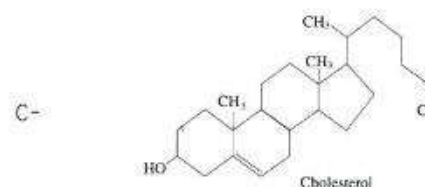
1-Fatty acid



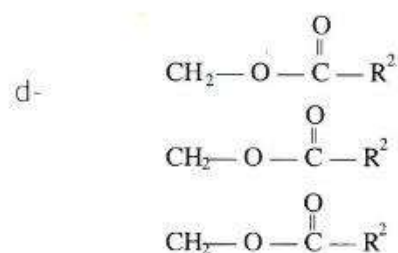
2-Glucose



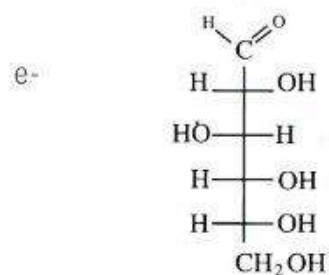
3-Glycerol



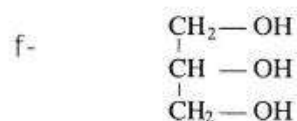
4-Cholesterol



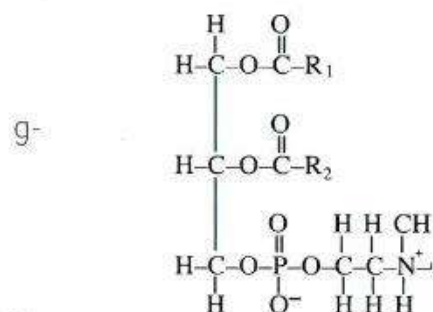
5-Triglyceride



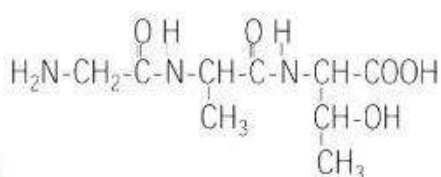
6-Amino acid



7-Phospholipid



VII-Consider the following compound



2-The melting point of a triglyceride increases with the increase of the number of:

- a-the double bonds in the glycerol
- b-oxygen atoms in the fatty acid
- c-double bonds in the fatty acid
- d-carbon atoms in the glycerol

3-The main role of proteins in food is to provide:

- a-the energy requirement for the functioning of the body.
- b-vitamins
- c-glucose
- d-aminoacids

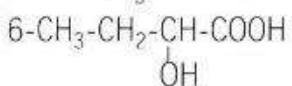
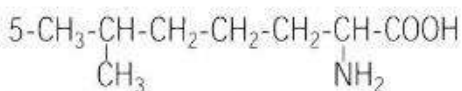
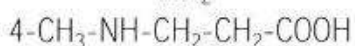
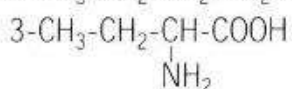
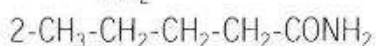
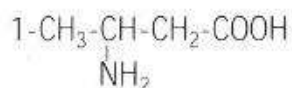
4-A balanced diet:

- a-is rich in vitamins
- b-has low fat content
- c-contains the six groups of foods
- d-contains carbohydrates, lipids, and proteins

IV-Classify the following as monosaccharides, disaccharides, or polysaccharides:

Amylose, glucose, lactose, sucrose, cellulose, galactose, fructose, starch, glycogen.

V-Which of the following structures represent an α -amino acid



Unit 2



Perfumes and Cosmetics

Chapter 8: Perfumes

Chapter 9: Cosmetics

- a-How many peptide bonds can you identify?
- b-Is it a mono peptide, dipeptide, tripeptide, or protein?

VIII-Short answer questions

- 1-State two differences between sucrose and fructose.
- 2-What chemical element occurs in proteins but not in carbohydrates, fats, and oils?
- 3-Why don't the proteins of an egg white return to their original form as the cooked egg cools?
- 4-What happens when a protein is denatured?
- 5-Suppose you have some pure cellulose and some pure starch in separate containers but you don't know which is which. What simple test could you do to decide which is the starch and which is the cellulose?
- 6-Name some foods that contain large amounts of fibers.
- 7-Carbohydrates, lipids and proteins are three classes of nutrients.
 - a-Which class of substances provides energy most rapidly?
 - b-Which class can be used as building materials in the body?
 - c-Which is used to store energy?

IX-Thinking Problems

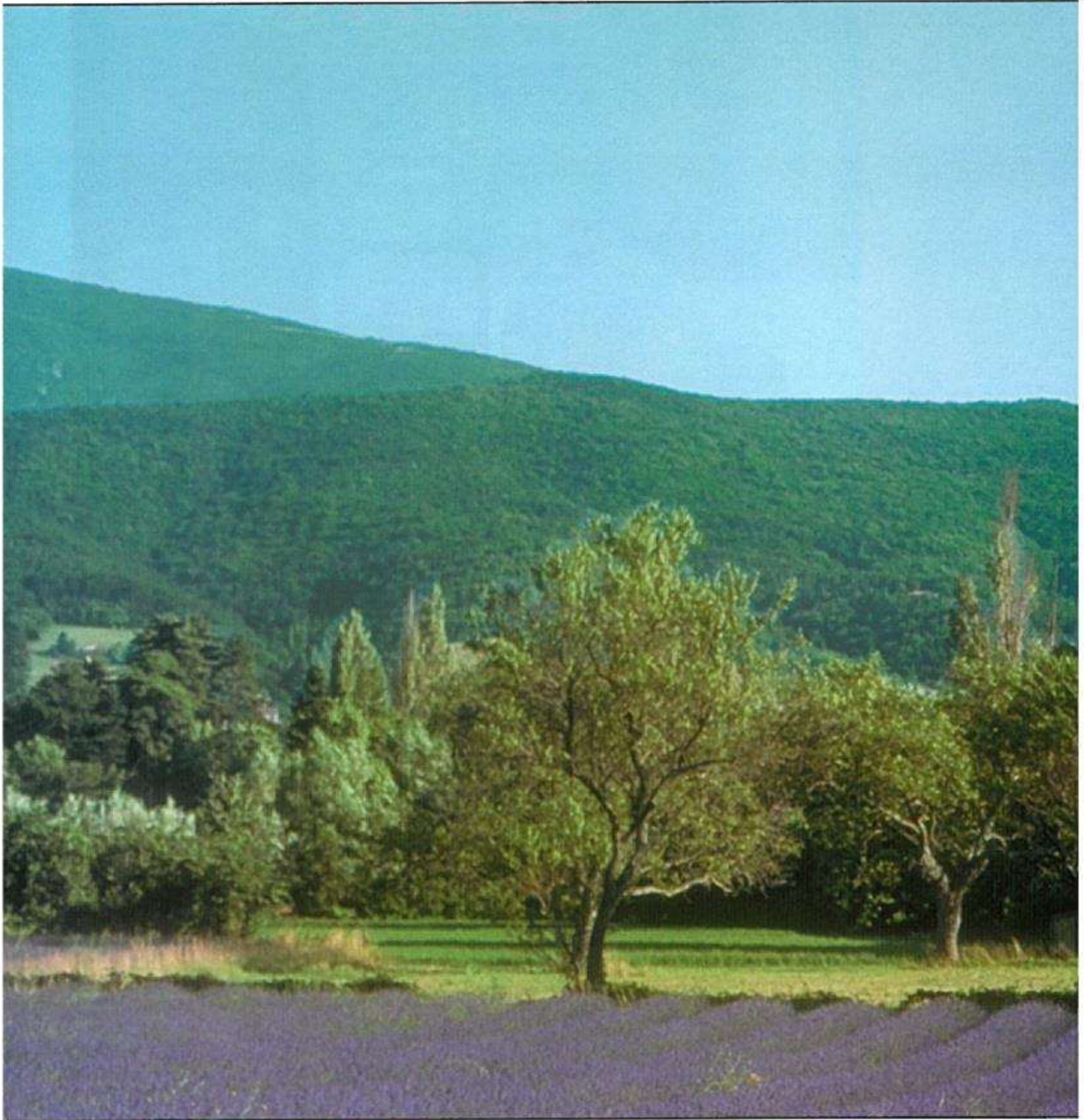
- 1-By consulting the table of milk composition, explain why it is considered as a perfect food and is recommended in the diet of most people. Do you recommend milk to people who are deficient in Lactase? Why?
- 2-What is the main role of fibers in a diet?
- 3-Why is it recommended to consume fresh foods instead of canned food?
- 4-Give one advantage and one disadvantage for cooking foods.
- 5-Nutritionists recommend eating fresh fruits instead of canned fruits and raw or steamed vegetables instead of canned or boiled vegetables. Which class of vitamins are they really concerned with? Why does their advice sound?
- 6-Describe, how we can distinguish between cellulose and starch on the basis of:
 - a-Their molecular structures
 - b-Their chemical behavior
 - c-The way our body uses them.
- 7-Describe the difference between amylose and amylopectin.
- 8-Food labels occasionally claim that the product is made with hydrogenated vegetable oils "to preserve freshness". How does hydrogenation of an oil help preserve freshness?
- 9-Of beef fat, vegetable butter, and coconut oil,
 - a-which has the highest total percentage of fully saturated fatty acids?
 - b-Which has the lowest?
 - c-What nutritional hazard to human health is present in beef fat but not in vegetable butter or coconut oil?

Perfumes 8



Objectives

- Define perfumes and describe their composition.
- Identify the raw materials used in the manufacture of perfumes.
- Describe the extraction of an essential oil.



The jasmine extract contains around 200 fragrant substances. Now a day, molecules having the fragrant of jasmine are being synthesized.

- Carbonyl group: Aldehyde ($\text{R}-\overset{\text{O}}{\underset{\text{||}}{\text{C}}}-\text{H}$) and Ketone ($\text{R}-\overset{\text{O}}{\underset{\text{||}}{\text{C}}}-\text{R}$)
- Ester group ($\text{R}-\overset{\text{O}}{\underset{\text{||}}{\text{C}}}-\text{O}-\text{R}$)
- Phenol group ($\text{—}\text{C}_6\text{H}_4\text{—OH}$)

The following table includes examples of fragrant molecules extracted from plants.

Origin	Name of the molecule	Structural formula	Functional group
Lemon, orange	Limonene		Alkene group
Mint	Menthol		Alcohol group
Wood roses	Linalol		Alcohol group
Cinnamon	Cinnamic aldehyde		Aldehyde group
Magnolia	Geranylacetone		Ketone group
Lavender	Linalyl acetate		Ester group
Thyme	Thymol		Phenol group

Table 8.1: Origin, name, structural formula and functional group of some fragrant molecules.

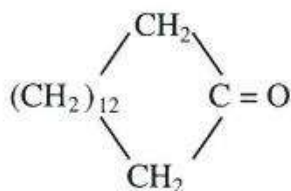


Fig.8.4: Structural formula of cyclopentadecanone (musk).

2-Fixatives

Fixatives are natural or synthetic substances with a strong fragrant. Fixatives are used to retain the other volatile fragrant molecules for long time (by making a complex). One example of fixatives is cyclopentadecanone (musk).

B- Solvent

Most fragrant substances give the desired fragrant only when they are diluted. Ethanol is chosen as the solvent of perfumes because it



Fig. 8.3: Some perfumes

8.2 Perfume composition

A perfume is composed of a mixture of fragrant oils (essence and fixatives) dissolved in ethanol. They constitute about 10% to 25% of the solution.

A- Fragrant oils

In the past, raw materials used in the manufacture of perfumes were mainly of plant origin and animal origin.

Today, for the protection of animals and because of the high price of the fragrant plants, synthetic fragrant substances have replaced natural ones.

Fragrant oils include essential oils or essence, and fixatives.

Fragrant oils should be volatile in order to reach our nose (the olfactory receptors), and they should be:

-slightly soluble in water, because they should dissolve in the mucus that cover the sensorial cells.

-very soluble in lipids, because they should pass through the fatty layer of the membrane of the cell in order to reach the olfactory receptors.

1-Essence or essential oils

Essence or essential oils are of plant origin, they can be extracted from different parts of the plant: the flowers (jasmine, rose), the flowers and the leaves (lavender), and the seeds (anise). The aroma of the molecules of the essence or the essential oil is associated to a variety of functional groups such as :

- Alkene group ($\text{C}=\text{C}$)
- Alcohol group (R-OH)

In addition to essential oil and fixative, modern perfumes contain also small percentage (2-7%) of natural or synthetic fragrant substance to assist the essential oil.

For devising the odors of today's cosmetics and perfumes, the modern perfumer has available a set of some 4000 aromatic plant and animal substances and another 2000 synthetic organic chemicals.



Fig. 8.1: Vases of cosmetics, oils, and ointments were sealed in the tombs of the kings of ancient Egypt to make them more presentable afterlife.

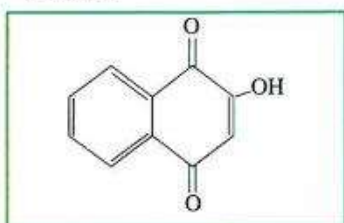


Fig. 8.2: Chemical structure of henna. Henna is a hair dye extracted from the henna plants of Africa, India, and the Middle East.

Musk is a brown fragrant substance secreted by the glands of the male musk deer to attract the female, it's a sexual pheromone.

The word perfume comes from the Latin "per" which means through and "fumus" meaning smoke.

Very early in the history of human kind people used cosmetics and perfumes that were prepared from natural products. The oldest cosmetics were found in Egyptian tombs (3000 B.C). The Egyptians used some perfumed products to protect their skin from dehydration caused by the hot and dry climate of Egypt. They also used kohl to color their eyebrows, some green substance to color their eyebrows, and henna (substance which is still used these day) to color their hair. By the middle of the first century Romans used cosmetics such as Kohl, blush, and pumice-stone to clean their teeth.

Perfumes were consumed for religious purposes, and their use was limited to essential oils from plants and some animal secretions (musk). The revolution in perfume production is due to the Arabs who invented distillation, the process that permitted the extraction of the essential oils from plants. These oils became the main constituents of perfumes. The chemistry of perfumes was born, far later, toward the nineteenth century, due to the development of chemical processes that permitted analysis, identification, and synthesis of the natural extract constituents.

8.1 What are perfumes?

They are blends of fragrant natural and/or synthetic substances dissolved in a volatile solvent: ethanol. Modern perfume may contain more than 100 constituents.

Perfumes are applied on the skin to induce a feeling of well-being and satisfaction.

Perfume reaches the nose in three phases:

1-The first impact, the top note, comes from components that vaporize easily and move to our nose quickly. The top note is the fragrance of the perfume that makes the first impression.

2-The most noticeable odor, the middle note, is produced by compounds that vaporize less readily than the components of the first note, the middle note determine the identity of the perfume.

3-The end note is a residual, longer lasting scent, which comes from components that are the least volatile in a perfume. The end note of a perfume is the fragrance that lingers.

cles aches, nausea; soothes bug bites; and others.

-Peppermint: Alleviates digestive problems; cleans wounds; decongests the chest; relieves headaches and muscle pain; and others.

-Tea tree: Fights fungal and bacteria infections; useful for skin conditions such as acne, insect bites, and burns; and others.

-Chamomile: Reduces swelling; treats allergic symptoms; relieves stress, insomnia, and depression; useful in treating digestive problems; and others.

-Thyme: lessens laryngitis and coughs; fights skin infections; relieves digestive problems; and others.



Fig. 8.7 : Some aromatherapy products.

Adapted from Web sites Alternative Therapy-Aromatherapy:

http://health.yahoo.com/health/Alternative_Therapy/Aromatherapy

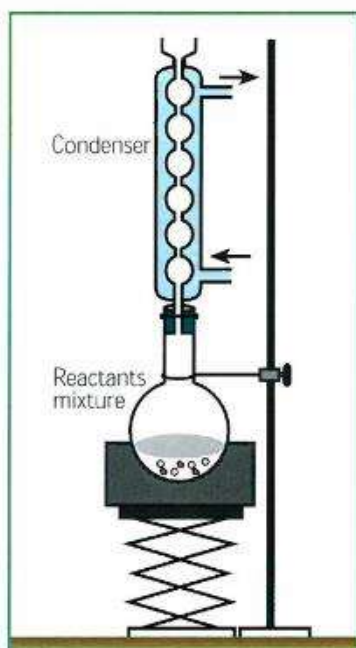


Fig. 8.6 :Reflux Set up.

Fragrant oils of a perfume contain often an ester function, therefore, esters are very important in perfume and aroma industry.

Activity

Activity 8.1. Preparation of an ester used in perfume

Objective:

Synthesis of the ester benzyl acetate ($\text{CH}_3\text{-COO-CH}_2\text{-C}_6\text{H}_5$), one of the components of jasmine essence.

Materials:

- Glacial acetic acid (30mL)
- Phenylmethanol (20mL)
- Concentrated sulfuric acid (12N)
- Saturated sodium chloride solution (100mL)
- Saturated bicarbonate solution (50mL)
- Anhydrous magnesium sulfate
- Ethanol
- Spatula
- Balance
- Round bottom flask (250mL)
- Condenser
- Heating mantle
- Crystallizing dish
- Graduated cylinders (50mL-100mL)
- Separatory funnel
- Erlenmeyer flask 250mL
- Micropipette
- Funnel
- Boiling chips
- Cotton glass

Procedure

- 1- In a 250mL round bottom flask add, with caution, acetic acid (30mL), phenylmethanol (20mL), sulfuric acid (20 drops), and boiling chips.
- 2- Assemble the reflux set up shown in figure 8.6.
- 3- Let the water circulates into the condenser, then heat up the reaction mixture slowly to reflux for 20 minutes.
- 4- Stop the reflux and remove the heating mantle. Let the reaction mixture first cool down (while the condenser is still on) to room temperature, then cool it in a crystallizing dish containing cold water.

5-Add to the reaction mixture 100 mL of sodium chloride solution, stir the mixture then transfer it into the separatory funnel (without the boiling chips).

6-In the separatory funnel the organic phase is at top and it contains the ester (benzyl acetate), and some acetic acid and the alcohol in excess. The aqueous phase is at the bottom and it contains sulfuric acid, and most of the untreated acetic acid.

7-Remove the aqueous phase.

8-Add 50mL of the sodium bicarbonate solution to the organic phase (in the separatory funnel) to neutralize it.

9-Stir the mixture, leave it to rest, and remove the aqueous phase.

10-Pour the organic phase into an Erlenmeyer flask and add 5 portions of the anhydrous magnesium sulfate using a spatulas. Shake the mixture slowly, then filter it through a funnel with cotton glass into another Erlenmeyer. This filtrate contains mainly the ester benzyl acetate.

Work under the fume hood and use goggles and gloves when handling concentrated acid solutions.

Review Questions

- 1-Define perfume.
- 2-Give the composition of perfume.
- 3-Give the percentage of fragrant oils (essence, fixative) and alcohol (ethanol) in a perfume.
- 4-Identify the raw materials used in the manufacture of perfumes.
- 5-Name the functional groups that are associated with aroma.
- 6-Describe the extraction of an essential oil by steam distillation.
- 7-Describe the solvent extraction of an essential oil.
- 8-Define cologne.

Research activity

Write a report on the preparation of rose water and /or orange flower water.

is an organic solvent that dissolves the fragrant oil and does not react with it.

8.3 Colognes

Cologne is a much more dilute version of a perfume, with concentrations of the fragrant oils running about a tenth of those used in the perfume. Colognes are usually much less expensive than perfumes.

8.4 Extraction of essential oil

The two principal methods for the extraction of an essential oil are: steam distillation and solvent extraction.

To describe these two methods let us take the example of the extraction of the essential oil present in the peels of lemons and oranges. The characteristic fragrant of this essential oil is mainly due to the molecule limonene.

1-Solvent extraction, or extraction with an organic solvent

The orange peels are crushed into very small pieces and soaked in an organic solvent (cyclohexane) for 30 minutes. The essential oil will be extracted with the organic solvent and can be recovered by evaporating the solvent

2-Steam distillation

In this method the steam distillation set up represented in figure 8.5a is needed. A mixture of distilled water and crushed peels of oranges are heated to reflux. The distillate collected is composed of two non miscible phases. The organic phase, containing the essential oil, is separated from the aqueous phase by decantation using a separatory funnel (figure 8.5b).

Cologne, a shortened form of eau de cologne (from the French for "water of cologne"). The term itself refers to the city of Cologne in Germany where an Italian settled in 1709 and began manufacturing a lotion based on citrus fruit.

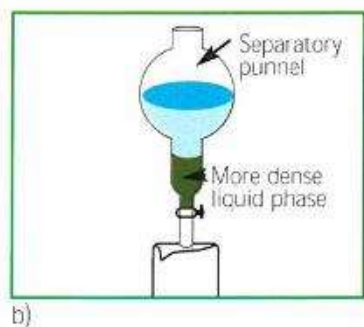
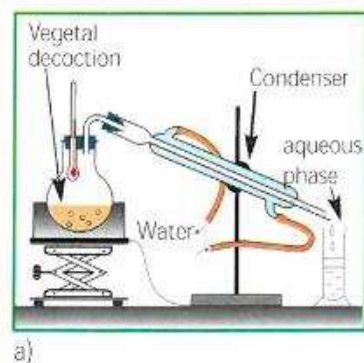


Fig. 8.5 a) Set up for steam distillation. b) separatory funnel.

Aromatherapy

Aromatherapy is a therapy that uses essential oils for treatment. Some doctors believe it has no significant effect on health, whatever relief it confers, they speculate, stems from emotional response to aromatherapy's scents, rather than any physiological effects.

Although medical science can find no physical reason for this therapy, aromatherapy has been used as a comforting ritual to reduce stress, enhance relaxation, and relieve anxiety, to respite insomnia, it has been also used as remedy for impotence, to ease the pain of arthritis, and others.

How the treatments are done?

Aromatherapy treatment relies on the use of highly concentrated essential oils extracted from various healing herbs. The volatile oils are mixed with a vehicle, usually a vegetable oil, such as soya oil, or diluted in alcohol before being applied to the skin, sprayed in the air, or inhaled.

These products are used by :

Inhalation, to treat problems with respiration.

Diffusion into the air, to calm the nerves, to enhance a feeling of well-being, and even to improve respiratory conditions.

Massage, aromatic oil rubbed into the skin may be either calming or stimulating, depending on the type of oil used.

Bathing, to induce a feeling of well-being.

Hot and cold compresses, for muscle aches or pains, bruises, or headaches.

Warning :

Never take aromatherapy oils internally. They are extremely potent and many can be poisonous.

Do not use high concentration of these oils by massage because they may irritate the skin. Some of these oils used by bathing may induce skin allergy.

Whatever the truth of the matter, aromatherapists assign specific properties to each essence. Here are typical claims for some of the more common essential oils:

-Lavender: Heals burns and cuts; destroys bacteria; relieves depression, inflammation, spasms, headaches, respiratory allergies, mus-

Cosmetics

9



Objectives

- Define cosmetics and describe their general composition.
- Identify the composition and the active ingredients of some cosmetics.
- Describe the general properties of some cosmetics.
- Recognize the risk associated with the use of some cosmetics.



Emulsifiers are in general esters of fatty acids. They reduce the surface tension between the two phases, water and oil, that constitute the emulsion. Thus, an emulsifier molecule has a hydrophylic end and a lipophylic end.

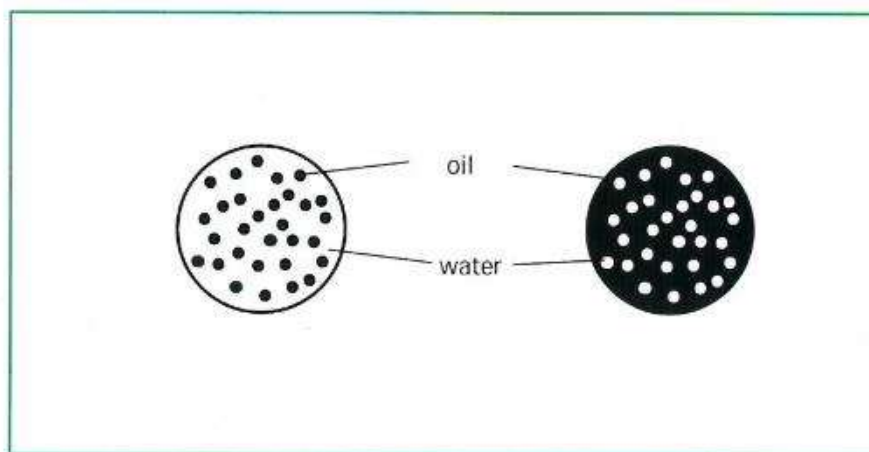


Fig. 9.2: W/O and O/W emulsions:

The effect of the two different types of emulsions is quite different. Water evaporates from O/W emulsion leaves a film of the oily ingredients and causes cooling. On the other hand, a W/O emulsion permits direct immediate contact of the oil phase with the skin. No cooling effects occur because the evaporation of the emulsified water is much more gradual.

B- Preservatives

Preservatives are chemicals used to protect cosmetics from spoilage caused by micro-organisms. Most cosmetics contain preservatives because they are used in direct contact with human body. In the case of shampoo, it is particularly important to ensure that the product contains no pathogenic organisms, especially those capable of damaging the eyes. Examples of preservatives used in shampoos are: Formaldehyde (H-CO-H), which normally constitutes 0.05% of a liquid shampoo, and thymol, which is a bactericide used in anti-dandruff shampoos.

C- Humectants

Humectants are chemicals used to protect emulsions and toothpaste from "drying-up". The most important humectants are glycerol, and sorbitol $\{\text{C}_6\text{H}_8(\text{OH})_6\}$.

These compounds have an affinity for water, they protect emulsions from drying up by preventing water molecules from evaporation because they attract water molecules and prevent them from leaving the emulsion.

D- Colorants

Colorants are chemicals that dissolve or disperse into a medium and color it. Colorants can be organic or inorganic. The

9.3 Composition of cosmetics

The main ingredients of cosmetic products are: active elements, vehicle or excipient, preservatives, humectants, colorants, and perfumes.

Ingredient	Weight %
Sodium lauryl sulfate (anionic surfactant)	27
Cocoamidopropyl betaine (amphoteric surfactant)	5
Diethanolamide (foam regulator)	3
Preservative	Trace
Perfume	Trace
Colorant	Trace
Deionized water	up to 100

Table 9.1: The composition of a shampoo. The list of the ingredients of a cosmetic is often given in decreasing order of their concentration.

A- Vehicle or Excipient

A vehicle or an excipient is a neutral substance that takes in the active ingredient and assures its transfer. In general, vehicles make up the highest percentage ingredient of a cosmetic. Mineral oils, synthetic oils, natural oils, creams, water, and thickeners such as talc, wax, and gum are used as vehicles for cosmetics. For example, the vehicle used in lipsticks is a mixture of mineral oil (paraffin oil), and natural oil (castor oil). The vehicle used in blush is talc which is magnesium silicate ($\text{Mg}_3(\text{OH})_2\text{Si}_4\text{O}_{10}$), and the vehicle used in mascara is gum, which is a polysaccharide. Creams used as vehicles are emulsions.

What is an emulsion?

An emulsion is a mixture of two non-miscible liquids. It is called water oil (W/O) emulsion when a small quantity of water is dispersed into oil and it is called oil/water (O/W) emulsion when small quantity of oil is dispersed into water. W/O emulsifiers and O/W emulsifiers are used to stabilize W/O or O/W emulsions.

Mineral oils for cosmetic use are high-boiling fractions obtained from crude oil distillation that are purified and refined by treatment with sulfuric acid. An example is paraffin.

Synthetic oils are esters usually obtained by direct reaction of fatty acids with alcohols.

Ingredient	Weight %
Almond oil	35
Beeswax	12
Lanolin	15
Spermaceti (from whale oil)	8
Rose water	30

Table 9.2: Composition of a skin moisturizer cream (emulsion (W/O) of rose water in oily mixture).

9.1 What is cosmetic?

A cosmetic is a substance applied directly to the human body to make it more attractive, by cleaning, beautifying, protecting, promoting attractiveness, or altering the appearance (presumably for the better). Examples are: shampoos, make ups, toothpaste, perfumes, moisturizing creams, sunscreens, slimming products... Since they are applied directly to human body, they must not be toxic. In the past, the distinction between cosmetics and drugs was somewhat not clear. Traditionally, medicines, by definition, alter body functions but cosmetics do not. For example, in the case of antiperspirant, it was considered a cosmetic and not a drug although it stops the secretion of sweat glands. Today the distinction is more clear. Cosmetics are used to make us more attractive, while medicines are used specifically for treating diseases and relieving pain.



Fig.9.1: The basic requirements for baby care products are similar to those for adults. However, baby care products contain the minimum requirement of colorants, perfumes, and preservatives to prevent any possible allergy and irritation.

Bath salts are generally intended to soften the water and produce a delicate fragrant atmosphere for bathing.

Some cosmetics may assure more than one function, for example toothpaste can be considered as a hygiene product and a care product.

9.2 Principal formulation of cosmetics

Today cosmetic industry is well developed. Cosmetic products are not anymore exclusive for women, men are increasingly using them (lotions, after-shaves and deodorants). There are many products designed for babies in the market (baby lotion, moisturizer, cologne, talc etc...).

Cosmetics can be classified as hygiene products, care products, and well-being products.

A-Hygiene products

Hygiene products, such as shampoos, bath shower gels, toothpaste and oral rinses are used to clean human body.

B-Care products

Care products, such as moisturizers, anti-age products, sunscreen products, conditioners, and baby care products are used to protect human body.

C-Well-being products

Well-being products, such as hair dyes, slimming products, coloring cosmetics, perfumes and hair gels are used to beautify human body.

<i>Ingredient</i>	<i>Weight (%)</i>	<i>Function</i>
<i>Water</i>	<i>37</i>	<i>Solvent and filler</i>
<i>Glycerol (Humectant)</i>	<i>32</i>	<i>Protect toothpaste from drying-up</i>
<i>CaHPO₄ (Abrasive)</i>	<i>27</i>	<i>Inhibit mechanically the accumulation of plaque</i>
<i>Sodium lauryl sulfate (Surfactant)</i>	<i>2</i>	<i>Cleaning and foaming agent, and inhibitor of enzymes that produce decay</i>
<i>CMC (Binder)</i>	<i>1</i>	<i>Thickening agent and stabilizer</i>
<i>Fluorides(NaF) and other additives</i>	<i>1</i>	<i>Enamel hardener, sweeteners, flavor, and preservatives.</i>

CMC is carboxymethyl cellulose

Table 9.3: Composition of a modern toothpaste.

Today, there are a number of toothpastes with various functions. Accordingly, they may contain a number of active ingredients:

- Abrasive such as (TiO₂ and CaCO₃), inhibits mechanically the accumulation of plaque.

- Surfactant, such as sodium lauryl sulfate, cleans and foams.

- Flavor, such as menthol, freshen the breath.

- Anti-carries, such as NaF and Na₂PO₃F. Fluorides reduce caries by 20 to 30%.

- Anti-plaque, such as chlorohexidine, inhibits plaque accumulation by killing the bacteria that converts our food to plaque.

- Anti-tartar, such as pyrophosphate, inhibits the fixation of calcium phosphate on the plaque.

2-Oral Rinses

Oral rinses are solutions used to kill the bacteria, and to freshen the breath. Oral rinse solution is normally composed of: alcohol (ethanol), detergent, flavor, humectant, and water. The composition of an oral rinse is shown in table 3.9.

A-Toothpaste, oral rinses, and whiteners

Toothpaste, oral rinses, and recently whiteners are cosmetics used to keep our teeth clean and healthy. The role of these cosmetics is related to the main problems encountered with our teeth: dental caries, plaques and tartar build-up, and bad breath due to bacteria.

Teeth are covered by a layer of enamel that serves both to grind food and to protect the tooth. Bacteria of the mouth continuously convert some of the sugars of our food into polysaccharide film called plaque. Plaque sticks easily to enamel and becomes a home to other bacteria, that in turn convert the plaque into an acid that erodes the calcium and the phosphate of the teeth. When enough erosion has occurred microorganisms can pass through the weakened barrier. The result is dental caries, better known as tooth decay or cavities. Further more, when the calcium and the phosphate, normally present in saliva, cover the plaque they form the tartar (calcium phosphate), where many microbes live and cause gum problems.

The whitening agent in "extra whitening" toothpaste is sodium bicarbonate.

Fluoride ions reduce dental caries by transforming the enamel (hydroxyapatite) into a more resistant one (fluoroapatite). Fluoride can also kill some bacteria that reduce the formation of acid



Fig.9.4: Some cosmetics used for teeth.

1-Toothpaste

Toothpaste or dentifrice can be defined as a substance that is used to clean and maintain the teeth and to freshen the breath. Toothpaste is a combination of water, detergent, humectant, binder, flavor, abrasive, and other ingredients, such as fluoride, that protect against decay.

The composition of a modern toothpaste is shown in table 9.2.

following table shows some organic and inorganic chemicals used as colorants for some cosmetics.

Chemical	Type	Color	Cosmetic
Tetrabromofluorescein	Organic	Red	Lipstick
Ultramarine blue	Inorganic	Blue	Eye shadow
Charcoal	Inorganic	Black	Eye shadow
Titanium oxide (TiO_2)	Inorganic	White	Cheek powder (useful in hiding the natural color of the skin)

Table 9.2: Name, type, and color of chemicals used as colorants in cosmetics

E - Perfumes

Perfumes are used to add pleasant odors to cosmetics. Although these compounds contribute less than 1% to the total weight of the cosmetic, the cost of these fragrances make up about 25% of the total value of all the raw materials used in its manufacture. In some cases, such as in soaps, perfumes are used to cover up the bad smell of the raw materials. In other cases, some perfumes with herbal fragrance, such as thymol and eugenol, assist preservatives because they have anti-bacterial activity. This anti-bacterial activity is related to the phenol group.

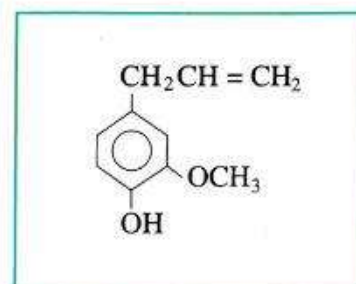


Fig. 9.3: Structural formula of eugenol extracted from clove plant.

F- Active ingredients (Active elements)

Active elements or active ingredients are the chemicals responsible for the action of the cosmetic. Such as the colorant in lipstick and the surfactant in shampoo.

Cosmetics that have more than one function to accomplish contain more than one active ingredient.

9.4 Composition and active ingredients of some cosmetics

To understand the characteristics and the role of a cosmetic, we should know first its composition and its active ingredients.

In this part we are going to discuss the general composition and the active ingredients of some common cosmetics, such as toothpaste, oral rinses, tooth whiteners, antiperspirants, and sun-screen products.

tains protein, along with cholesterol, steroids and other lipids. It has no pronounced odor but, can easily become disagreeable. The bacteria that accumulate in the nearby strands of hair can degrade the contents of the apocrine fluids into foul-smelling products.

Antiperspirants are products used to inhibit sweating, and to keep the body relatively dry.

The most widely used active ingredient in antiperspirants are aluminum chlorohydrates, $\text{Al}_2(\text{OH})_4\text{Cl}_2$ and $\text{Al}_2(\text{OH})_5\text{Cl}$. These compounds release aluminum cations which seem to reduce wetness by physically closing the ducts of the eccrine glands for as long as several weeks. This makes the medium unsuitable for the bacteria to live in and to decompose the organic fluid secreted by the apocrine glands.

In general antiperspirants contain: active ingredient, vehicle, and perfume. They may contain other ingredients depending on how they are formulated (e.g. solution, spray...). The composition of an alcoholic solution of antiperspirant is represented in figure 9.7.

Ingredient	Weight %
Active ingredients (50% solution)	40
Ethanol	45
Humectant	2.5
Perfume	0.5
Water	up to 100

Table 9.5: Composition of an alcoholic solution of antiperspirant.

C- Sunscreens

The ultraviolet spectrum that reaches the earth is between 280 and 400 nanometers (nm), and can be divided into two regions:

-UV/A (320-400nm), produces slow tanning and long-term aging of the skin.

-UV/B (280-320nm), is more powerful of the two, and produces tanning and sunburns. Repeated exposure to the UV/B radiation over a longer periods of time produces skin cancers, especially among fair-

The best way to prevent the degrading of the sweat glands secretion into foul-smelling products by bacteria is to wash our body daily.

Tanning is one of the means for the skin to shield itself against the harm of UV radiation. When the skin is exposed to UV light, it produces substantial amount of the dark pigment melanin which screens out part of the radiation and helps minimizing the damage

Tooth whitening preparations, once exclusive to the dental profession, are now being marketed directly to the general public. However, it is advisable to use them under medical supervision. Peroxides are toxic and their decomposition into free radicals may cause cancer.

Deodorants differ from antiperspirant in that deodorants directly attack odors themselves. They mask odors with fragrant ingredients that cover the offending aroma and with antibiotics that eliminates the bacteria.

Ingredient	Weight %	Function
Ethanol	10	Solvent for the flavors, cleaning agent, kills the bacteria.
Glycerol	10	Humectant, inhibits crystallization around the closer.
Sodium lauryl sulfate (surfactant)	2	Foaming and cleaning agent.
Menthol (flavor)	0.25	Breath-freshener
Water	Up to 100	Vehicle for the ingredients

Table 9.4: Composition of an oral rinse

The active ingredients in a classical oral rinse are: alcohol (ethanol), surfactant, and flavor.

Some oral rinses contain special active ingredients such as: fluoride, anti-plaque, and anti-tartar.

3-Tooth whiteners

The active ingredients in most of tooth whitening products are carbamide peroxide (10%) or calcium peroxide. These peroxides react with water to release hydrogen peroxide (H_2O_2), the chemical responsible for the whitening activity.

B- Antiperspirant products

In order to have good understanding of these products we must first define sweat and its origin.

Two types of sweat glands are present in the human body. The most prevalent (more than two million) are the eccrine sweat glands, which have a primary function of controlling body temperature. The eccrine glands secrete a slightly acidic, very diluted solution of inorganic ions (largely sodium and potassium, and chloride), lactic acid ($CH_3-CHOH-COOH$), some urea ($H_2N-CO-NH_2$), and a little glucose. This secretion ordinarily has no odor.

The apocrine glands, the other type of sweat glands, are found primarily under the arms. They release viscous and milky fluid that con-

The AHA's are citric glycolic, lactic, malic, and tartaric acids. Common natural sources for these acids are, among others, citrus fruits, sugar cane, soured milk, apples, and grapes.

Some cosmetic products fill in wrinkles and appear to make them vanish. Example of composition: water 70%, stearic acid 20%, glycerol 10%.

Some anti-age products are biological products that stimulate cellular activity. They are designed to improve the skin texture and to retard or slow down the appearance of wrinkles.

Vitamin A, vitamin E, vitamin C, and the organic acids, α -hydroxy acids (AHA), are examples of active ingredients used in such cosmetics. Vitamin A (retinol) and its derivatives stimulate the formation of collagen and elastin, the proteins responsible for skin texture and elasticity. Vitamin E is an antioxidant used for maintaining a good skin texture. Vitamin C is an antioxidant, it protects the skin, stimulates the formation of collagen, and activates blood circulation in skin vessels. The AHA (alpha hydroxy acid) preparations have been reported to help slowing down the formation of wrinkles.

C- Astringent substances

Astringent substances are used to tighten the pores and to reduce secretions of the skin. Aluminum salts are examples of astringent chemicals used generally in antiperspirants. Ethanol is another example used generally in astringent lotions used by oily-skinned consumers and teenagers for cleansing and acne treatment.

These astringent lotions are hydroalcoholic solutions with an alcohol content from 20-70%. Solutions with lower level content of alcohol are developed for sensitive skins. While those with higher level content of alcohol are for oily skins.

<i>Ingredient</i>	<i>Weight %</i>
<i>Deionized water</i>	<i>50.90</i>
<i>Ethanol</i>	<i>40.00</i>
<i>Glycerol</i>	<i>4.00</i>
<i>Sorbitol</i>	<i>2.00</i>
<i>Menthol</i>	<i>0.10</i>
<i>Witch hazel extract</i>	<i>3.00</i>
<i>Fragrance and color</i>	<i>trace</i>

Table 9. 6: Composition of hydroalcoholic astringent lotion

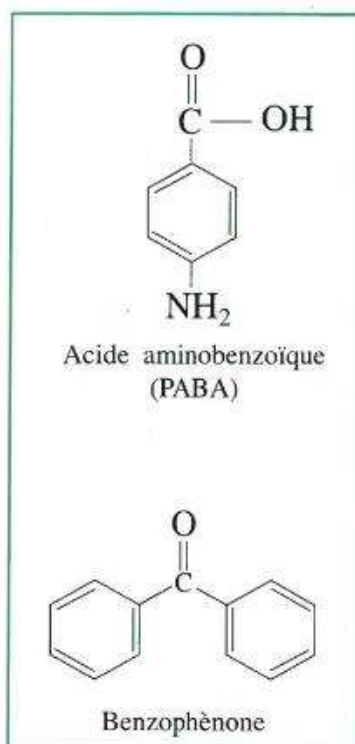


Fig. 9.5: Among the more widely selective active ingredients used in suntan lotions are p-aminobenzoic acid (PABA) and benzophenone. They do a good job of absorbing UV-B radiation while letting UV-A radiation pass through to produce tan. For this reason lotions containing PABA and similar compounds are considered tanning lotions rather than sunscreens. Recently it has been reported that PABA is toxic and may cause cancer.

haired, light skinned people. However, one beneficial effect of the absorption of small amounts of UV/B radiation is the generation of vitamin D.

To minimize the risk from UV radiation and enhance its benefits, we can use sunscreen products, or simply try to expose our skin to sun in the early morning or in the late afternoon.

Sunscreens are products used to prevent amounts of UV radiation from reaching the skin. The most effective active ingredients used in sunscreen products are oxides, especially zinc oxide (ZnO) and titanium dioxide (TiO₂). These oxides scatter all radiation, letting none through the skin.

The degree of protection of any sunscreen product is indicated with a number on its label. Products with higher number, provide higher protection.

9.5 General properties of some cosmetics

Most cosmetics are designed to interact with the largest organ of the body the skin.

The skin has two major layers: the dermis, the inner layer, and the epidermis the upper layer. The dermis contains nerves, blood vessels, hair follicles, and sweat glands. It also contains two kinds of protein, collagen and elastin, that give the skin strength and elasticity. The epidermis (the skin outer layer) consists of several tiers of cells. It contains the pigment producing cells and is basically free of blood vessels. The outermost layer of the skin (stratum corneum) is a layer of 25 to 30 tiers of dead cells, containing mainly the protein keratin. Its main function is to protect us against the outside world (pollution, temperature).

In this part we are going to study the general properties of slimming products and some skin care products such as moisturizing (hydrating) and softening products, anti-age products and biological products stimulating cellular activity, and astringent substances.

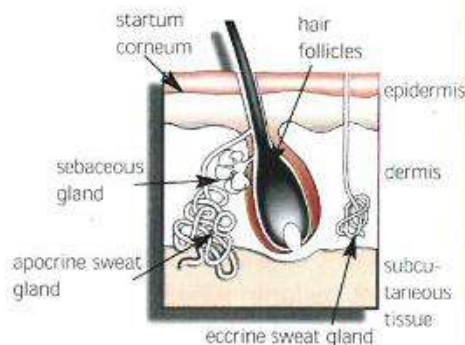


Fig. 9.6 : Cross-section of the skin.

Overall, only about 10% of the retail cost of the average cosmetic goes toward its raw materials: colorants, detergents, perfumes, moisturizers, solvents, and the like. The remainder is paid for packaging, advertising, research, labor, and similar expenses, and of course, profit.

9.7 Economical aspects

Consumers spend a lot of money on chemicals designed to make them more attractive. In the United States, for example, the amount spent each year by every person for such products as shampoos, toothpaste, perfumes, make ups, hair care products and others, is around 80\$ making the annual total 20 billion \$. The annual total spent on cosmetics comes to 30 billion \$ in West Europe and to 15 billion \$ in Japan.

Worldwide, the annual total comes to about 70 billion \$, or roughly 13\$ a year for each and every one of the earth's 5.5 billion inhabitants.

France is considered the first exporter for perfumes, make ups and toiletries products. It is estimated that several 10 billion Franks are invested in this industry (52.7 billion Franks in 1993).

Cosmetic industry is important to economy for three reasons:

- 1) the importance of the research related to this industry
- 2) more jobs are made available by this industry
- 3) the importance of profit in this industry

Review Questions

1. Define cosmetics.
2. Explain why the distinction between cosmetics and drugs was not clear in the past.
3. List the three classes of cosmetic products and give examples of each.
4. List the main ingredients of a cosmetic.
5. Define a vehicle or an excipient. What is the general composition of a cosmetic?
6. Define emulsion and emulsifier.
7. Name substances that can be used as vehicles in cosmetics.
8. Define the role of preservative in a cosmetic, and name one example added to shampoos.
9. Define humectant and describe its role in protecting emulsions.
10. Identify the type of colorants used in cosmetics.
11. Identify the different roles of perfumes in cosmetic products.
12. Give the composition of a toothpaste and identify the active ingredients.
13. Give the composition of an oral rinse and list its active ingredients.
14. Name one active ingredient used in tooth whiteners.
15. Give the composition of an antiperspirant product and



Fig.9.7: Some skin care cosmetics

A- Hydrating (moisturizing) and softening substances

Moisturizers and softeners are applied on the skin to maintain the outermost layer of the skin moisturized and soft. The condition of this layer depends mostly on its water content. A water content of 10% is just about right for this layer, too little dries it out, producing flaking and cracking.

Moisturizers (hydrating substances) are products designed to replace the water that evaporates from the skin. They are usually emulsions, either O/W or W/O. These emulsions contain normally glycerol as humectant.

When the moisturizer is rubbed on the skin, the emulsion (O/W or W/O) breaks-down and adds water from its own composition to the skin. The oil covers the skin by a thin film and lowers the rate of water's evaporation. The humectant helps in binding water molecules to the skin and in resisting their evaporation from it.

The purpose of softeners (emollients) is to soften the lifeless cells of the outermost layer of the skin. One good example of softeners is lanolin which is a mixture of esters extracted from wool.

B- Anti-age products and biological products stimulating cellular activity

With age the skin loses its elasticity and becomes thinner and drier, wrinkles appear especially on the face and around the eyes, and irregular pigmentations (age spots) are formed. These changes occur because of intrinsic and extrinsic factors. Most intrinsic factors are probably unavoidably genetic. They can be somewhat optimized by maintaining overall good health and good blood circulation (by exercising, eating a well balanced diet with adequate vitamin and mineral, and keeping cholesterol profile as good as possible). Extrinsic factors are mainly due to repeated exposure to ultraviolet (UV) radiation. Other external factors are exposures to air pollution and to unfavorable weather conditions like cold, heat, or dryness.

The body supplies protection to the skin with the sebum (oily substance) secreted by the sebaceous glands of hair follicles.

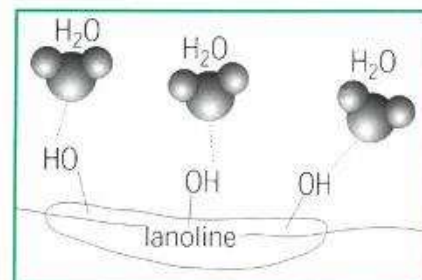


Fig 9.8: The hydroxyl groups of lanolin form hydrogen bonds with water and keep the skin moist. The fat parts of the molecule are soluble in the protein and the fat layers of the skin.

D- Slimming products

Excess weight has been proven to contribute to an increasing risk of developing many medical problems including high blood pressure, high cholesterol, heart diseases, and diabetes. The consumption of excess fatty food and calories play a significant role in the development of excess weight. Reducing the consumption of fatty and high calorie foods is one potential way of losing weight.

Different cosmetic products designed to reduce body fat are now in the market. Slimming products are chemicals that can either be applied directly on the skin, or taken internally.

Those that are applied directly on the skin are claimed to reduce cellulite, fat under the skin, and are called anti-cellulite. While, those that are taken internally may modify the function of the body by stimulating the formation of fat burning hormones, or blocking the absorption of the fat in the intestines.

Care should be taken when using slimming products. Some creams may irritate the skin, and those taken orally may affect the normal function of the body.

9.6 Risks of using cosmetics

Medicines and cosmetics are products used on human body for different purposes. While drugs require elaborate safety testing prior to receiving approval from the concerned authorities, cosmetics do not.

In fact, a drug is submitted to years of testing to determine its efficacy and long term toxicity before it is commercialized.

Although the majority of cosmetics come in direct contact with the skin and might be absorbed by it, they are usually submitted only to rapid dermal tests for allergic reactions before they are commercialized. In spite of this, some allergic reactions to these products are still encountered by some people.

Some cosmetics, used innocently for long time, have been reported to be harmful or to contain harmful ingredients for humans. It has been reported recently, that PABA, the active ingredient of most commercialized suntan products, is harmful and it may cause skin cancer.

Misuses of some cosmetics may also be harmful. Tooth whiteners should be used with caution because the active ingredient is a peroxide, and peroxides have been reported to be harmful and may cause cancer.

It is also advisable to avoid the use of cosmetics that alter the normal function of the body such as slimming products (taken orally) because their long-term effect is not known yet.

It is advised not to use kohl because it contains lead.

Research on the effect of chemicals is going on. Therefore, some chemicals which are used in cosmetics may not be considered harmful for humans at the present time but may turn out to be harmful in the future.

ture accompanying their formation. As a result, as new hair grows out, the "permanent" process has to be repeated.

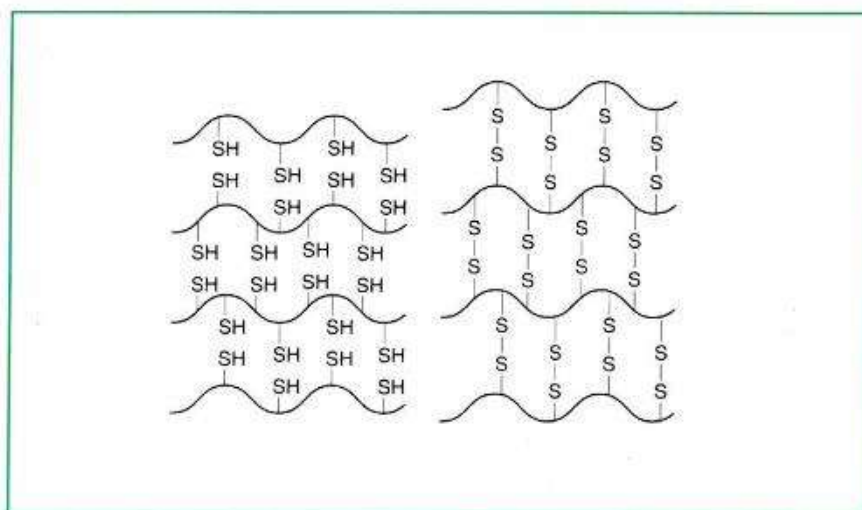


Fig.9.9 b: Oxidizing agents form disulfide linkages at new locations.

Adapted from: Stoker, H.S.(1998), Organic and biological chemistry .
Boston: Houghton Mifflin Company. P.302.

3-Anti-age products are used to:

- a-reduce cellulite.
- b-stop the wrinkles.
- c-treat acne
- d-retard the appearance of wrinkles.

4-Glycerol is often used as humectant in cosmetics because it:

- a-dissolves oil in water.
- b-has an affinity to water.
- c-stops the proliferation of bacteria.
- d-masks the bad smell.

5-Sunscreens are used to :

- a-give color to skin.
- b-protect the skin from the harm of UV radiation.
- c-moisturize the skin.
- d-give total protection from UV radiation.

III-Describe the role of the following substances:

1. Fluoride (NaF) in toothpaste.
2. Titanium dioxide in sunscreen.
3. Lanolin in softener.
4. Paraffin oil and castor oil mixture in lipstick.
5. Aluminum salts in antiperspirants.
6. Talc in face powder.
7. Abrasive in toothpaste.
8. Humectant in oral rinse.
9. Calcium peroxide in tooth whitener.
10. Oil in moisturizer.

name one active ingredient.

16. Name active ingredients used in sunscreens and explain how they protect the skin.

17. Identify the composition of moisturizers.

18. Justify the need for using a moisturizer.

19. Give an example of skin softener.

20. Name some substances that are used in anti-age products.

21. Define the role of an astringent substance and give some examples.

22. Define a slimming product.

23. Do drugs and cosmetics follow the same testing regulation before being commercialized?

24. Identify the risks associated with the use of some cosmetic products.

25. Explain why cosmetic industry has a positive effect on a country's economy.

Research activities

1. Write an essay on the controversy regarding the use of deodorants versus antiperspirants.

2. Write an essay on the risks of using cosmetics (for example hair dye).

3. Estimate the money your family spends on cosmetics per year. Deduce how much each person spends.

4. Find the brand names of cosmetics made in Lebanon.

Document 9

Denaturation of human hair: Permanent

The process used in waving hair, that is, in a hair permanent, involves reversible denaturation. Hair is protein in which many disulfide (-S-S-) linkages occur as part of its tertiary structure; 16%-18% of hair is the amino acid cysteine. It is these disulfide linkage that give hair protein its overall shape. In the permanent process, hair is first treated with a reducing agent (ammonium thioglycolate) that breaks the disulfide linkages in the hair, producing two sulfhydryl (-SH) groups:

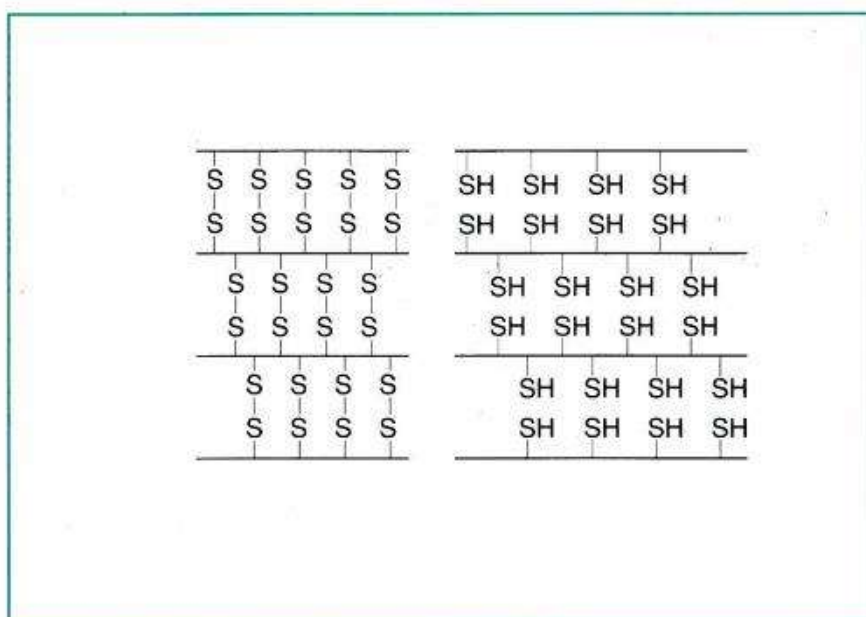


Fig. 9.9 a: Reducing agents break the disulfide linkages producing SH groups.

The reduced hair, whose tertiary structure has been disrupted, is then wound on curlers to give it a new configuration. Finally, the reduced and rearranged hair is treated with an oxidizing agent (potassium bromate) to form disulfide linkages at new locations within the hair:

The new shape and curl of the hair are maintained by the newly formed disulfide bonds and the resulting new tertiary struc-

unit Questions

I-Label each of the following statements as true or false, and correct the false one.

1. Cologne fragrant oils make up 10 to 25% of the ethanol solution.
2. Perfumes fragrant oils have the same volatility.
3. In general, fragrant oils are slightly soluble in organic solvents and very soluble in water.
4. Essential oils are extracted from plants.
5. The perfume natural raw materials are of plant origin and of animal origin.
6. Fixatives give perfumes their characteristic odor.
7. Cosmetics are substances used to make humans more attractive.
8. Perfumes are hygiene products.
9. Perfumes are added to soaps to cover some bad smell.
10. Preservatives are used in cosmetics to prevent their dehydration.
11. Emulsions are the vehicles used in moisturizers.
12. Humectants have great affinity to water.
13. Thymol is used in shampoo as preservative.
14. Aluminum salts are the active ingredients in sunscreens.
15. Fluorides are anti-plaque agents used in toothpaste.
16. Moisturizers (hydrated substances) are used to disinfect the skin.
17. Anti-age products are used to slow down and retard the appearance of wrinkles.
18. Vitamin A is an astringent agent.
19. Astringent agent used for the treatment of acne is ethanol.

II-Choose the right answer

1-Essential oils are:

- a-extracted from olives
- b-of animal origin.
- c-extracted from plants.
- d-petroleum derivatives.

2-The active ingredients in a classical oral rinse are:

- a-ethanol, humectant and flavor.
- b-ethanol, flavor and surfactant.
- c-ethanol, surfactant and preservative.
- d-ethanol, surfactant and abrasive.

6-The composition of a brand of a baby cream is the following:

Ingredient	Weight %
Mineral oil	15
Softener	5
Emulsifier	5.5
Glycerol	2
Water	72.5
Perfume, preservatives etc.,	Trace

a-Identify the vehicle in this cosmetic.

b-What is the role and the type of the emulsifier?

c-What is the role of the glycerol?

d-Why does this cream contain preservative?

e-What is the role of this cream?

IV-Relate the chemicals in column A to the corresponding cosmetic in column B where they are used.

A	B
1. Abrasive (SiO_2)	a-Antiperspirant
2. Aluminum salts	b-Softener
3. Lanolin	c-Anti-age product
4. Talc ($\text{Mg}_3(\text{OH})_2\text{Si}_4\text{O}_{10}$)	d-Tooth paste
5. Ethanol	e-Sunscreen
6. Vitamin A	f-Cheek powder
7. Formaldehyde ($\text{H}_2\text{C}=\text{O}$)	g-Oral rinse
	h-Shampoo
	i-Slimming product

V-Thinking problems

- There is a variety of antiperspirants on the market.
 - Describe their role and their mode of action.
 - Are you for or against their use? Explain.
- What are the possible risks associated with the use of slimming products?
- Tooth paste is considered as hygiene and care product. Explain why.
- Slimming products are considered as cosmetics and not as drugs. Explain why.
- The following table contains the ingredients and the weight percentages of a brand of lipstick. Complete this table by giving the function of each ingredient.

Ingredient	Weight %	Function
Castor and paraffin oils	50	Gives the stick of the lipstick its stiffness by elevating the melting point of the oily mixture.
Lanolin	25	
Beeswax	18	
Colorant	4-8	
Perfume and flavor	Trace	

Unit 3



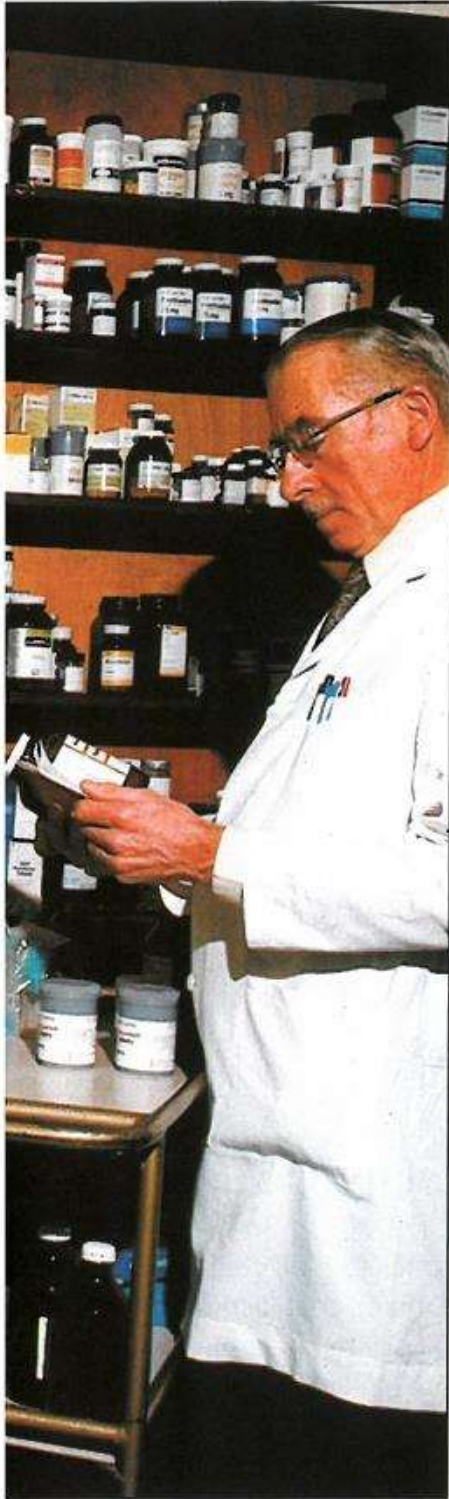
Painting 1860: The pain

Current Medicinal Drugs

Current M



Medicinal Drugs 10



Objectives

- Define some current medicinal drugs.
- Identify the brand name, the generic name, and the possible side and overdose effects of some current medicinal drugs.

The superscript ® is the symbol of "registered name".

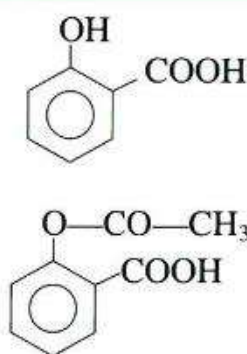


Fig.10.5: Chemical structures of salicylic acid and aspirin (acetylsalicylic acid).

tion of the muscles and the joints.

Table 10.1 represents the generic name, the trade name and the pharmaceutical action of some of the most widely used mild analgesics (antipyretic and anti-rheumatic) in Lebanon.

Generic name	Trade name	Pharmaceutical action
Aspirin	Aspirin®	Analgesic, anti-pyretic, and Anti-rheumatic
Paracetamol or Acetaminophen	Panadol® Tylenol®	Analgesic and anti-pyretic

Table 10.1: Generic name, trade name, and the pharmaceutical action of some of the most widely used mild analgesics in Lebanon.

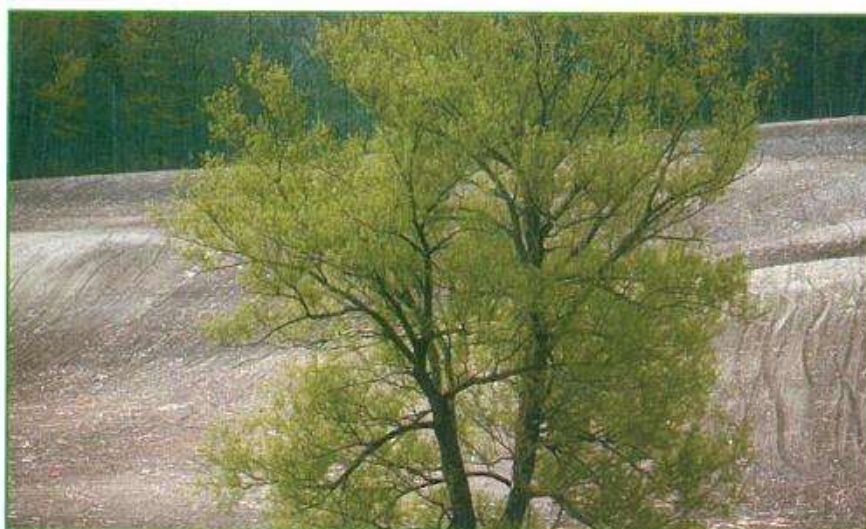


Fig. 10.4: The bark of the willow tree contains salicin, which can be converted to salicylic acid and then to aspirin (acetylsalicylic acid). The name of an entire family of analgesic compounds (salicylates) is derived from the Latin term for willows (salix).

Aspirin is an effective pain reliever, but it causes severe damage to the stomach leading to significant bleeding. It is estimated that 2% to 10% of all those who take aspirin suffer occasionally from some form of stomach distress or nausea. Some of the major risks in taking aspirin routinely come from its effect on the blood's ability to clot. It inhibits the formation of the blood's platelets. In addition aspirin is more toxic than we may realize, it has LD₅₀ of 1.5g/Kg (orally in mice and rats).

Tylenol and panadol have been proved to be a reasonable analgesic and anti-pyretic substitutes for aspirin, especially for children and patients who can not tolerate aspirin.

10.5 Anesthetics

Anesthetics are substances that produce anesthesia, a state of being unable to feel pain, heat, cold etc...

Modern surgical procedures would not be possible without anesthetic agents to block the traumatic pain that would otherwise be experienced by patient.

Anesthetics are classified into local anesthetics and general anesthetics.

Local and general anesthetics are reversible depressants of the nerves, which means that when the drug is eliminated by metabolism or excretion its action is terminated and the nerves resume completely normal function.

A- Local anesthetics

Local anesthetic drugs are widely used to provide temporary pain relief on localized regions of the body. They act directly on nerve cells to block their ability to transmit impulses down their axons. They can be either administered by injection close to the nerves to be blocked, or applied topically to the surface of the site to be treated.

Local anesthetics are used for minor surgery such as treatment of a superficial skin lesion, in dentistry, and others.

One of the oldest local anesthetics used by injection is cocaine, its synthetic derivative lidocaine, is currently the preferred one. Ethyl chloride is an example of local anesthetic used as a spray.



Fig. 10.7 Ethyl chloride is a local analgesic commonly used as a spray on injuries in sporting events.

Other examples of local anesthetics include procaine, etidocaine, mepivacaine, bupivacaine.

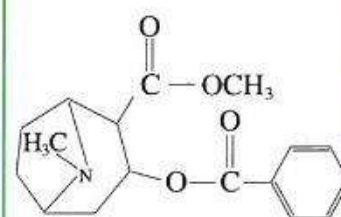


Fig. 10.6 Chemical structure of cocaine.

stearate) are present in small amounts. They prevent the ingredients from sticking to the machinery used to compress the drug into a tablet.

- Disintegrators such as dry starch, expand when they become wet, speed the breakdown of the tablet and aid in absorption.

- Coating has various uses. They solve problems of taste, appearance, and stability by keeping the contents from contact with air, and they can be designed to control the location and rate of release of the drug.

- Fillers such as sucrose, lactose, or some other carbohydrates are used to increase the size of the tablet to a reasonable volume.

Capsules are usually made of the active ingredient of the drug and a gelatinous container. The gelatinous container has no flavor, and it keeps the contents from contact with air. It disintegrates rapidly, thus the contents are quickly released. The drug can be in powdered form so that there is no delay in disintegration after it is consumed and therefore it can be readily dissolved and absorbed.

Although capsules circumvent problems of taste, solubility, and disintegration, pharmaceutical manufacturers prefer to market their products in tablet form because capsules are more expensive.

10.3 Characteristics of some currently used drugs

Drugs may be classified according to their chemical structure or pharmaceutical activity. For convenience, we will organize our discussion, in this unit, in terms of pharmaceutical activity rather than chemical structure.

Thus drugs will be classified according to what they do, for example, analgesics (relieve pain), tranquilizers (reduce anxiety) etc...There are at least 100 categories of drugs. We will discuss here only a few, the most commonly used drugs such as analgesics, anesthetics, anti-acids, anti-inflammatory, antibiotics, tranquilizers, and anti-depressants.

10.4 Analgesics

Analgesics are substances that reduce or eliminate pain.

Pain is often the first indication of malfunction or injury, and decreasing it may be the first item of concern. A vast array of analgesics is available, both over the counter and by prescription. Analgesics are divided into three groups:

- Narcotic, which are used to reduce or eliminate moderate and severe pain, morphine and codeine are the prototype.
- Anti-pyretics, which are used to lower or eliminate fever.
- Anti-rheumatics, which are used to reduce or eliminate inflamma-



Fig 10.3: The German chemical firm Bayer, who had the exclusive rights to fabricate Aspirin in the United States, has made its fortune from this drug.

Thalidomide was used in Europe as a tranquilizer from 1957 to 1962. Large numbers of its consumers were pregnant women, and in 1962, serious birth defects were traced to it.

Because of the thalidomide disaster, many testing regulations for drugs have been put into effect since the early 1960s.



Fig.10.1: Drugs can be formulated into tablets, capsules, liquids, sprays, ointments, and suppositories.



Fig. 10.2: Billions of tablets and capsules are manufactured each year.

10.1 What are Medicinal Drugs?

Medicinal drugs are substances used for treating diseases or relieving pain.

Drugs come from several sources. Some are synthesized (aspirin, tranquilizers and antihistamines). Some are obtained by fermentation or by microbiological cultures (antibiotics such as penicillin, streptomycin, tetracycline). Others are obtained through extraction from animal sources (insulin and hormones) or plant sources (alkaloids such as opium).

A drug is considered effective if it is nontoxic and if it has no side effects such as an allergic reaction or diarrhea. At the very least, the range between the therapeutic and toxic doses should be wide. The effectiveness of a drug should not be diminished by substances present in human body, such as gastric acid. A drug must be easily administered in reasonable sized doses, be stable so that it can be stored for long periods at various climatic extremes, and inexpensive.

Few drugs can meet in full with these conditions at the same time. As with almost everything in life, some compromises must be reached.

10.2 Drug formulations and drug additives

Drugs can be formulated into tablets, capsules, liquids, sprays, ointments, suppositories, and so on. In general drugs are composed of active ingredient and inert ingredients. The inert ingredients are drug additives. Drugs active ingredient is the same for different brands, but the inert ingredients may vary with the brands and the formulation. For example, when a drug is formulated into a tablet, inert ingredients such as fillers, binders, and a coating are added. These additives contribute to the tablet size, shape, flavor, and appearance. Whereas drug formulated into a capsule is free of these additives.

Tablets and Capsules

The majority of drugs are formulated into tablets or capsules. Tablets are made by mixing the ingredients (active and inert) and compressing them in a mechanical device under high pressure.

The main additives in tablets are binders, lubricants, disintegrators, coatings, and fillers.

- Binders such as starch paste provide adhesiveness to the tablet.

- Lubricants such as soaps (calcium stearate and magnesium

Activity 10.1

Activity 10.1: Identifying the pH of anti-acids solutions.

Objective:

To identify the pH of a number of liquid anti-acids.

Material:

- Three samples of liquid anti-acids.
- pH paper.
- Glass rod.

Procedure:

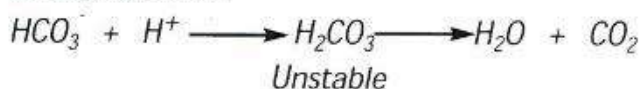
- 1-Determine the pH of the three samples.

Question:

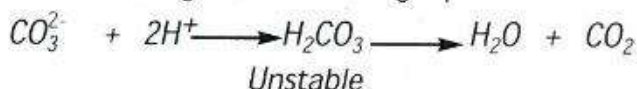
What can you deduce?

How do anti-acids react?

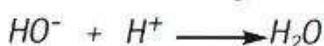
Anti-acids are weak bases that neutralize the excess of gastric acid. Sodium bicarbonate reacts with the gastric acid according to the following equation:



Calcium carbonate and magnesium carbonate react with the gastric acid according to the following equation



Aluminum hydroxide and magnesium hydroxide react with the gastric acid according to the following equation



Activity 10.2

Activity 10:2 Comparing the effectiveness of different brands of anti-acids.

Objective:

To compare the effectiveness of different brands of liquid anti-acids.

Material:

- Three samples of liquid anti-acids.
- Distilled water.
- Hydrochloric acid 1N.
- Bromothymol blue.
- Graduated cylinder 100ml.
- Three Erlenmeyer flasks 250mL.
- Burette.
- Pipette.

Procedure:

- In the table provided, record the brand name and the active ingredient(s) listed on the label of the anti-acids.
- Using the pipette take 5ml of an anti-acid solution and transfer it to a clean Erlenmyer flask, then add 100mL of distilled water.
- Add few drops (2 or 3 drops) of the indicator to the Erlenmeyer flask. The solution turns blue.
- Using the burette add the hydrochloric acid while swirling the flask. Record in the data table the volume of acid added to obtain a yellow color.
- Repeat the same procedure for the two other anti-acids.

Anti-acid	Active ingredient(s)	Volume of HCl

Question:

Which anti-acid is more effective?

Ethyl chloride ($\text{CH}_3\text{CH}_2\text{Cl}$) is a gas at room temperature (boiling point 12°C), but it can be liquefied under pressure in a sealed container. When the liquid ethyl chloride is sprayed on the surface of the skin, it quickly evaporates and cools the surface so quickly that it freezes the tissues near the surface, making them insensitive to pain. It is often sprayed on a wound or injuries in sporting event and it can also be used for minor surgery.

Because the termination of local anesthetic action ultimately depends on the movement of drug into the systematic circulation, side effects and toxicity may occur in organs and tissues perfused by blood. The nervous system and muscles are potential targets. In this case it may cause tremor and restlessness, respiratory failure, general depression and drowsiness.

B- General anesthetics

General anesthetic agents are reversible depressants of the central nervous system that produce a loss of sensation and consciousness.

General anesthetics are classified into inhaled and intravenous anesthetics.

Inhaled anesthetics can generally induce unconsciousness with few minutes after inhalation of the drug. Diethyl ether was the first (1840s) anesthetic used successfully during surgical procedures on humans, other examples of inhaled anesthetics include nitrous oxide and halothane.

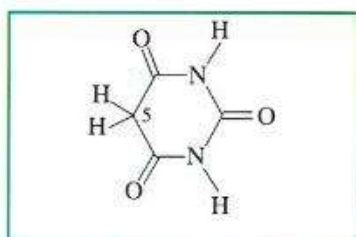


Fig.10.8: Chemical structure of barbituric acid

Other examples of intravenous anesthetic include benzodiazepines, ketamine, etomidate, and propofol.

Name	Chemical formula
Nitrous oxide	N_2O
Halothane	$\text{CF}_2\text{-CHBrCl}$
Isoflurane	$\text{CF}_2\text{-CHCl-O-CHF}_2$
Enflurane	$\text{CHFCl-CF}_2\text{-O-CH}_3$
Methoxyflurane	$\text{CHCl}_2\text{-CF}_2\text{-O-CH}_3$
Diethyl ether	$\text{CH}_3\text{-CH}_2\text{-O-CH}_2\text{-CH}_3$

Table 10.2: Name and chemical formula of some general inhaled anesthetics.

Intravenous anesthetic agents are administered by injection. These agents are used either alone or are administered in conjunction with inhaled anesthetic agents.

Penthiobarbital is an example of intravenous anesthetic, which belong to the group of synthetic derivatives of barbituric acid.

None of the general anesthetics currently available are free of problems. In general, the side effects of anesthetic agents are dose related, and include depression of respiratory drive and depressed cardiovascular function, nausea, and vomiting.

An overdose of general anesthetics may lead to coma and death.

10.6 Anti-acids

Anti-acids are substances used to treat problems caused by excessive gastric acid, HCl.

Many over the counter anti-acids are available in the market, they contain basic active ingredients such as sodium bicarbonate NaHCO_3 , calcium carbonate CaCO_3 , magnesium carbonate MgCO_3 , aluminum hydroxide Al(OH)_3 , magnesium hydroxide Mg(OH)_2 , and magnesium trisilicate $\text{Mg}_2\text{Si}_3\text{O}_8$.

Many persons prefer effervescent anti-acids. Effervescent anti-acids contain both sodium bicarbonate and citric acid or tartaric acid. Thus, when they are dissolved in water the acid and the base react to give carbon dioxide, which evolves as gas and produces the bubbling and effervescent action.

Table 10.3 represents the brand name and the active ingredient of some commonly anti-acids used in Lebanon.

Brand name	active ingredient
Maalox	Magnesium hydroxide and aluminum hydroxide
Milk of magnesia	Magnesium hydroxide
Rennie	Calcium carbonate and magnesium carbonate

Table 10.3: Brand name and active ingredient of some commonly used anti-acids in Lebanon.



Fig. 10.9: Examples of some commonly used anti-acids in Lebanon.

High doses of magnesium salts may have a laxative effect. In fact milk of magnesia Mg(OH)_2 is sold for that purpose. Aluminum salts may be constipating, again depending on the dose. Therefore, several anti-acids are formulated with combinations of magnesium and aluminum salts to prevent such side effects.

Anti-acids that contain sodium should be avoided by individuals on low-sodium diets (people suffering from high blood pressure).



Fig.10.10: Aspirin, Advil, and Voltarene are examples of non-steroidal anti-inflammatory drugs.

Advil is not a good choice for individual with high blood pressure or heart disease because it may cause sodium and water retention

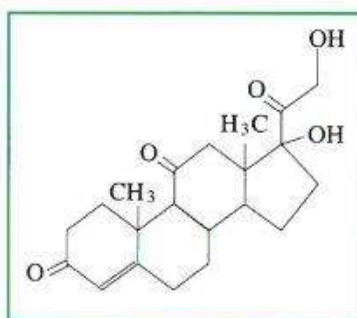


Fig. 10.11: Chemical structure of cortisone.

10.7 Anti-inflammatory drugs

An anti-inflammatory drug is a substance that reduces or eliminates inflammation. Inflammation is the reaction of tissues to injuries, it affects the connective tissues and constitutes a reaction of defense of the organism against external aggressions, which may result from blows, foreign bodies, chemicals, electricity, heat and cold (thermal causes), microorganisms, and surgery (traumatic causes). Anti-inflammatory agents can be divided into two main groups: steroidal anti-inflammatory agents (substances with structure based on steroid nucleus), and non-steroidal inflammatory agents. Table 1.4 represents the generic name, the trade name, and the side effects of some commonly used non-steroidal anti-inflammatory drugs.

Generic name	Trade name	Possible side effects
Aspirin	Aspirin®	Stomach distress and nausea
Ibuprofen	Advil®	Nausea, vomiting, diarrhea, digestive hemorrhage, itching, dermal eruption, asthma attacks, anemia, acute renal insufficiency
Diclofenac	Voltarene®	

Table 10.4 Generic name, Trade name, and side effects of some non-steroidal anti-inflammatory drugs.

Salicylates such as aspirin, salsalate, and diflenac are often used to treat arthritis and other inflammatory disorders.

An overdose with aspirin is often characterized by a marked increase in respiratory rate, which causes the overdosed person to appear to pant (quick breath, gasp). Cortisone (Cortisone®), hydrocortisone or cortisol (Hydrocortisone®), and their synthetic derivatives are examples of steroidal anti-inflammatory drugs. The side effects of steroidal anti-inflammatory drugs include fluid retention, unwanted hair, a moon-face appearance, and mental depression. In its anti-inflammatory action, cortisone also suppresses the body's ability to fight infection and response to stress. An overdose of cortisone and hydrocortisone may lead to overweight and obesity, hypertension, decrease in size of the muscles (amyotrophia), muscular paralysis due to extreme potassium depletion in blood (hypokalemia), and increase in the rate of glucose in blood (hyperglycemia).

20. List the advantages and the disadvantages of administering a combination of antibiotics.
21. Describe the resistance of bacteria to antibiotics.
22. Identify the brand name and the generic name of two antibiotics.
23. Describe the overdose effects of an antibiotic.
24. Define tranquilizers.
25. Identify the brand name and the generic name of two tranquilizers.
26. Describe the overdose effects of a tranquilizer.
27. Define anti-depressant drugs.
28. Identify the brand name and the generic name of two anti-depressant drugs.
30. Describe the overdose effects of an anti-depressant drug.

Research Activities

1. Write a report on general safety rules to follow when using medicinal drugs.
2. Write a report on alternative medicines such as herbal medicines.
3. Write a report comparing different types of pain relievers in terms of price, content....
4. Write a report on Aspirin and its alternatives including advantages and disadvantages of both.
5. Write a report on the use of anesthetics in surgery throughout history.
6. Write a report on the use of antibiotics in animal feed.
7. Conduct research and write a report on the diseases that are treated by antibiotics.
8. Write a report on the abuse of tranquilizers and alternatives to treatment by tranquilizers.
9. Write a report on diseases treated by tranquilizers.
10. Write a report on diseases treated by anti-depressant drugs.
11. Write a report on the abuse of anti-depressant drugs.

D- Combination of antibiotics

Many infections can and should be treated with a single antibiotic. However, physicians often use a combination of two or three anti-biotics:

- In case of emergency, when the invading microorganism is unknown.
- To treat a mixed infection.
- To obtain a more intensive effect, and to limit the side effects.
- To prevent emergence of resistant bacteria.
- To permit the use of a lower dose of one of the antibiotics.

Some times the combination of two antibiotics is considered a poor strategy because the two antibiotics may be antagonistic toward one another.

10.9 Tranquilizers

Tranquilizers are substances used to sedate without inducing sleep. They are used to relieve anxiety, excitement, and restlessness.

Anxieties can be classified into normal and pathological. Normal anxieties are of short duration and are usually event-related and not under the subject's control. Under severe stress, normal individuals may experience periods of discomfort of minor aches and pains, irritability, or sadness. However, research suggests that pathological anxiety is not an exaggeration of normal anxiety but may stem from inheritable traits. Generally the treatments of normal and pathological anxieties involve the same drugs.

All tranquilizers must be taken only by prescription. Benzodiazepines such as diazepam and chlorodiazepoxide are among the most prescribed drugs. Table 10.5 represents the generic name, the trade name, and the overdose effects of some commonly prescribed tranquilizers.

Generic name	Trade name	Over dose effect
Diazepam	Valium®	State of intoxication (drunkenness), drowsiness, coma depression in respiratory drive
Chlorodiazepoxide	Librium®	muscle rigidity, coma, and death

Table 10.5: Generic name, trade name, and overdose effects of some tranquilizers.

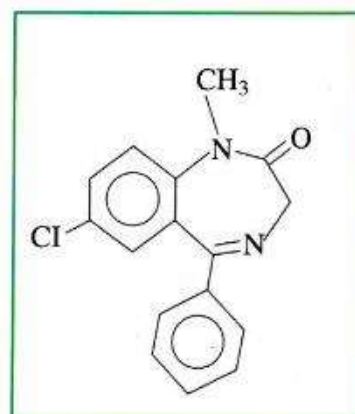


Fig.10.14: Chemical structure of valium.

Driving should be avoided when taking tranquilizers.

1961. The development of new antibiotics is a continuing goal of the pharmaceutical industry.

B- Broad spectrum and narrow spectrum antibiotics

Antibiotics can be classified into broad spectrum and narrow spectrum antibiotics.

Broad-spectrum antibiotics are effective against a wide variety of microorganisms. While, narrow spectrum antibiotics are effective against a specific microorganism.

Physicians usually prescribe a narrow spectrum antibiotic when the invading microorganism is identified and a broad-spectrum antibiotic when the invading microorganism is unknown. Although the use of broad spectrum antibiotics seems to be a logical approach to treat infections, however their use should be restricted because they might kill some beneficial bacteria such as the intestinal floras.

C- Bacteria resistant to antibiotics

Bacteria have been proven to develop resistance to antibiotics. A bacterium, that can not any more be eradicated by the antibiotic, is called "resistant".

Bacteria resistance to antibiotics can be natural or acquired. Acquired resistance can be against many antibiotics or limited to one antibiotic. It is usually the result of genetic modifications.

Bacteria resistance to antibiotics can be explained by different mechanisms:

-Some resistant bacteria have the capacity to produce enzymes, which modify or break the molecules of the antibiotic, and render it inactive.

-In other cases, the bacteria are capable of growing in the presence of the unmodified antibiotic. In such case the bacteria either modify their permeability to the antibiotic or change the structure of their site of action so the antibiotic will not be able to kill them, or they acquire a state of tolerance.

Resistance is one reason for continuing research efforts to develop new antibiotics.

Bacteria that are resistant to penicillin have developed the enzyme penicillinase, which transforms the penicillin to an ineffective antibiotic.

10.10 Anti-depressants



Fig.10.15: Depression is one of the most underdiagnosed and undertreated of the medical illness. About 10% of the adult population can expect to experience a period of depression serious enough to require medical intervention during lifetime.

Other examples of anti-depressant drugs include fluoxetine (Prozac) and imipramine (Tofranil).

Anti depressants are drugs used for treating depression. Typical symptoms of depression include continuous sadness, loss of interest and pleasure in activities, crying spells, emotional lability, and feeling guilt, worthless, and hopelessness.

Barbiturates, such as phenol barbitol, secobarbital (Segonal®), and amobarbital (Amital®) are the most prescribed anti-depressants.

The possible overdose effects of barbiturates are unconsciousness, coma, respiratory paralysis, and death.

Review Questions

1. Define a medicinal drug.
2. List the different formulations of drugs.
3. Describe the role of drug additives in tablets.
4. Compare the advantages of drugs in tablet and capsule form.
5. Define and classify analgesics.
6. Give the brand name and the generic name of two common analgesics.
7. Define and classify anesthetics and give one example of each class.
8. Classify general anesthetics and give one example of each type.
9. Identify the possible side effects and overdose effects of general anesthetics.
10. Identify the methods of application and the possible side effects of local anesthetics.
11. Define anti-acids and list four active ingredients.
12. Give the brand name of two anti-acids
13. Describe the chemical composition of effervescent anti-acids.
14. Define anti-inflammatory drugs.
15. Identify the brand name and the generic name of a number of anti-inflammatory drugs.
16. Describe overdose effects of an anti-inflammatory drug.
17. Define and classify antibiotics.
18. Outline the history of antibiotics.
19. Differentiate between broad spectrum and narrow spectrum antibiotics.

10.8 Antibiotics

Antibiotics are chemicals produced by one microorganism (natural antibiotics), or by chemical synthesis to kill or inhibit the growth of other microorganisms. Antibiotics that kill bacteria are called bactericidal, those that kill fungi are called fungicidal.

Examples of commonly used antibiotics include penicillin such as benzyl penicillin (Penicillin G®), sulfa drugs such as sulfadiazine (Adiazin®), and other antibiotics such as streptomycin (Streptomycin®), erythromycin (Erythrocin®), and tetracyclin (Tetracycl®).

An overdose of penicillin G may cause dysfunction of the brain (encephalopathie): dizziness, hallucination, troubles in the perception, intense contractions in the heart muscles (myoclonic) and in limb muscles.

Diarrhea (sometimes persistent) is a frequent side effect of antibiotic therapy. Normally, a heavy concentration of beneficial bacteria lives in the large intestine (intestinal floras). The use of antibiotics upset these intestinal floras and cause diarrhea.

A- History of antibiotics

The history of antibiotics is relatively recent. It started in 1907, when the German scientist Paul Ehrlich, while he was searching for a treatment for the African sleeping sickness, discovered the salvarsan, a substance that has a lethal effect on disease-causing organisms. This substance contained arsenic.

The German Gerhard Domagk discovered in 1931 that the sulfonamide "Prontosil", which has been synthesized as a red dye for wool, showed antibacterial action. Domagk used the sulfonamide Protonsil, called also "sulfa drug", on infected animals. He also used it to treat his daughter of a bacterial infection.

In 1929 the British Alexander Fleming discovered that the mold *Penicillium notatum* produced a substance lethal to the bacterium *Staphylococcus aureus*. The discovered substance was called penicillin. Fleming was unable to isolate penicillin.

Penicillin was first used in 1941 on an Oxford policeman who was suffering from general infection. His condition showed astonishing improvement after four days of treatment, but he eventually died, because supplies of the drug have run out.

The first penicillins came into general use after world war II. One of the most widely used, ampicillin, was not available until

The development of antibiotics is one of the greatest achievements in the medical sector.

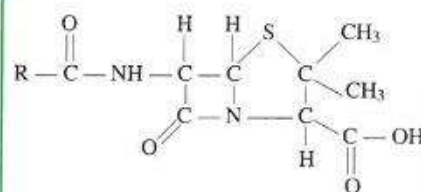


Fig. 10.12: Chemical structure of penicillin.

Antibiotics react on the microorganisms by different mechanisms. They can react on the bacterium membrane, on the cytoplasm membrane, or on the nucleus.

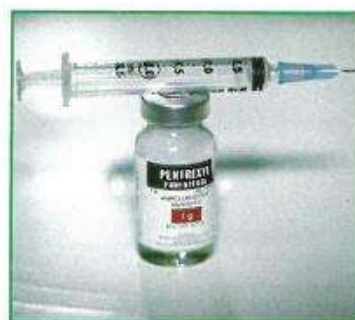


Fig. 10.13: In 1929 the British Alexander Fleming discovered the first antibiotic "Penicillin".

3-Nitrous oxide is the active ingredient of a (n):

- a-local anesthetic used by injection.
- b-inhaled anesthetic.
- c-intravenous anesthetic.
- d-local anesthetic used as a spray.

III-Relate the drugs in column A to their generic name in column B.

Column A

- 1-Aspirin
- 2-Panadol
- 3-Milk of magnesia
- 4-Librium
- 5-Valium
- 6-Amital
- 7-Penicillin G
- 8-Advil

Column B

- a-Ibuprofen
- b-Diazepam
- c-Chlorodiazepoxide
- d-Aspirin
- e-Amobarbital
- f-Paracetamol
- g-Sulfa drug
- h-magnesium hydroxide
- i-Benzylpenicillin

IV-Short answer questions

1. What are the two commonly used over the counter analgesics? How do their actions differ?
2. What class of analgesics is usually used to treat severe pain?
3. What cause the effervescent action of some anti-acids?
4. What are the principal drug additives in a tablet?
5. What is the role of starch paste in the production of drug tablets?

V-Thinking problems

1. What are the risks in using Aspirin?
2. List the advantages and the disadvantages of the use of anesthetics.
3. People who are under the effect of tranquilizers or anti-depressant drugs are not allowed to drive, why?
4. With the exception of mild analgesics, drugs must be bought by prescription, why?
5. Read the following drug leaflet, then answer the questions:

DI-ANTALVIC FOR ADULTS

QUALITATIVE AND QUANTITATIVE COMPOSITION

Dextropropoxyphene

unit 3 Questions

I-Label each of the following statements as true or false, and correct the false ones.

1. Medicinal drugs are substances used for treating disease and eliminating pain.
2. Analgesics are substances used to reduce anxiety.
3. Anesthetics are substances used to alleviate fever.
4. Antibiotics are substances used to reduce inflammation.
5. Valium and Librium are examples of anti-depressant drugs.
6. Penicillin and sulfa drugs are examples of antibiotics.
7. Paracetamol (Panadol)[®] is an anti-inflammatory drug.
8. When a drug is formulated into a capsule or a tablet it contains the same ingredients.
9. Analgesics are classified into anesthetics, anti-pyretic and anti-rheumatic.
10. The possible side effects of general anesthetic are dose related and include nausea and vomiting.
11. The ingredients of an effervescent anti-acid are sodium bicarbonate and sodium hydroxide.
12. Cortisone is an anti-inflammatory with no side effects.
13. A bactericidal is an antibiotic.
14. Broad-spectrum antibiotics are used to kill a large number of bacteria.
15. The pH of an anti-acid is higher than 7.
16. The combination of two antibiotics could be antagonistic.
17. Local anesthetics have no side effects.

II-Choose the correct statement.

1-Aspirin is an:

- a-antibiotic, anti-inflammatory, and analgesic.
- b-anti-inflammatory, anti-pyretic, and antibiotic.
- c-analgesic, anesthetic, and anti-pyretic.
- d-analgesic, anti-inflammatory, and anti-pyretic.

2-Penicillin is an:

- a-analgesic.
- b-anti-inflammatory.
- c-antibiotic.
- d-anti-pyretic.

When antibiotics are chosen based on the most likely type of bacteria, this is called "empiric therapy". Generally, empiric therapy is designed to cover a number of different organisms. Once the laboratory data return with more specific information about the infecting bacteria, the antibiotics can be adjusted to cover the organisms involved in the infection. Treatment of a specific organism is termed "directed therapy". At this point the physician must decide whether to continue the same antibiotic that was started empirically at the onset therapy or whether a different antibiotic should be used. Physicians may wish to change to antibiotics that are more specific for the particular organism, especially if they are better tolerable or safer for the patient.

Once antibiotic treatment is started, it is very important that patients continue to take their antibiotic therapy as prescribed. One of the major factors in selecting out resistant organisms is patient discontinuing their antibiotic therapy too soon or taking the medication intermittently. This has been a particular problem with tuberculosis and has resulted in emergence of multi-drug resistant bacteria.

John Segreti/1999 Grolier interactive Inc.

hydrochloride.....30mg
Paracetamol.....400mg

Excipients: magnesium stearate

Per capsule

Capsule shell: gelatin, titanium oxide, indigotin, yellow iron oxide, sulfuric anhydride.

PRESENTATION FORM

Box containing 20 capsules under thermoformed alveolar patches.

PHARMACOTHERAPEUTIC CATEGORY

Peripheral analgesic/opioid

(N: central nervous system)

PERSON RESPONSIBLE FOR PLACING THE PRODUCT ON THE MARKET

NAME AND ADDRESS

Laboratories HOUDE

1, terrasse Bellini

92910 Paris La Défense Cedex

Tel :01 40 81 42 00

MANUFACTURE'S NAME AND ADDRESS

USIPHAR

F.60200 Comiègne

CASE(S) IN WHICH THIS DRUG CAN BE USED

(THERAPEUTIC INDICATIONS)

This medicine is a composition of two active ingredients: it is an analgesic (which relieves pain)

This medicine is recommended for the treatment of moderate or severe pain which not relieved by aspirin or paracetamol alone.

CAUTION!

CASE(S) IN WHICH THIS DRUG MUST NOT BE USED

This medicine should not be used in the following cases:

- children under the age of 15 years,
- known allergy to paracetamol and/or dextropropoxyphene,
- severe kidney or liver disease,
- breast feeding (see section on Pregnancy and Breast-feeding)

Unless your doctor decides to the contrary, this medicine should not generally be used in the event of treatment with carbamazepine (a compound used to treat epilepsy).

IF YOU HAVE ANY DOUBT, ASK YOUR PHYSICIAN OR PHARMACIST FOR ADVICE.

SPECIAL WARNING

Avoid the consumption of alcoholic beverages.

Prolonged use at high doses may give rise to a state of dependency.

Therapeutic uses of antibiotics

Antibiotics should be used only when an infection susceptible to the antibiotic is present or suspected. While this may sound obvious, many times doctors will prescribe and/or patients will demand antibiotics for conditions that are not caused by organisms that can be treated with these types of drugs. A common example is the patient who demands an antibiotic for a cold. The common cold is due to a variety of viruses that do not respond to any currently available antibiotics. Use of an antibiotic in this situation does not benefit the patient and exposes him or her to potential toxicity from the drug. More importantly this practice helps select bacteria resistant to antibiotics by killing off nonresistant bacteria. It is critical for doctors and patients alike to be aware that not all infections need antibiotic for therapy.

If the patient appears to have an infection, the physician needs to determine its type and location, to begin to figure out the type of organisms involved. Appropriate culture and radiographic studies are obtained as necessary. If the physician believes that treatment with antibiotic is indicated, an antibiotic covering the most likely organisms associated with the infection is chosen. For example, the types of bacteria commonly causing a urinary tract infection are quite different from those associated with sore throat. The physician must also decide whether the patient can be treated with antibiotics administered orally or intravenously or whether the patient requires admission to the hospital or can be treated on an outpatient basis.



Fig. 10.16: Examples of some commonly used antibiotics.

- e-What are the side effects of this drug?
- f-Is it safe to drive if you are under the effect of this drug?
- g-Is it an over the counter drug?
- h-In case of an overdose, what should you do ?
- i-What drugs should you avoid when you take this drug?

TREATMENT DURATION

Do not use for prolonged periods.

IN ALL CASES, ALWAYS STRICTLY COMPLY WITH YOUR DOCTOR PRESCRIPTION

MANAGEMENT OF OVERDOSE

Take medical advice in the event of overdose or accidental poisoning.

RISK OF WITHDRAWAL SYNDROME

Risk of dependency in the event of prolonged use at high doses.

ADVERSE REACTIONS

Like all active products, this medicine may, in certain people, give rise to varying degrees of unpleasant effects.

The following may occur in certain rare cases: skin rash (red lesions on the skin) or allergic reaction: stop the treatment and inform your doctor.

This medicine may, in certain people, give rise to varying degrees of unpleasant effects:

- nausea, vomiting,
- or abdominal pain, constipation, dizziness, minor visual disorders, headaches, fatigue, sleepiness, disorientation,
- hypoglycemia (lowering of blood sugar levels, resulting in sweating, problems with concentration or mood, malaise),
- hepatitis (liver impairment, with possible yellow coloring of the eyes or skin).

INFORM YOUR PHYSICIAN OR PHARMACIST OF ANY ADVERSE REACTION THAT IS NOT MENTIONED IN THE LEAFLET

CONSERVATION

DO NOT EXCEED THE EXPIRY DATE WHICH IS CLEARLY INDICATED ON THE PACKAGING.

PACKAGE INSERT REVISION DATE

October 1996

LABORATOIRES

HOUE

Y712770

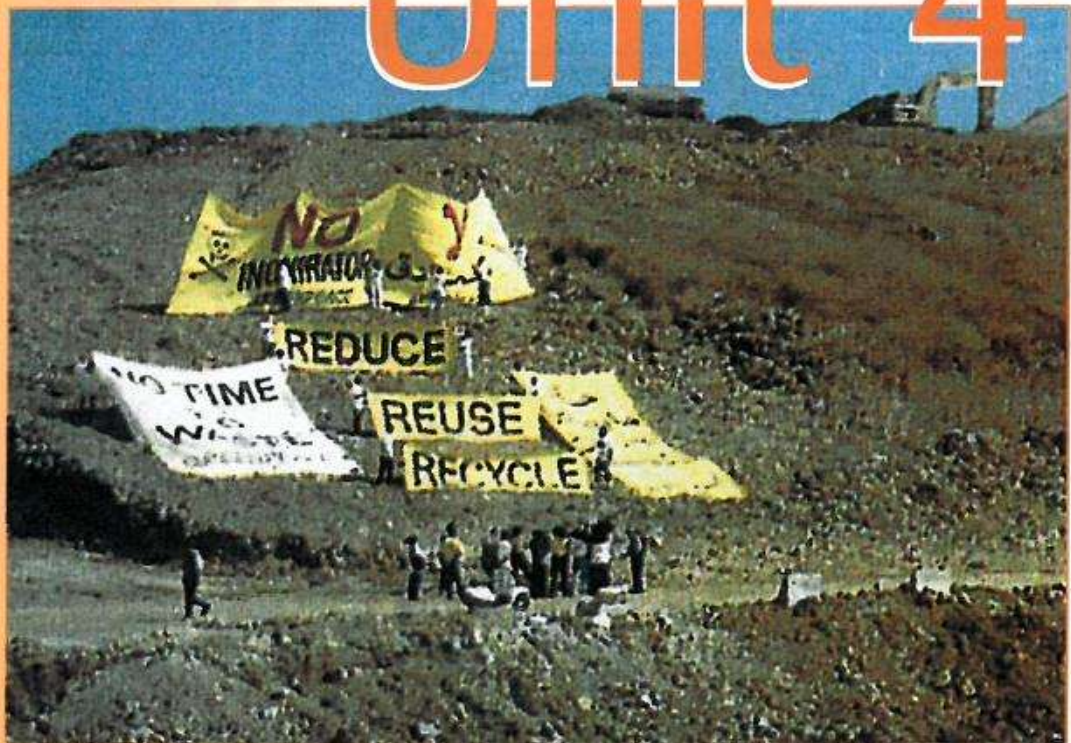
a-What is the pharmaceutical activity of this drug?

b-What are the active ingredients?

c-What is the role of magnesium stearate?

d-What should be the characteristics of the capsule shell?

Unit 4



Treatment of Wastes

Chapter 11: Water and Soil Pollution

Chapter 12: Solid and Hazardous Wastes

PRECAUTION FOR USE

Do not exceed the recommended dose.

If you are receiving treatment with an anti-depressant or a tranquilizer, consult your doctor before taking this medicine.

IF YOU HAVE ANY DOUBT, ASK YOUR PHYSICIAN OR PHARMACIST FOR ADVICE.

DRUG INTERACTIONS AND OTHER INTERACTIONS

IN ORDER TO AVOID POSSIBLE INTERACTIONS BETWEEN SEVERAL DRUGS, SYSTEMATICALLY INFORM YOUR PHYSICIAN OR PHARMACIST OF ANY OTHER CURRENT TREATMENT.

In particular, inform your doctor or pharmacist if you are receiving a treatment of epilepsy which contains carbamazepine.

PREGNANCY AND BREAST-FEEDING

Your doctor may consider it necessary for you to receive this medicine during pregnancy. In all cases, you should comply with the dosage and duration of treatment which have been prescribed for your case.

This medicine passes into breast milk.

When administered in a breast-feeding mother, it may cause respiratory pauses or lowered tone in the infant. Consequently, this medicine is contraindicated in breast-feeding mothers.

DRIVERS AND MACHINE OPERATORS

This medicine may cause dizziness and drowsiness.

SPORTMEN

Warning: take care, this specialty contains an active product which can induce in a positive way during antidoping controls.

HOW TO USE THIS MEDICATION

DOSAGE

Reserved for the use in adults from the age of 15 years.

Follow your doctor prescription.

Generally, 4 capsules per day at regular intervals. Capsules should be swallowed with a large glass of water, and taken for preference at the time as a meal or snack.

MODE AND ROUTE OF ADMINISTRATION

Oral route.

FREQUENCY AND TIME OF ADMINISTRATION OF THE DRUG

Doses should be taken in intervals of six hours (or a minimum of 4 hours), capsules should be taken during a meal or snack.

In the event of renal failure, doses should be taken at 8 hours intervals.

Soil Pollution 11



Objectives

- List pollutants of water and soil and identify their sources.
- Describe methods of water and soil treatment.
- Describe health effects of water pollution.
- Describe economic effects of soil pollution.

Fluoridation is a last step in the treatment of water in some countries. At very low concentration, up to about 1 ppm (parts per million), fluoride ions (F^-) can reduce tooth decay, causes of osteoporosis (bone-calcium loss among older-adults), and hardening of arteries.

3- Flocculation: suspended particles are removed by the addition of coagulant agents, such as ferric and aluminum sulfates. These substances form gelatinous metal hydroxides which trap and settle the suspended particles.

4- Hardness removal: excessive hardness, which renders the water unsuitable for many industrial purposes, is removed by precipitating of the Ca^{2+} and Mg^{2+} ions.

5- Settling and filtration of products of steps 3 and 4.

6- Post-chlorination: the chlorine concentration is adjusted so that to protect water from bacterial infestation.

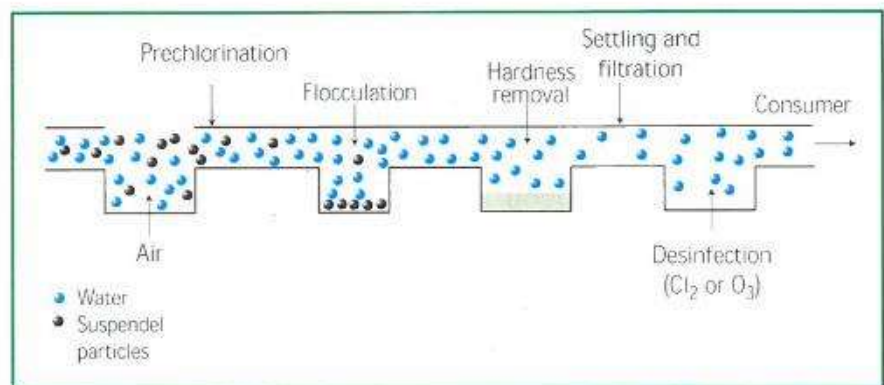


Fig. 11.2: Municipal water treatment plant diagram

B- Sewage (wastewater) treatment

The major goal of wastewater treatment is to reduce its content of suspended solids, oxygen-demanding materials (mainly organic material, the decomposition of which leads to oxygen depletion), dissolved inorganic compounds (particularly compounds of phosphorus and nitrogen), and harmful bacteria. The treatment of wastewater can be divided into three stages.

- **Primary treatment:** This stage involves four main steps:
 - large objects are removed by screens.
 - heavier materials settle out in a grit chamber.
 - suspended solids are removed in a sedimentation tank.
 - bacteria and odors are destroyed by chlorination.
- **Secondary treatment:** In this stage dissolved organic matter is broken down and removed by microorganisms.
- **Tertiary treatment:** The primary and secondary treatment methods are adequate for normal sewage. However, other treatments are necessary for sewage with complex industrial

most toxic pesticides that were widely in use in the past are the chlorinated hydrocarbons (DDT).

Fertilizers contain phosphates and nitrates, which cause water pollution. Fertilizers drain from the land and enter the water supply by way of runoff. Nitrates and phosphates contribute to eutrophication. In sufficient concentrations nitrates are toxic to humans and animals.

Water pollution from agricultural sources can also reach the groundwater by way of infiltration in areas of agricultural activities

C- Sewage water (domestic waste):

The two major sources of domestic pollution are detergents and human organic wastes.

Synthetic detergents are generally non-biodegradable and are not removed by accepted waste treatment methods. They persist when discharged into streams. Another pollution problem arises from detergent builders such as phosphate compounds (STPP).

The second major cause of domestic pollution is human organic wastes. These are the main source of microorganisms causing water biological pollution.

11.2 Water treatment methods

Today, many rivers are both a source of municipal water and a place to dump wastewater. So water must be cleaned twice: before and after we use it. Pre-use cleaning, called water treatment, occurs at a treatment plant. Post-use cleaning of water occurs at a sewage treatment plant.

A- Municipal water treatment

The treatment plant is the place where water from rivers or wells is purified to make sure that the water produced presents no hazards to the consumer. Water treatment steps are the following:

1- Aeration: Contact of water with air removes volatile and odorous substances such as hydrogen sulfide, carbon dioxide, methane, methane thiol (CH_3SH), and bacterial metabolites. This removes undesirable taste and odor. Contact with oxygen also removes iron by oxidizing the soluble iron (II) to insoluble iron (III).

2- Pre-chlorination: Chlorine, a powerful disinfecting agent, is added to kill disease-causing organisms.

Water and



Thermal pollution is an example of physical pollution of water. It occurs when cooling water from factories and power plants are discharged into bodies of water causing their temperatures to increase.

Usually fish are severely affected by a pH of 5 and can not survive in a pH of 4.5 or less.

The oil spills in the Gulf area in 1983 during the Iran-Iraq conflict, and in 1991 during the Gulf war, resulted in enormous damage to the entire area, especially to the marine life.

11.1 Water pollutants and their sources

Water pollution is the degradation of the quality of water by the discharge of polluted substances into rivers, streams, lakes, and oceans. It can be chemical, physical, or biological.

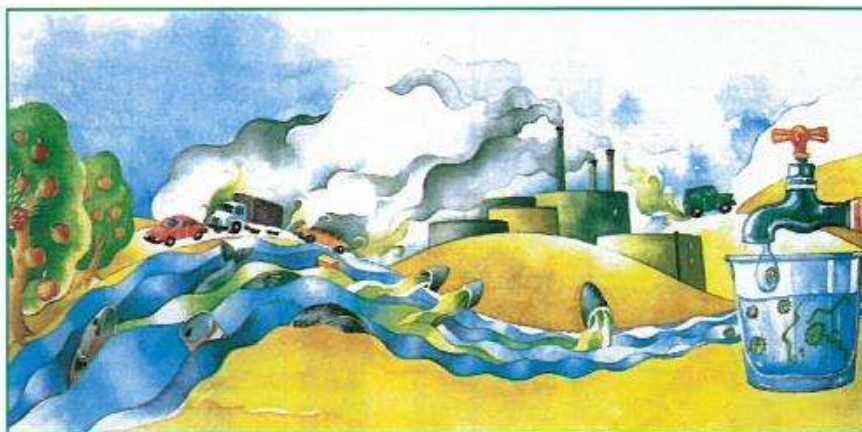


Fig. 11.1: Water pollution originates from different sources

Water pollution originates from different sources: industry, agriculture, and domestic wastes.

A- Industry:

Many industries generate a lot of organic and inorganic wastes that are often discharged in rivers and seas. Acids, bases and salts as well as heavy metals such as lead, mercury, and cadmium are examples of inorganic pollutants. Petroleum waste and residues of organic chemical industries, such as plastics, pesticides, and paints are examples of organic pollutants.

Acid rain is another source of water pollution resulting from industry. Acid rain is the term used for rain with a pH of less than 5.6 making lakes and rivers unable to support fish and plants. Another problem related to acid rain is that water of low pH is able to dissolve toxic minerals from the ground including those which contain aluminum and heavy metals. Those would not normally be sufficiently soluble in water to be harmful.

Another industry related source of pollution is oil spills. They may arise from leaks and discharges from oil transporting ships.

B- Agriculture:

In modern agriculture, pesticides and fertilizers are essential in producing food and other crop products required for the world population. They are also sources of water pollution.

Most pesticides are chemically non-biodegradable and as a result of extensive use they accumulate in the environment. The

wastes. In the tertiary treatment of wastewater, the following are removed:

- Plant nutrients such as phosphates and nitrates.
- Non-biodegradable organic chemicals such as the chlorinated hydrocarbons used in the manufacture of pesticides.
- Industrial inorganic pollutants such as dissolved heavy metals.

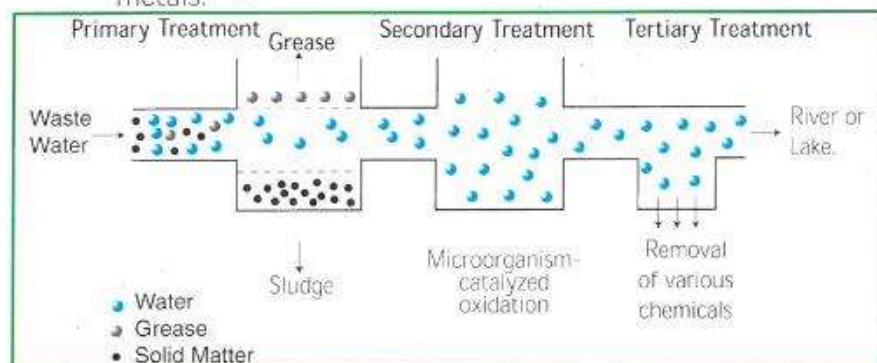


Fig. 11.3: The stages of treatment of wastewater.

11.3 Description of some steps in the treatment of water

A- Removal of organic substances

Most of the biodegradable organic matter can be removed from water through biological oxidation by microorganisms. This is done in one of two processes.

- In the first, water is passed through a bed of stones where bacteria gather and multiply until they have the capacity to breakdown most of the organic matter in the wastewater.
- In the second process, air is blown into the container of water mixed with sludge containing bacteria that breakdown the organic matter. The aeration replaces the oxygen removed in the bacterial action, and thereby speeds up the process.

Non-biodegradable organic matter that resists normal bacterial action is removed by a special method called adsorption. The wastewater is passed through beds of activated charcoal which adsorbs as much as 98% of the organic material.

B- Removal of inorganic substances

Precipitation and coagulation methods have been used for years to treat industrial wastes. Chemicals are added to precipitate dissolved materials or to coagulate fine suspended particles. The precipitate or coagulate is then allowed to settle out in a sedimentation tank.

is bubbled into the water or sodium hypochlorite, NaOCl or calcium hypochlorite, Ca(OCl)_2 , are added to water.

Ozone (O_3) can also be used to disinfect water if is bubbled through it for about 10 minutes. The disadvantage is that, since the lifetime of ozone molecules is short, this method would not protect the water once it leaves the treatment plant. Also, ozone can be toxic if not used properly.

11.4 Health effects of water pollution

The World Health Organization (WHO) has estimated that 80 % of sickness and disease in developing countries is caused by the contamination of water by organisms and sanitation. Water-borne diseases include cholera, typhoid, dysentery, bilharzia, and leprosy.



Fig. 11.6: Polluted water is the main cause of diseases in developing countries

Excess nitrate ions (NO_3^-) in drinking water is a potential health hazard. Special bacteria can reduce them to the nitrite ions (NO_2^-), which combine with the hemoglobin in blood and hence preventing the proper absorption and transfer of oxygen to cells and leading to respiratory failure. Nitrites are also considered carcinogenic.

Heavy metals like lead, mercury, cadmium, and arsenic are toxic even in small amounts. Their ions attach themselves to proteins in our bodies prohibiting them from functioning properly. The effects of their poisoning is severe. They include damage of the nervous system, liver, and kidney; probable mental retardation, birth defects, and death.

Non-biodegradable organic substances may cause cancer when accumulated in the body.

The problem of contaminated water is not just confined to developing countries. One-quarter of all mediterranean beaches are considered dangerously filthy. Nearly all its mussels and oysters are unsafe to eat.

Heavy metals are particularly hazardous because they can build up in the food chain and become more dangerous as they accumulate in the body.

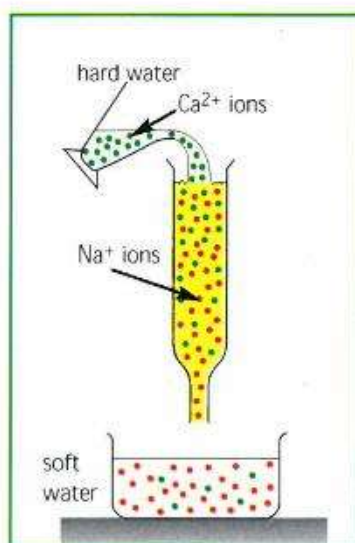


Fig. 11.4: Ion exchange water softener.



Fig.11.5: Calcium hypochlorite is often used to disinfect swimming pools.

- Sodium aluminate and aluminum sulfate are commonly used to coagulate alkaline substances in water.
- Phosphate ion can be removed by the addition of sufficient calcium hydroxide, $\text{Ca}(\text{OH})_2$, so that insoluble calcium phosphates such as $\text{Ca}_3(\text{PO}_4)_2$ and $\text{Ca}_5(\text{PO}_4)_3\text{OH}$ are formed as precipitates that can be readily removed.
- Heavy metals can be removed by adding sulfide ions to form the insoluble metal sulfides.
- Nitrogen compounds (e.g. nitrates and nitrites) can be removed by converting them to nitrogen or ammonia. In the case of ammonia, this is followed by "air stripping" the gas from water. Water is vigorously aerated causing ammonia to pass from the solution to the gas state.

C- Removal of calcium

Excess calcium can be removed from water by one of the following methods:

- Addition of phosphate ion to precipitate the calcium ion in a process analogous to that previously discussed for phosphate removal.
- Addition of sodium carbonate, Na_2CO_3 , causes the calcium ions to precipitate as calcium carbonate.



- Addition of slaked or hydrated lime (CaO and $\text{Ca}(\text{OH})_2$) raises the pH of water, by increasing the concentration of the hydroxide ion and converts the bicarbonate ion (HCO_3^-), naturally present in the water, into carbonate ions.



The carbonate ions can then react with the calcium ions to precipitate CaCO_3 .

- Another method for removing calcium from hard water is the ion-exchange method. Hard water is passed through an ion-exchange resin. Hardness-causing cations are retained on the ion-exchange resin, and cations that do not cause hardness (usually Na^+) are released from the resin into the water to take their place.

D- Water disinfection

For decades, chlorine has served as a water disinfectant. When it is added to water, it kills pathogenic microorganisms. Chlorination takes place in one of two ways: either chlorine gas, Cl_2 ,

Most soils are composed mainly of air, water, minerals, and organic matter. The organic matter is derived from the decomposition of animal and plant residues and is called humus. Soil mineral matter contains sand (silicate minerals), clay (silicates of aluminum, potassium, and magnesium cations), and limestone (calcium carbonate).

Lead arsenate was the main pesticide used in the past.

11.5. Soil pollutants and their sources

The increasing amounts of pesticides and other agricultural chemicals applied to soils plus industrial and domestic waste-disposal practices led to increasing concern over soil pollution. Soil pollution is the buildup of pollutants, which have adverse effects on plant growth and animal health. Pollutants are substances that accumulate in the soil medium and change the natural soil equilibrium.

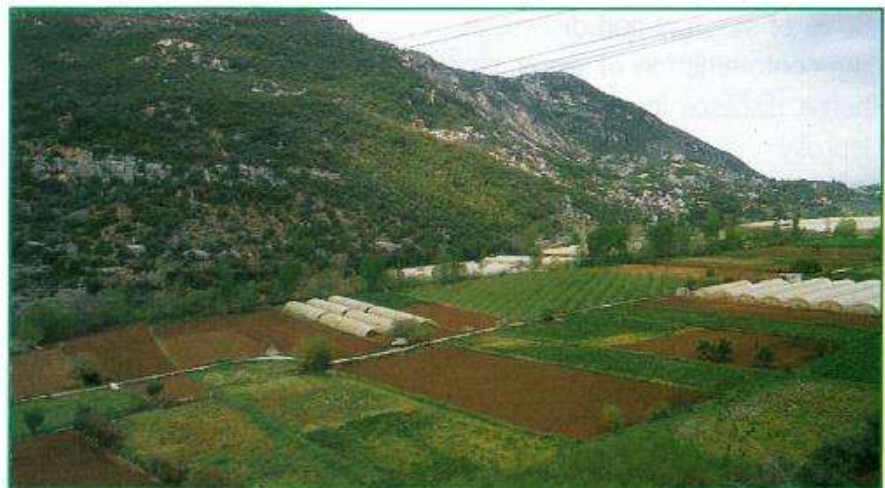


Fig. 11.7: Soil pollution may cause soil infertility.

Soil pollutants can originate from many sources such as industrial wastes, waste disposal, agriculture, and acid rain.

Some soil pollutants are:

Nitrogen compounds: the main inorganic N compounds detected in soils include nitrates, nitrites, and ammonium. An important source of the nitrogen compounds is excessive fertilization of agricultural lands and from land disposal of wastes and sewage effluents. Adverse effects of nitrogen encompass both human health and the terrestrial environment, and these occur when excess of nitrates and their reduction product nitrites are present.

Salts: the irrigation of arid lands often leads to pollution with salts. Irrigation water usually contains dissolved salts. As the water evaporates or is used by plants, the salt will accumulate on the soil as a whitish deposit. Sometimes, saline water is used for irrigation and as a result salt is accumulated in the soil. When this happens, soil becomes infertile.

Heavy metals: arsenic (As) and lead (Pb) have accumulated in the soil as a result of years of spraying crops with lead arsenate. An increase of lead content in soils is also due to its use as an additive in gasoline. Cadmium (Cd) sources can be industrial sludges and

effluents, fertilizers, and radioactive waste disposal. Mercury (Hg) sources are mainly from chlorine manufacturing and Hg mining. Heavy metals attached to soil will end up in the food we eat.

Organic compounds: non-biodegradable organic substances such as pesticides (chlorinated hydrocarbons), and organic solvents (benzene, toluene, xylene, dichloromethane, and trichloroethylene) are considered soil pollutants. The hazards of such compounds result from their persistence in soils (they have long half-lives) and their accumulation in the food chain and consequently in the human body

11.6 Treatment of polluted soil

The objective of treatments is to reduce the risk related to the presence of pollutants in soil to an environmentally acceptable level. Treatment does not necessarily mean that pollutants have to be removed. In general, soils are typically remediated to reduce the potential risk to human health and to reduce a source of groundwater and surface water contamination.

Remediation technologies serve to immobilize contaminants, separate them from the soil, or destroy them. These technologies are used to treat contaminated soil in-situ (in place) or after the soil is excavated and transported for treatment or to be disposed of.

A-Immobilization technologies:

Immobilization technologies used to immobilize contaminants in soils, sludges, or sediments include stabilization/solidification and vitrification.

1- Stabilization/Solidification processes treat waste to improve handling and the physical characteristics of the waste.

Stabilization usually involves adding materials which insure that the hazardous constituents are maintained in their least mobile or toxic form. Solidification implies that treatment produces a solid block of waste material so that the contaminants are mechanically locked within the solidified matrix.

2-Vitrification is the thermal treatment process that melts soil or sludge into a glass-like material.

B- Separation technologies:

Contaminants can be separated from soils based on their volatility, their solubility in water, or their electrical charge. Separation technologies produce concentrated contaminants which must be further treated to destroy or immobilize the contaminants.

1- Soil vapor extraction (SVE) has been used to extract volatile organic contaminants from soil into special units for treatment.

The goal of thermal desorption is to separate, not to incinerate, the contaminants. Thermal desorption units operate at temperatures lower than incinerators and, often, by using a sweep gas in the thermal processor which contains little or no oxygen.

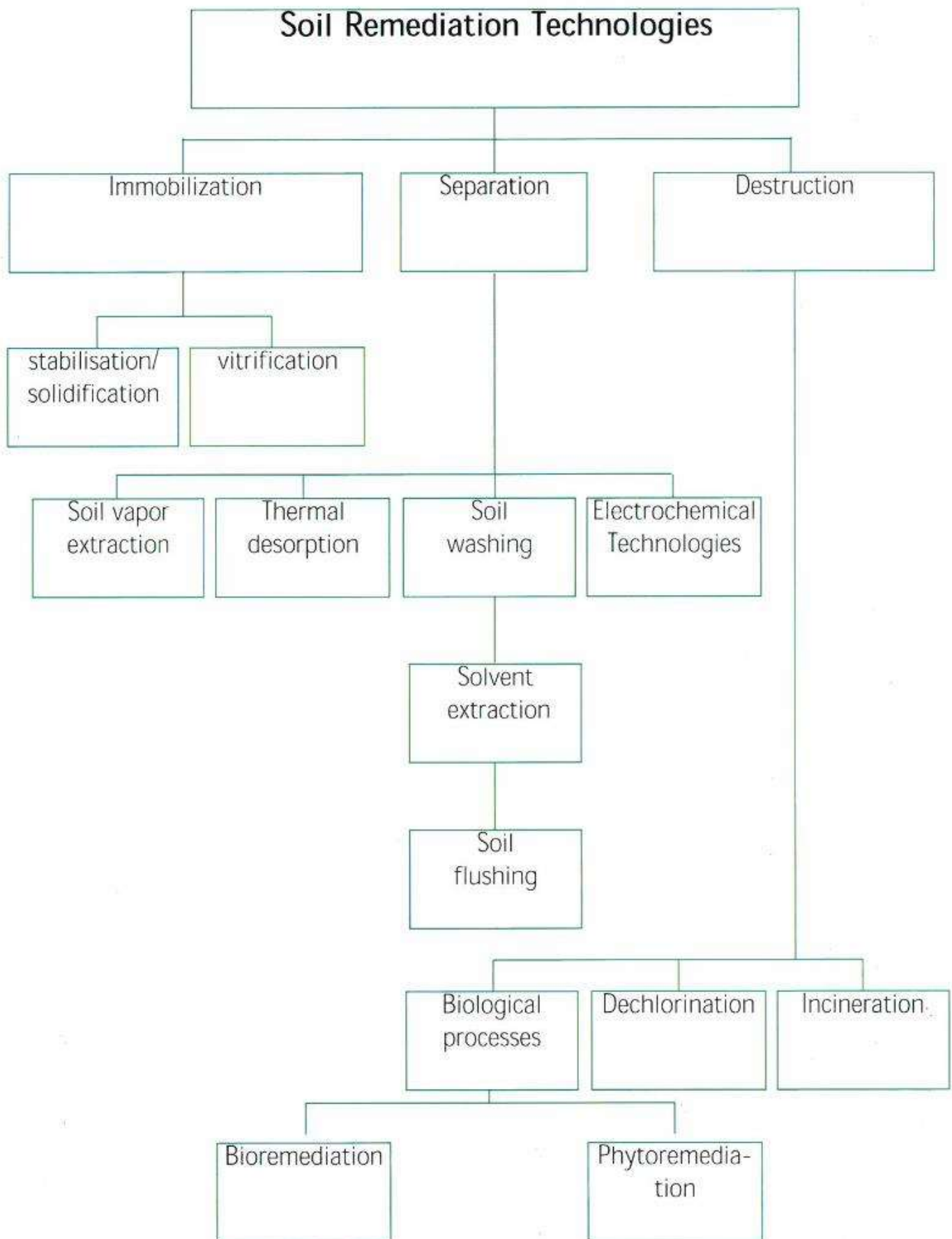


Fig. 11.8: Soil pollution remediation technologies

Solid and H



2- Dechlorination is the process used to dechlorinate chlorinated aromatic compounds by using chemical reagents. Soils and sludges containing PCBs, dioxins, and chlorobenzenes have been treated by dechlorination.

3- Incineration destroys organic materials by burning them to form carbon dioxide, water, sulfur dioxide, nitric oxide, ash, and/or products of incomplete combustion. Incineration does not destroy metals. They concentrate in the ash or enter the off-gas treatment system. Because of the relatively high cost to handle and heat soil to incineration temperatures, other technologies, such as SVE or thermal desorption, are sometimes used to separate organic contaminants from soil so that just the contaminants can be incinerated in vapor or liquid phase. However, concentrated liquid or solid hazardous wastes are often incinerated directly.

11.7 Economic effects of soil pollution

Soil pollution has been linked to soil infertility and reduced crop yield. It affects the plants' absorption of nutrients from the soil. This leads to economic loss especially if the country depends on agriculture. Surface and ground water can also be polluted as a result of soil pollution. This leads to further expenses for water treatment. Some plants absorb certain toxic substances and accumulate them in the food chain causing many diseases to human beings and consequently medical expenses. A related economic effect is the high cost of soil remediation techniques.

Review Questions

- 1- List some water pollutants.
- 2- Identify sources of water pollution.
- 3- Describe water treatment and sewage treatment methods.
- 4- Explain how calcium, organic substances, and inorganic substances can be removed from water.
- 5- Explain how water can be disinfected.
- 6- List some pollutants of soil.
- 7- Identify sources of soil pollution.
- 8- Describe methods of treatment of polluted soil: immobilization, separation, and destruction of pollutants.
- 9- Describe the economic effects of soil pollution.

Research Activities

- 1- Organize a field-trip to a water treatment plant and write a report about the visit.
- 2- Write a research report on different types of water filters

Document 11

Phytoremediation

The term phytoremediation refers to processes which use plants to treat polluted soil by concentrating or degrading contaminants. Remediation occurs in the rhizosphere, or soil that surrounds and is influenced by the roots of plants. Although plant-based treatment of soils has been proposed since the mid-1970s, phytoremediation is still a developing technology.

Phytoremediation has been used to treat many types of organic and inorganic contamination, primarily in laboratory or field trials: chlorinated solvents, nitrotoluene, PAHs (polynuclear aromatic hydrocarbons), chlorinated and organophosphate pesticides, excess nutrients such as nitrate, ammonium, and phosphate, and metals such as Cd, Cr, Co, Pb, Ni, and Zn. Effective treatment of lead requires that the soil be amended with chelating agents to make lead more bioavailable.

One of the mechanisms by which plants treat soil and shallow ground water is phytoextraction.

In phytoextraction, plants remove contaminants from soil and concentrate them in the plant tissues. Ideally, the plants used for phytoextraction can accumulate and tolerate high concentrations of contaminants and grow rapidly. Certain plants, which typically grow in areas with naturally high levels of metals in soil, have unusual ability to accumulate metals. These plants are harvested and landfilled, or incinerated to further concentrate the metals in the ash. A third option in the future, depending on economics may be to treat the metals as an ore and recover the metals content. Phytoextraction has also been used to remove certain organic contaminants, such as chlorinated solvents, from soil. Plants can metabolize or store an organic contaminant within plant tissue. The roots of plants can also release enzymes into the soil, which degrade certain contaminants.

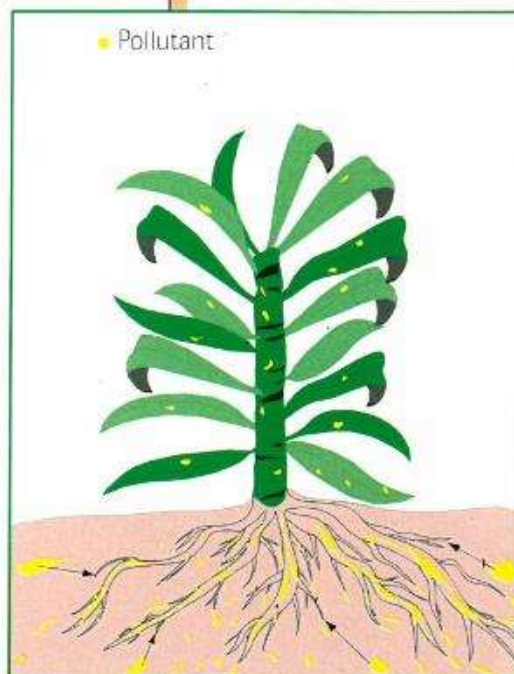
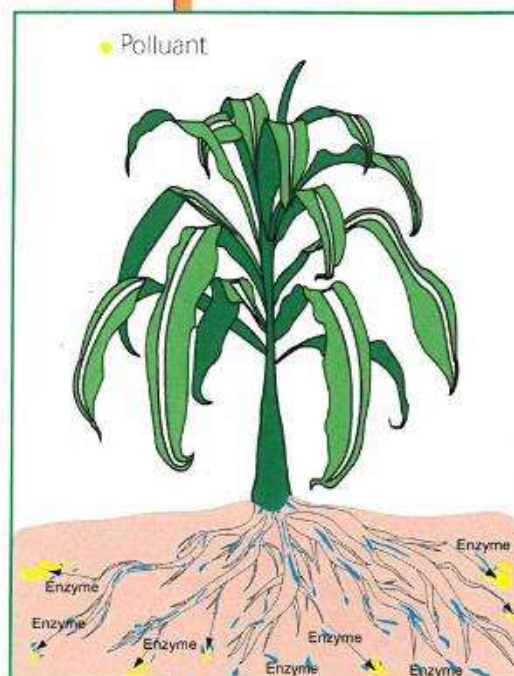


Fig 11.10: Phytoremediation technology uses plants to treat polluted soil

2- Thermal desorption refers to the process of heating soil or sludge to volatilize contaminants and remove them from the soil. The off-gases are then collected and treated. Depending on the operating temperature of the unit, thermal desorption can remove volatile organic compounds such as benzene, petroleum hydrocarbons, certain pesticides, PCBs (polychlorinated biphenyls), and dioxin. It has also been used to remove mercury from soil.

3- Soil washing, solvent extraction, and soil flushing separate organic and inorganic contaminants from soil into an extracting fluid or groundwater. The generated liquid stream then requires further treatment to remove or destroy the contaminants. These three treatment technologies are related but each works differently.

-Soil washing processes use aqueous solutions to remove contaminants from excavated soil. Soil washing reduces the volume of contaminated material: the fine particles (to which contaminants are adsorbed) are washed out of the soil and disposed as sludge.

-Solvent extraction processes use organic solvents to dissolve contaminants and remove them from soil.

-Soil flushing is an in situ process. Aqueous solutions are leached through soil to flush contaminants out of the soil and into the groundwater. A soil flushing system must include extraction and treatment or in situ treatment of the contaminated groundwater.

4- Electrochemical technologies are based on applying a direct current between electrodes placed in soil, ionic contaminants will migrate toward the electrodes based on their charge. Electrochemical technologies have been used to treat inorganic soil contaminants such as lead, arsenic, nickel, mercury, copper, zinc, chromium, uranium, thorium, and radium. Electrochemical processes have also been successfully tested on organic soil contaminants such as benzene, toluene, xylene, and phenol. Related processes have also been used to treat extracted groundwater.

C- Destruction technologies:

Technologies such as enhanced biodegradation, dechlorination, and incineration can destroy organic contaminants and certain inorganic contaminants. As elements, metals can not be destroyed.

1- Biological processes are of two types: bioremediation and phytoremediation.

-Bioremediation refers to a group of treatment methods designed to enhance the natural microorganism degradation of organic contaminants.

-Phytoremediation refers to processes which treat soil using plants to concentrate or degrade contaminants. It has been used to treat many types of organic and inorganic contamination.

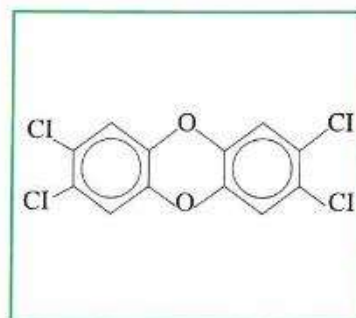


Fig. 11.9: Dioxin is a class of toxic compounds. The most popular, known simply as "dioxin" is tetra-chlorodibenzo-p-dioxin.

Destruction technologies can be used to destroy pollutants while in the soil or after they are separated.

Hazardous Wastes 12



Objectives

- List types and identify sources of solid and hazardous wastes.
- Describe methods of disposal and treatment of solid and hazardous wastes.
- Describe the role of chemistry in creating and reducing problems associated with solid waste.
- Describe health effects of hazardous waste.

12.1 Solid wastes and their sources

Solid wastes are the useless solid or semisolid materials that result from human activities. Solid waste has been increasing dramatically over the past several decades because of the population increase and the change in people's life-style. Lack of space for dumping solid waste has become a problem throughout the world.



Fig.12.1: High standards of living has dramatically increased the amount of solid waste.

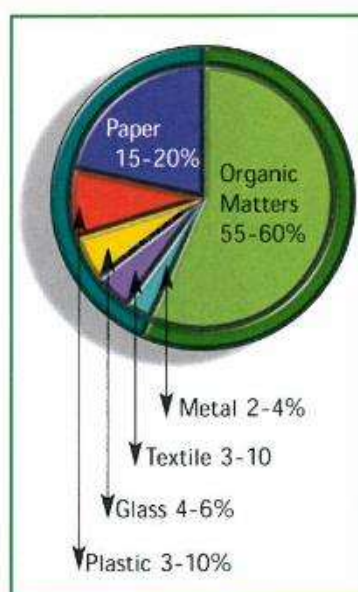


Fig. 12.2: Composition of domestic solid waste

Solid waste include things such as paper, metals, glass, plastics, textiles, yard waste (weeds, leaves, twigs,...), and food waste. Solid waste originates from different sources. They can be:

- Agricultural wastes: consist of crop residues and animal waste.
- Commercial wastes: typically consist of paper, metals, and rubber.
- Industrial wastes: may contain large quantities of residues of industries such as food processing, plastics, paints, textiles, and cement.
- Residential wastes: are wastes associated with the packaging, preparation, and consumption of food and drink, waste paper, and discarded domestic items, some of them are bulky items such as cardboard boxes and appliances.
- Other solid wastes, mostly found in Lebanon, are slaughterhouse wastes, construction wastes, and discarded tires and cars.



Fig. 12.3: Discarded cars constitute a major part of the solid wastes in Lebanon.

12.2 Methods of solid waste disposal

The most common method of solid waste disposal is simply to dump them outside the city or village limits. To minimize the volume of wastes, the dump is often burned or leveled off by a bulldozer. These dumps usually attract populations of rodents and other pests and emit foul odors. Dumping and burning wastes is no longer an acceptable practice from an environmental or health per-

spective. More convenient methods of disposing of solid wastes are sanitary landfills and incineration.

A- Landfills

A modern sanitary landfill is typically a clay-lined depression in which each day's wastes are covered with a layer of soil. New landfills have specific bottom layers to trap contaminated water leaking through the buried wastes. In addition, monitoring systems are necessary to detect methane gas production and ground water contamination. In some cases, methane produced by rotten garbage is collected and used to generate electricity. The water that leaches through the site must be collected and treated. As a result, new landfills are becoming increasingly more complex and expensive. Selection of new landfill sites is also becoming a problem, as it should be based on an understanding of groundwater geology, soil type, and sensitivity to local citizens' concerns.



Fig. 12.5: Example of a modern landfill

B- Incinerators

In incinerators municipal solid waste is fed into a furnace where it falls onto moving grates and is burned at temperatures up to 1300 °C. In addition to heat, the products of incineration include the normal products of combustion, carbon dioxide and water, as well as oxides of sulfur and nitrogen and other gaseous pollutants, ash, and unburned solid residue. Heat produced can be used to generate electricity.

Primary risks of incineration involve air pollution and production of toxic ash and solid residue. Emissions of fly ash and



Fig. 12.4: Borj Hammoud dump

Gases are generated in landfills through anaerobic decomposition of organic solid waste. If a significant amount of methane is present, it may be explosive. Proper venting eliminates this problem.

Incinerators drastically reduce the amount of solid waste up to 90% by volume and 75% by weight.

On an individual level, we can all attempt to reduce the amount of waste we generate. The result might be a significant reduction in municipal solid waste.

The practice of recycling solid waste is an ancient one. Metal implements were melted down and recast in prehistoric times.

other particles are often trapped by special filters. However, small amounts of pollutants are still released into the atmosphere. Small concentrations of heavy metals are also present both in the air emissions and residue. Many cities have difficulty in disposing of incinerator ash. The cost and siting of new incinerators are also major concerns facing many countries.

12.3 Solutions to solid waste problems

Four methods are used as possible solutions to solid waste problems. These are reduction, recycling, reuse, and composting.

A- Reduction

The most fundamental method to reduce waste is to prevent it from ever becoming waste in the first place. One way is to use less material when making a product. Another way is to convert from heavy packaging materials to lightweight ones. Companies can also reduce waste by making consumer products in concentrated form. These concentrated products can then be packaged in smaller containers.

B- Recycling

Solid wastes contain large quantities of recyclable materials such as paper, glass, plastic, and metals. Recycling have grown rapidly during the past several years, however a small but growing proportion of generated solid waste is recycled. In addition to pollution reduction, recycling will also conserves resources. One problem associated with recycling is a technical problem and is of particular concern when recycling plastics. There are many types of plastic polymers and technologies used to recycle them differ from one type to another. Industry is researching new technologies that will allow mixing of different plastics. Until such technology is developed, sorting different plastic products before recycling will be necessary.

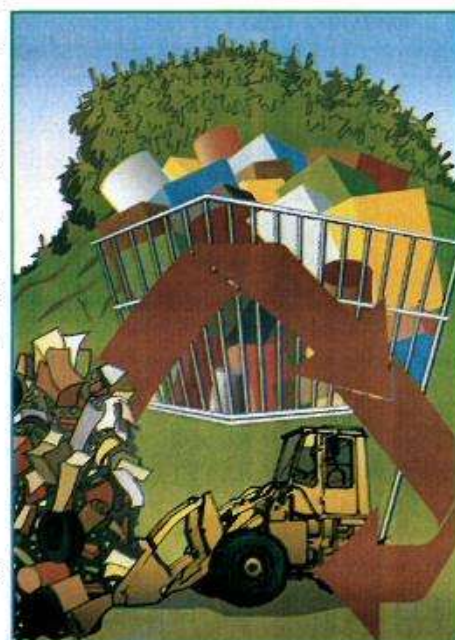


Fig. 12.6: Waste and recycling

C- Reuse

Some discarded materials are "resources out of place" that can be collected and reused for the same or for different purposes. By "reuse" is meant processing manufactured items (e.g. collecting, sterilizing, and refilling glass bottles) so that they can be used again. Crushed glass and shredded rubber can be used in road construction.

D- Composting

Composting is the process by which organic solid waste is degraded by aerobic microorganisms. It is made by building a long pile consisting of alternate layers of soil, manure, and vegetable materials such as weeds and tree leaves. The plant matter is then degraded biologically to humus, which can be used to enrich soil and improve its structure. This process may take up to six months.

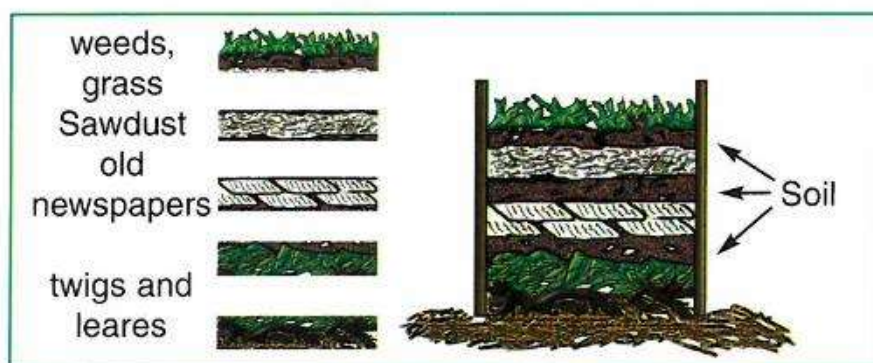


Fig. 12.7: Composting of organic wastes

12.4 Role of chemistry in creating and reducing the solid waste problem

The chemical industry is steadily developing. It has created many new and useful materials but, at the same time a number of problems have been associated with this industry. Examples are massive amounts of toxic waste and pollution of air and water. Many of these problems have proven hard to solve. The chemical industry has the responsibility to reduce and solve these problems. Chemists are working on making new products that are as free from hazard as possible. They are also working on devising new methods for treating wastes and for the recovery of some chemicals for reuse.

12.5 Hazardous wastes and their sources

Hazardous wastes are defined as wastes that pose a potential hazard to humans or other living organisms. They are considered hazardous because they may be:



Fig. 12.8: Chemists are working on making safer products.

A hazardous waste product is often an ordinary product which is present in the wrong place, at the wrong time, or mixed with the wrong substances.

Research Activities

- 1- Write a report on solutions to domestic waste.
- 2- Write a report on types of landfills and problems associated with each of them.
- 3- Write a report on using energy from incineration plants.
- 4- Write a case study on accidents involving hazardous wastes (Chernobyl, petroleum spills, chemical spills).

- discharge of hazardous liquids into sewers, rivers, and streams
- sealing wastes in drums and burying them in shallow ground.

Accidents are also a major source of contaminating the environment. Examples include accidents in nuclear plants (Chernobyl), accidents in chemical plants (Bhopal in India), and accidents of ships and trucks transporting crude or refined oil leading to dangerous oil spills.

12.6 Disposal and treatment of hazardous wastes

Most industrial hazardous wastes end up being disposed of on or in the land. Several technologies are under consideration as alternatives to land disposal. Any of these methods involve physical, chemical, or biological treatments.

A- Landfill:

has been the most common way of disposing of solid hazardous wastes and some liquids. The greatest environmental concern is the generation of leachate that causes groundwater contamination. Hazardous waste landfills constructed under current regulations provide control over such leachate.

B- Deep-well disposal of liquids

consists of their injection under pressure into underground geological formations and salt caverns. The disadvantage of using deep-well disposal is the potential contamination of groundwater.

C- Thermal treatment

involves heating waste in incinerations. Under controlled conditions, hazardous waste incinerators burn 99.99% of certain hazardous materials. The relatively high cost of incineration and the concerns for safety of surrounding areas in case of accidents have kept incinerations from becoming a major form of disposal.

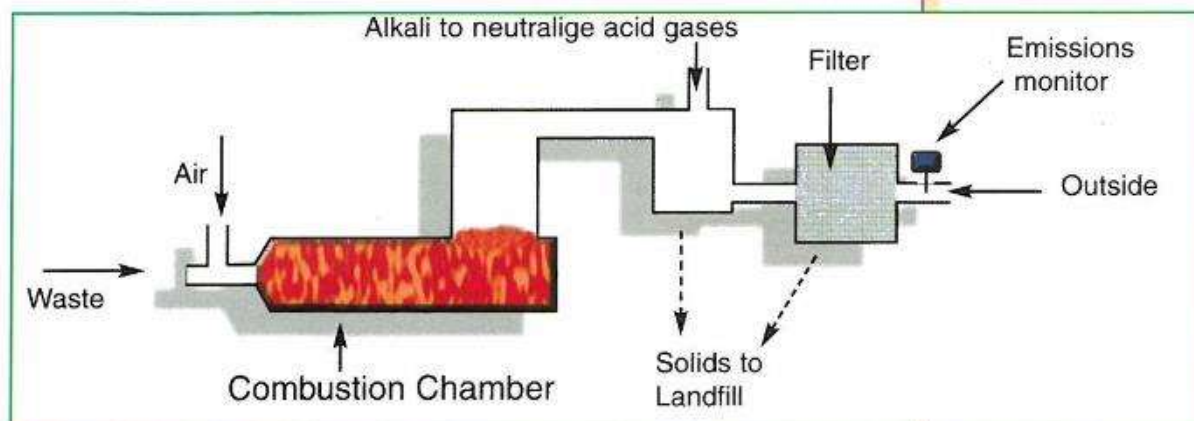


Fig. 12.10: A toxic waste incinerator

K- Waste reduction:

perhaps the wisest solution to the hazardous waste problem is cutting off the production of toxic wastes at the source (source reduction). While some industries have significantly reduced the amount of hazardous waste generated, source reduction is still largely untried.

12.7 Hazardous wastes and health

In recent years the health aspects of hazardous substances have received increased attention. Establishing the medical consequences of exposure to hazardous wastes is extremely complicated and linking a particular chemical to a specific injury or disease is also more complicated because of the lack of toxicity data on most hazardous wastes.

Although assessing environmental contamination from hazardous wastes and determining health effects is extremely difficult, what little is known is cause for concern. Most hazardous-waste dumps, for example, contain dangerous and toxic chemicals along with heavy metal residues. All of these substances can damage human health as a result of low-level, long-term exposure. Some of these substances can cause immediate poisoning (e.g. arsenic and mercury) and others (including radioactive wastes) are thought to cause cancer, birth defects, and genetic damage.

Review Questions

- 1- Give examples of solid wastes.
- 2- Identify major sources of solid waste.
- 3- Describe solid waste disposal methods.
- 4- Identify the advantages and disadvantages of solid waste disposal methods.
- 5- Describe some solutions to the solid waste problem.
- 6- Identify the role of chemistry in creating and reducing the problems resulting from solid waste.
- 7- Define hazardous waste.
- 8- Give examples of different types of hazardous wastes.
- 9- Identify sources of hazardous wastes.
- 10- Describe some methods for treatment and disposal of hazardous wastes.
- 11- Identify the health problems arising from hazardous wastes.

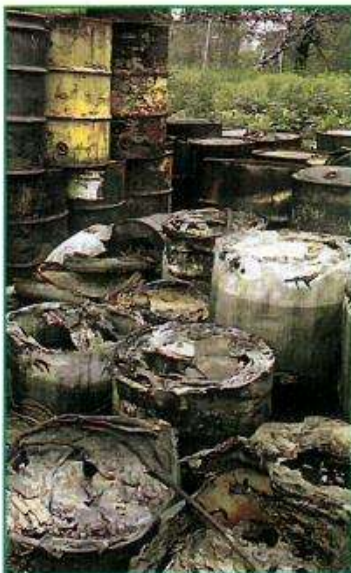


Fig. 12.9: Improper storage of toxic waste

The term asbestos refers to a family of six naturally occurring silicate minerals that are fibrous in character. The most commonly used form of asbestos, chrysotile, has the formula $\text{Mg}_3\text{Si}_2\text{O}_5(\text{OH})_4$. It is now recognized that asbestos is a human carcinogen. It causes mesothelioma, an incurable cancer of the lung, abdomen, and heart.

- Flammable
- Corrosive
- Explosive
- Toxic
- Radioactive

While the term "toxic" and "hazardous" are often used interchangeably to refer to wastes, yet there is a difference. Toxic commonly refers to narrow group of poisonous substances that cause death or serious injury to humans and animals by interfering with normal physiology. Hazardous, the broader term, refers to all dangerous wastes, including toxic ones, that present an immediate or long-term effect on human health or cause an environmental risk.

Many specific chemicals in widespread use are hazardous. Hazardous wastes usually consist of mixture of specific chemicals. They can be in the form of sludge, liquid, gas, and solid and they may include:

- Asbestos
- Flammable liquids, such as gasoline
- Explosives, such as dynamite and ammunition
- Compressed gases, such as sulfur dioxide and hydrogen
- Corrosive substances, such as sulfuric acid and caustic soda
- Poisonous substances, such as cyanides
- Heavy metals, such as lead and mercury
- Hospital wastes
- Radioactive substances, such as plutonium, cobalt-60, and uranium

Hazardous wastes contaminate the environment in two major ways, the way they are disposed of and from accidents.

Many ways of hazardous waste disposal can pollute both the soil and the groundwater as they leach down the ground. Examples are:

- ordinary household chemical wastes (e.g. small amounts of left-over paint, not quite empty pesticide spray cans, unknown contents of old bottles with no labels, and small amounts of leftover garden and pool chemicals) are usually disposed of either by pouring them down the drain and onto the ground or by dumping them in landfills not designed for hazardous wastes.
- used engine oil is particularly a problem when it is drained directly into the ground
- discharge or dumping of industrial hazardous wastes in improperly controlled landfills

Ions taken up by an ion exchange resin may be removed by treating the resin with concentrated solutions of acid, base, or salt (NaCl) to regenerate the resin.

Addition of powdered activated carbon to the waste treatment systems remove the non-biodegradable organic wastes.

D- Filtration:

It is a physical method used to remove particles from hazardous wastewater by coagulation and filtration.

E- Distillation:

This process can be used with hazardous wastes to separate liquids of different volatilities. Some specific wastes suited to treatment by distillation include aqueous phenolic wastes, waste oil, and organic solvents.

F- Fixation and solidification:

These are methods used to immobilize hazardous wastes by mixing them with materials such as cement. This either "fixes" the hazardous particles, in the sense of immobilizing them or making them chemically inert, or "solidifies" them into a solid mass. Solidified waste is sometimes made into solid blocks that can be stored more easily than a liquid.

G- Neutralization:

Acids and bases are frequently encountered in hazardous wastes. Adding an acid to a highly alkaline waste or a base to a highly acidic waste can neutralize such wastes.

H- Ion exchange:

It is a process by which anions or cations can be removed from a solution onto a solid resin. The greatest use of ion exchange in hazardous waste treatment is for the removal of low levels of heavy metal ions from wastewater.

I- Biological (aerobic and anaerobic) waste treatment:

Biodegradable organic wastes can be treated either by aerobic or anaerobic microorganisms. Aerobic treatment utilizes aerobic bacteria and fungi that require oxygen. It can be applied to hazardous wastes such as landfill leachates. Anaerobic waste treatment in which microorganisms degrade wastes in the absence of oxygen can be practiced on a variety of hazardous wastes. Most of the wastes for which anaerobic digestion is suitable consist of oxygenated compounds.

J- Recycling:

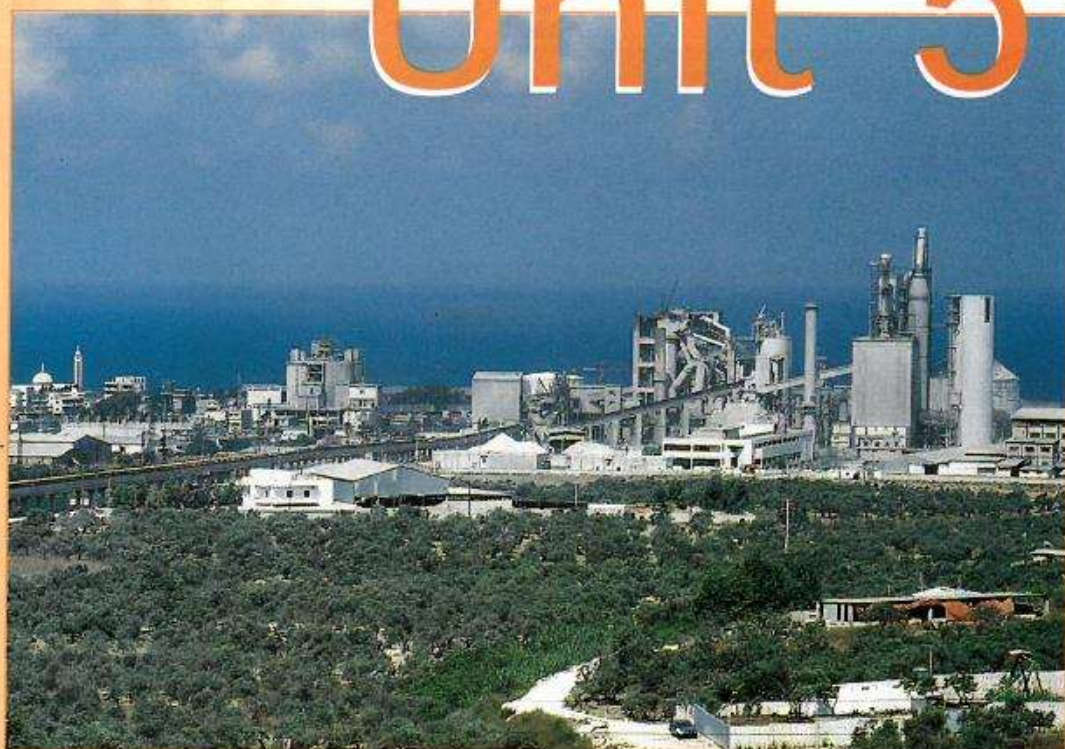
Used engine oils, organic solvents and some industrial wastes can be recycled using methods such as distillation, solvent extraction, and vacuum distillation. For example waste oil can be recycled to produce lubricating oils and other by-products.

8. Do you pollute the environment? If so, explain and give examples. What can you do as an individual to decrease pollution.

V - Refer to document 11 "phytoremediation" to answer the following questions:

- a-Define the term "phytoremediation".
- b-Give examples of pollutants that can be removed by using this technology.
- c-Explain how special plants remove contaminants from soil.
- d-Explain how to dispose of contaminated plants.

Unit 5



Chemistry and Economy

unit 4 Questions

I - Label each of the following statements as true or false, and correct the false ones.

1. Aerobic and anaerobic microorganisms can degrade non-biodegradable organic wastes.
2. Water pollution is the degradation of the quality of water.
3. Acid rain is a source of pollution for water and soil.
4. The two major sources of domestic water pollution are soaps and detergents.
5. Heavy metals can be degraded by incinerations.
6. Solid wastes include paper, textiles, glass, plastics, and pesticides.
7. Solid waste has increased in the past several years because of the change in people's life-style.
8. Chlorination is used to remove organic compounds.
9. Aeration of water is used to remove microorganisms.
10. Ion-exchange resin is used to remove Ca^{2+} and Mg^{2+} and heavy metals.
11. Flammable, corrosive, explosive, toxic, and radioactive substances are hazardous.
12. Soil pollutants are substances that accumulate in the soil medium and change the natural soil equilibrium.

II- Choose the correct statement.

1. In the primary treatment of sewage water:
a-large objects and suspended materials are removed.
b-organic matter is degraded by microorganisms.
c-excessive hardness is removed.
d-phosphates and nitrates are removed.
2. Separation of soil contaminants can be done by:
a-soil washing, soil flushing, and dechlorination.
b-solvent extraction, electrochemical technology, and incineration.
c-soil vapor extraction, thermal desorption, and electrochemical technology.
d-soil vapor extraction, solvent extraction, and biodegradation.

3. Which of the following is **not** usually considered a method of solid waste reduction:
 - a-using less material when making a product.
 - b-reusing discarded materials.
 - c-using light weight packaging.
 - d-making products in concentrated form.
4. Why is it desirable to solidify and fix hazardous wastes?
 - a-they will not explode.
 - b-they will degrade by themselves.
 - c-they become chemically inert.
 - d-they can be processed into a source of energy.

III- Relate pollutants of water in column A to their method of removal in column B.

Column A

- 1- odorous substances
- 2- microorganisms
- 3- calcium ions
- 4- suspended particles
- 5- chlorinated pesticides
- 6- biodegradable substances
- 7- alkaline substances

Column B

- a- adsorption on activated charcoal
- b- addition of sulfide ions
- c- coagulation by sodium aluminate
- d- aeration
- e- filtration
- f- chlorination
- g- screening
- h- ion-exchange
- i- biological treatment

IV- Thinking problems

1. Chlorine and ozone can be used for municipal water disinfection. Which do you think is better to be used? Why?
2. Nitrogen compounds in drinking water, such as nitrates, are dangerous to health especially to newborn babies. Explain how they can be removed from water.
3. Explain how adding $\text{Ca}(\text{OH})_2$ to hard water will cause calcium ions to precipitate.
4. Thermal desorption and incineration are two technologies used to treat polluted soil. Compare the two technologies.
5. What are some problems facing the installation of new hazardous waste incinerators?
6. Seawater is too salty to drink. Is seawater polluted by salt?
7. It is estimated that about 2.5 billion non-rechargeable batteries are thrown each year. Does this have any possible environmental effect? Explain.

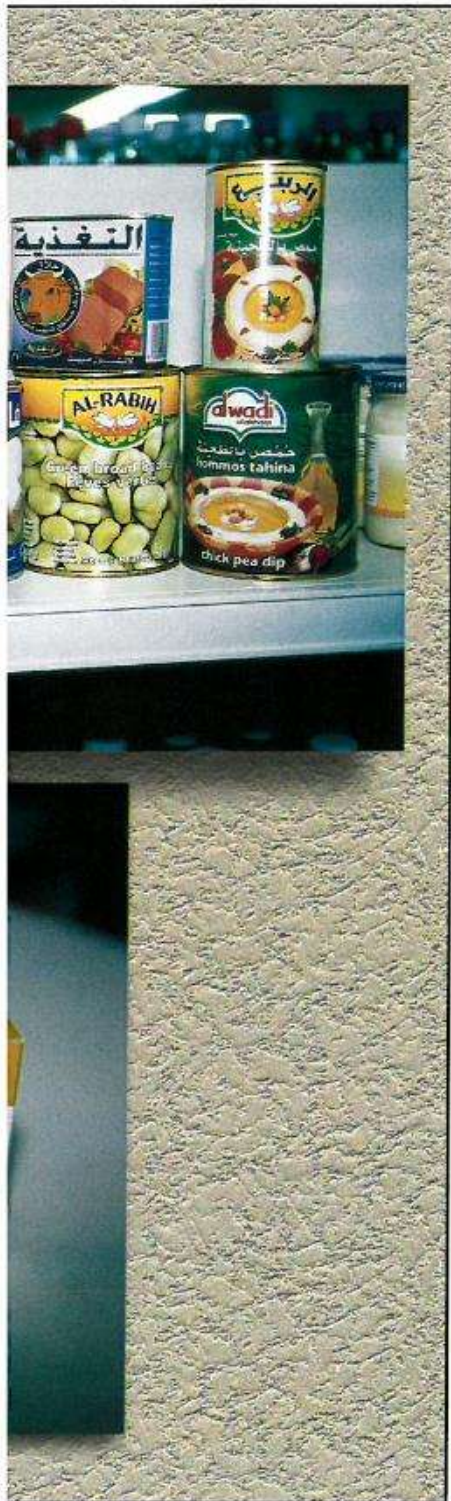
Document 12

Summary of laws relating to solid waste management in Lebanon

Summary	Year	Law/Decree No
Waste should not be dumped around houses, but be buried or removed by the municipality.	1931	7975
Refuse should be stored in plastic bags.	1971	425/1
Municipalities are responsible for the collection and the disposal of domestic wastes.	1974	8735
Household and Construction wastes (rubble) may not be dumped in public places or private land adjacent to roads and residential districts.		
It is an offence to drop litter in streets, government buildings and public areas.		
Only tightly closed containers should be used for the storage of refuse. Municipalities may not pile wastes on the roadside before it is collected.		
Wastes should not be transported in open vehicles, but in vehicles that are "tightly covered".		
Disposal sites must have the approval of the Health Council of the Mohafaza.		
No insecticides may be imported into Lebanon if it is banned for use in the country of origin.	1978	11/78
It was made the duty of every person to preserve the safety of the environment from pollution. A list of hazardous materials was published.	1988	64/88
Importation or possession of radioactive or poisonous wastes is prohibited. In extreme cases the death penalty could be applied.		
This law formally created the Ministry of the Environment	1993	216

Adapted from the final report for the Ministry of the Environment, "Lebanon: Assessment of the state of the Environment". November, 1995. p.210.

nd Economy 13



Objectives

- Name some chemical industries in Lebanon and describe their contribution to the economy.
- Describe the impact of the chemical industry in Lebanon on the environment.
- Identify the social, political, and economic effects of concentrating industries around cities.
- Identify the necessary measures for encouraging the industrial sector in Lebanon.
- Identify considerations that should be taken for investing in the chemical industry.
- Describe the effects of petroleum production in some Arab countries.

In Lebanon, there are two Portland cement plants (located in Chekka 15 km south of Tripoli) and one white cement plant (located in Sibline 10 Km north of Saida). The Sibline plant is the most modern plant. It was given, in 1994, authorization to increase its production capacity to 1.1 Mt/year (million tons per year). Nearly all the cement production is consumed internally due to the construction projects in Lebanon,

White cement has the same composition as Portland cement with a low percentage of iron oxide (Fe_2O_3).

Plant	1992 production	1993 production
Societe de Ciments Libanais, SAL (Chekka)(Portland Cement)	1,318	1,463
Cimenterie Nationale, SAL (Chekka)(Portland Cement)	514	909
Ciments de Sibline, SAL (White Cement)	295	341
Boury Cement (off-shore) (packaging of imported cement)		332
Total	2,127	3,045

Table 13.1: Cement production in Lebanon ('000t/year).

2-Ceramics industry

Ceramics are products that have varied properties and applications. There are two kinds of ceramics, traditional and new. Traditional ceramics are those used in construction (ceramic tiles, sanitary ware, fire bricks), in homes (cookery, fire china, porcelain vases), or in traditional industries (steel and iron, glass). New ceramics are used in electronics.

Ceramics industry in Lebanon produces only traditional ceramics. The raw materials are, in general, of natural origin (extracted from soil) and of low price. They are either clay that changes, upon heating to silicates, or combination of oxides.

In Lebanon, there are two major ceramics plants. They produce 2.2 Mm²/year of ceramic tiles, 8500 t/year of sanitary ware and 300.000 fire bricks. Most of these products are for export. Their value reached 13.3 million dollars in 1992.



Fig. 13.2: In Lebanon ceramics industry produces sanitary ware and tiles.

$$\text{Mm}^2 = 10^6 \text{ m}^2.$$

Chemistry



13-1 Chemical industry in Lebanon

Chemical industry in Lebanon is relatively new but it is diverse and dynamic. It occupies an important place in the Lebanese economy. Chemical industry is divided into organic and inorganic sectors.

The inorganic chemical industry includes cement, glass, ceramics, phosphate fertilizers, sulfuric acid, industrial gases (CO_2 , O_2 , N_2 , argon, and helium), and steel industries. The organic chemical industry includes food processing, plastics, detergent, disinfectant, cosmetics, leather tanning, paper, and pharmaceutical industries.

Lebanon exports an important part of its chemical products to the Arab and African countries. Some products are also exported to European countries.

A- Inorganic industry in Lebanon.

This sector is important because it is partly related to Lebanon's reconstruction program. It mainly produces building materials, such as ceramic tiles, glass, cement, and marble. Other products are phosphate fertilizers, sulfuric acid, and carbon dioxide, oxygen, nitrogen, argon, and helium gases.

These industries are geographically dispersed and are based on local raw materials. Most of the raw materials are provided locally such as limestone (CaCO_3), sand, and clay, which are used in the manufacture of cement, glass, and ceramics. Others are imported such as phosphate rocks and sulfur needed for the manufacture of fertilizers and sulfuric acid.

1-Cement industry.

Cement is a complex mixture of calcium and aluminum silicates obtained by heating the raw materials, limestone and clay, to very high temperatures. The cement products that are widely in use are Portland cement and its derivatives.



Fig. 13.1: Cement plants in Chekka produce the majority of the cement in Lebanon.

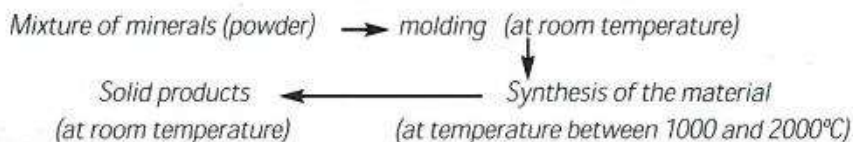
Portland cement is produced when a mixture of 80% limestone and 20% clay is heated to temperatures between 1000 and 1450°C. The products of the thermal decomposition of the raw materials then react to produce cement.

The approximate composition of Portland cement is:

50% C_3S , 20% C_2S , 12% C_4AF , 8% C_3A , and some impurities such as MgO , Al_2O_3 , Fe_2O_3 , Cr_2O_3 , TiO_2 , and CaO .

Where $\text{C}=\text{CaO}$; $\text{S}=\text{SiO}_2$; $\text{A}=\text{Al}_2\text{O}_3$, and $\text{F}=\text{Fe}_2\text{O}_3$.

Manufacture of traditional ceramics



Ceramics are hard, chemically inert, and good electric and thermal insulators. They also have high melting points, low densities, and low coefficients of expansion.

3-Glass industry

In Lebanon, glass industry produces flat glass and glassware (dishes, bottles, glasses, vases, ...). The flat glass industry has a capacity of producing 1.86 Mm²/year. The composition of classical glass is: 65-74% SiO₂, 13-15% Na₂O, and 8-9% CaO.

Glass is a solid material formed from the following raw materials:

-Silica (small granules of very pure sand) mixed with melting and stabilizing agents such as : Na₂CO₃, Na₂SO₄, CaCO₃, feldspar (sodium, potassium, or calcium aluminum silicate), and dolomite (calcium and magnesium carbonate)

-other ingredients used in the manufacture of glass are:

Colorants: transition elements cations (Cr³⁺, Co³⁺, Ni²⁺, Nd³⁺, Pr³⁺...)

Decolorants

Opalescents.

Refining agents to eliminate bubbled gases (As₂O₃, Sb₂O₃, ...).

4-Fertilizer industry

Fertilizers provide plants with the necessary nutrient elements (N,P,K) in the form of ions. They are classified according to their physical properties (solids or liquids) or chemical nature (simple or complex compounds). Simple fertilizers contain one major nutrient element. They are named nitrogen, phosphate, or potassium fertilizers. An example of a nitrogen fertilizer is ammonium nitrate (NH₄NO₃). Complex fertilizers may contain two or more nutrient elements. An example of a binary fertilizer is KNO₃ (NK), and an example of a ternary fertilizer is KCl-NH₄NO₃-Ca(H₂PO₄)₂ (NPK).

In Lebanon, there is one phosphate fertilizer plant in Selaata in the north. The raw materials (phosphate rocks) are imported from Jordan and Morocco. They are treated with sulfuric acid to give superphosphate (simple phosphate fertilizer). This plant produces 250,000 tons/year of which only about 7000 tons are consumed locally and the rest are exported.



Fig. 13.3: The fertilizer plant in Selaata produces 250,000 tons/year of superphosphate.

- Nahr Ibrahim, where a paper plant is discharging its wastes into the river.
- The coastal zone around Selaata, where the fertilizer plant discharges phosphates and sulfates to the sea.
- The coastal zone around Batroun and Chekka, where the cement plants are discharging cooling water to the sea (thermal pollution).
- The river in the area of Wadi Ghadir is polluted from soap and detergent factories and olive presses located in Choueifat, Ain Anoub, and Bchamoun.



Fig. 13.6: Ghadir river is highly polluted from the discharges of domestic and industrial wastes.

4-Soaps, detergents, and perfumes industries

The raw materials for these industries are imported except for olive oil, which is used for the preparation of baladi soap. Most of these industries are restricted to formulation and packaging. They represented 18% of the total sales for the industrial sector in 1985.

5-Pharmaceutical industry

The raw materials for this industry are imported. The industry is restricted to formulation and packaging of generics such as tablets, ointments, and injectable fluids. Only part of the production is consumed locally and the rest is exported. Pharmaceuticals represented 10% of the industrial sector sales in 1985.

13.2 Impact of the chemical industry on the environment

The chemical industry in Lebanon is located along the coast from Tripoli to Saida and on riversides (figure 13.7). The main sources of pollution are gas and liquid effluents.

A- Gas effluents

The main air pollutants from the chemical industries are:

- SO₂ and CO₂ from the combustion of fuels.
- Fine particles from cement plants.
- Fluorine from fertilizer industry.

B- Liquid effluents

In general, liquid effluents of the chemical industry are discharged in the sea and rivers. Figure 13.7 and Table 13.2 indicate that the chemical industry is imposing serious problems in a number of places:

- In the Berdouni river, and hence into the Litani river, effluents from food processing and paper industries, wineries, and tanneries are discharged from Zahle and Chitoura.
- Discharges from the sugar beet plant at Anjar add to the pollution in the Litani river.
- Effluents from tanneries and glue factories, at Machghara, are also discharged in the Litani river.
- The coastal zone, from the port (North Beirut) to Jounieh, is polluted by wastes discharged from tanneries, and beer, beverages, and fats factories.

There are no data on the quality of water in the Litani river, nor in the Karaoun lake.

Location	Type of Industry	m ³ /day	BOD* Kg/day	Other pollutants	Disposal route
Mount Lebanon					
Ain Aanoub	1 Olive press	15	1875	coloration	Stream
Baouchriya	Beer	400	49880	Caustic	Sewer
Bchamoun	3 Soap factories	3	22	NaOH	Stream
	2 Olive presses	20	2500	Coloration	Stream
Borj Hammoud	2 Tanneries	21	140	Cr ⁺ , As, sodium sulphide	Sea
Choueifat	Detergents	30	17	H ₂ SO ₄ , NaOH, phosphates	Septic tank, stream
Dbaiye	Soft drinks	300	105	Cl ₂ , Na ₂ CO ₃ , MgCO ₃	Sewer
Nahr Ibrahim	Paper rolls	100	15		River
Zouk Mkhayel	Cooking fats	700	413		Sea
	Pharmacological				
	Textile bleaching, dyeing	500	1930	Bleaches, dyes	settlement Sewer
Zouk Mosbeh	Textile, bleaching, dyeing	240	288	Bleaches, dyes	Streams
NORTH LEBANON					
Selaata	Fertilizer			Phosphates and sulphates	Sea
Chekka	Cement			Hot water	Sea
SOUTH LEBANON					
Ghazieh	-Tannery	150	1005	Cr ⁺ , As, As, sodium sulphide	sea
	-Soap			NaOH, NaCl	River
Saida	4 Tanneries	300	2010-	Cr ⁺ , As, sodium sulphide	Sea
BEKAA					
Anjar	Sugar	2,880	2466	Solides, Na ₂ CO ₃	River
Chtoura	Canned fruits	250	276	Organics, solids NaCl Na ₂ CO ₃ salts	Own sewer, primary treatment
Machghara	8 Tanneries	400	2680	Cr ⁺ , As, sodium sulphide	Channel
	5 Glue factories	10	42	Lime	Stream

Table 13.2: Estimates of effluents discharged by selected industries (1991)

*BOD refers to "Biochemical Oxygen Demand". It is the quantity of oxygen necessary for the biodegradation of the organic materials in the effluent. High values of BOD indicate high quantities of pollutants.

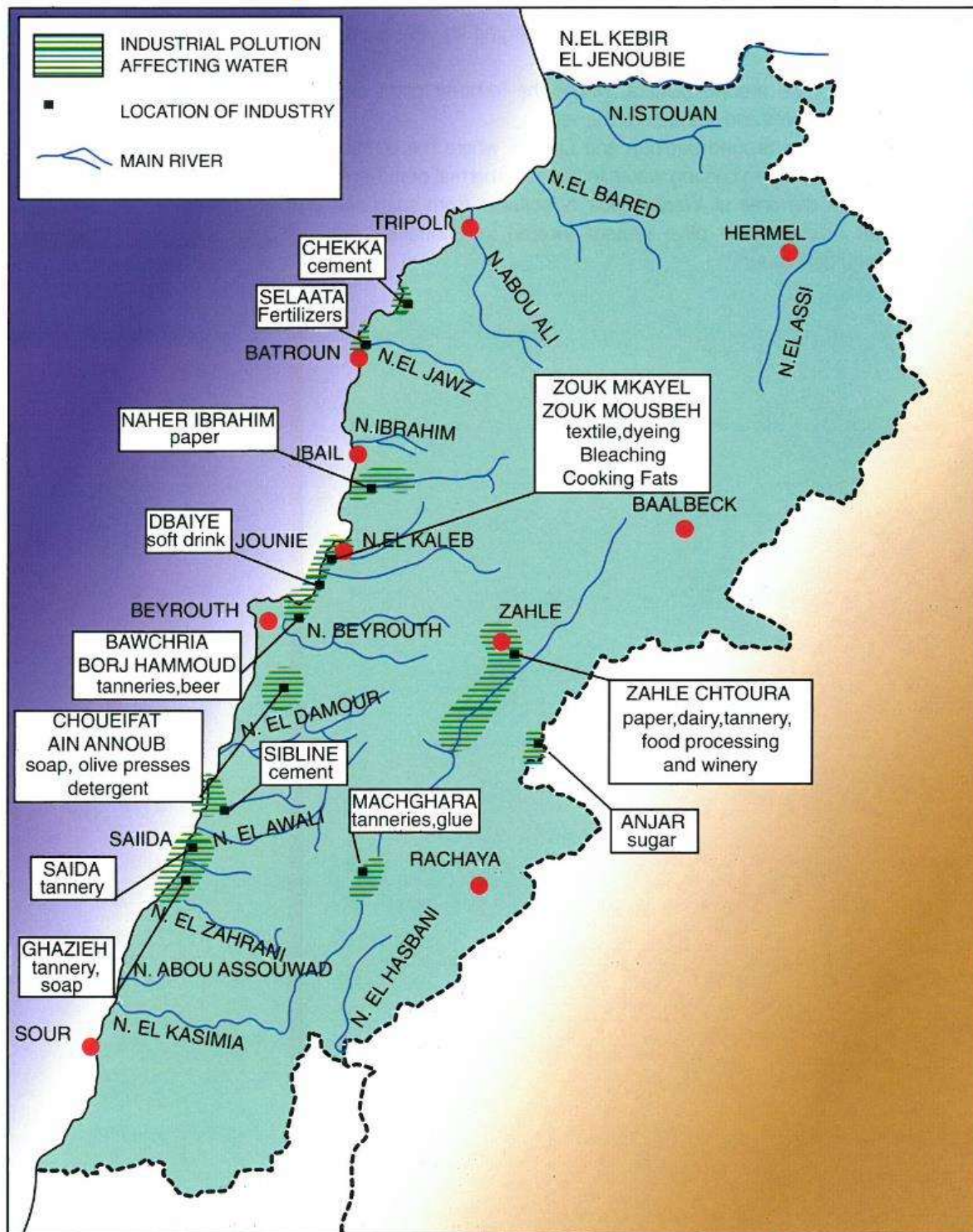


fig. 13.7: Surface water resources affected by industrial activity.

5-Gas industry

In Lebanon, there are many plants that produce gases, such as oxygen, nitrogen, argon, and helium, by the fractional distillation of liquefied air.

B- Organic industry in lebanon

Raw materials for most organic industries, such as plastics and pharmaceutical products are imported.

1-Food processing industry

In general, food-processing industry occupies an important place among the Lebanese industries. Together with textiles and garments industry and the manufacture of metal products and machinery (including construction materials) account for about 60% of all the industrial enterprises in the country.

Food-processing industry include milk and dairy products industry, industry using agricultural products as raw materials such as sugar, oil, juice, and caned and dried food industries, and the refining and hydrogenation of oil and other food products.

Food-processing industry in Lebanon is distributed in the different parts of the country. Most of the raw materials are produced locally and only a few are imported. Large quantities of products are exported to the Arab countries and to some European countries.

2-Paper industry

Paper industry in Lebanon is limited to the recycling of paper. A related industry, which is a highly developed one, is the printing industry. In 1992, paper and printing exports reached 13.5 million dollars.

3-Plastic industry

The polymers used as raw materials for the plastic industry are not provided locally. They are imported in the form of resins. The most imported polymers are polyethylene and polypropylene. This industry is restricted to melting and molding the thermoplastic polymers. Some plastic factories are using recycled plastics as the raw materials.



Fig. 13.4: Gas producing plant in Lebanon.



Fig. 13.5: Variety of foods produced in Lebanon.

Paper is made of wood. The main components of wood are polymers of high molecular masses such as:

- Carbohydrates: 60 to 80% (cellulose and hemicellulose).
- Phenol substances: 20 to 30% (lignine).
- Tannin and colorant substances.
- Resins
- Mineral constituents.
- Other minor constituents.

are exported to other Arab countries.

-Evolution in the infrastructure and in the education and health sectors.

-Development of agriculture and consequently the food processing industry.

-Investment of the surplus from petrol production in important projects in the entire world and especially in other Arab countries reflects positively on these countries economy.

Review Questions

1. Name some organic and inorganic industries in Lebanon.
2. Describe the contribution of some industries on the economic development in Lebanon.
3. Describe the impact of chemical industry on the environment.
4. Identify the social, political, and economic effects of the concentration of industries around the cities.
5. List the necessary measures that should be taken to encourage the industry in Lebanon.
6. Identify elements of the feasibility study of an industrial project.
7. Identify the social, political, and economic effects associated with the production of petrol in some Arab countries.

Research Activities

1. Write research reports on the economic importance of a number of chemical industries.
2. Write a report on manpower in the chemical industry compared to other industries.
3. Write a report comparing the role of the chemical industry in Lebanon, Arab countries, and industrialized nations.

unit 5 Questions

I- Label each of the following statements as true or false, and correct the false ones.

- 1- Prior to 1975, the rate of the industrial development in Lebanon was high (10% per year).
- 2- The majority of the industry in Lebanon belong to the private sector.
- 3- Raw materials for the ceramics industry are of natural origin and of low cost.
- 4- Most of the raw materials of the inorganic industry in Lebanon are provided locally.
- 5- Nitrogen, phosphate, and potassium fertilizers are produced in Lebanon.
- 6- The gas industry in Lebanon produces oxygen, nitrogen, argon, and helium from air.
- 7- Lebanon exports most of its cement production.
- 8- Lebanon exports most of its ceramics production.
- 9- The pharmaceutical industry in Lebanon is limited to formulation and packaging.
- 10- Lebanon produces paper by recycling used paper.

II- Choose the correct answer

- 1- Percentage of the number of the industrial plants in the Greater Beirut Area with respect to the total industries in Lebanon is:
 - a-17 %
 - b-37 %
 - c-57 %
 - d-87 %
- 2- Cement is formed essentially from:
 - a-calcium and aluminum silicates.
 - b-mixture of oxides.
 - c-clay and oxides.
 - d-silica.
- 3- Food processing industry is important because:
 - a-it is concentrated on the coastal zone.
 - b-most of the raw materials are imported.

Sugar industry

Table sugar or sucrose is extracted from beets or sugar cane. In Lebanon, sugar is obtained only from beets. The procedure for the production of sugar include the following steps:

- Beets are washed and cut to small pieces.
- Diffusion: is the operation by which sugar is extracted by the action of a hot water current (70 to 80°C). This results in the formation of crude juice that contains 12 to 15 % sugar. The residue, called pulp, is used for feeding animals.
- Purification: the crude juice contains 1 to 3 % impurities. They can be separated from the sugar by carbonation process. Purified clear juice is obtained.
- Evaporation: the purified juice is concentrated by evaporation to obtain a syrup.
- Crystallization: syrup is supersaturated by boiling under vacuum. This causes the appearance of white sugar crystals. These crystals are then separated from the syrup, washed by a jet of vapor, dried by hot air, cooled, and stored.



Fig. 13.10: The beet sugar plant in Bekaa.

13.3 Consequences of concentration of industries around cities

Concentration of industries around large cities poses an actual problem in several countries because of the social, political, and economic consequences. This problem results from the migration of the workforce, used to work in the agricultural sector in the villages, to the large cities where the industries are concentrated. Two direct consequences of this migration are:

1- Decrease in agricultural cultivated areas and consequently of agricultural production. This might affect seriously the future of this sector.

2- Concentration of people in zones, around great cities, lacking convenient infrastructure (such as electricity, water, and sanitary installations) creates, with time, social problems and consequently political ones.



Fig. 13.8: Residential areas around Beirut lack the necessary infrastructure.

It has been reported in 1994 that strong concentration of industries is in the Greater Beirut Area (57% of industrial units and 70% of the industrial workforce). Among twelve Cazas in Lebanon that account for 86% of total industrial employment, the two Cazas Baabda and Metn account for 42% of the total industrial employment.

The same report also notes that many smaller industrial establishments have been set up in residential areas in Greater Beirut Area which had never been intended for industrial activities.

13.4 Economic data and future plans

Economic data in Lebanon is available for the whole industrial sector and not for just the chemical industry.

Prior to 1975, industry in Lebanon had been growing strongly at the rate of 10% per year. The Ministry of Industry and Petrol estimated that in the last 6 years of the war (1985-1991) around 1500 enterprises stopped production and direct and indirect losses summed to 750 million dollars. Between 1990 and 1994, 30% of industrial units were established and this gave hope for the future of the Lebanese industry. The industrial sector in Lebanon is mainly private and the ministry of industry is aware of this fact. The ministry also recognizes the need for a coherent policy for the development of this sector.

The future of the chemical industry in Lebanon can not be separated from the future of other industries. In general, several measures should be considered to encourage investing in the industrial sector. These include:

- Tax exemptions on new investments.
- Customs protection for locally manufactured products and customs exemptions on raw materials and equipment imported to be used in the Lebanese industry.
- Credit facilities for investing in new industries.
- Subsidies for energy cost.
- Professional training and encouraging research.
- Finding markets for exporting goods.

Recently the state has started a plan to encourage food processing industries, mainly in Bekka and Akkar.

13.5 Investment in the chemical industry

Feasibility studies should be done prior to the installation of any industrial firm. Such studies include the following:

- Installation and supplying the basic equipment cost (capitalistic expenses).
- Amortization and redemption of the capitalistic expenses.
- Current expenses including costs of raw materials, labor, energy, and research.
- Study of local and foreign markets.
- Financial analysis elements: cash flow, gross profit, net profits, evolution of prices....
- Considerations for the environment.
- Choice of site which will take into consideration the above cited factors.

In general, inorganic chemical industry, in contrary to the organic chemical industry, is a heavy industry and requires high capital to be installed (building and equipment)

13.6 Effects of petrol production on some Arab countries

The discovery of petrol in some Arab countries, especially in the Gulf, has led to important social and economic consequences. The economic and social development is probably going to continue due to the large amount of petrol reserves in these countries, especially in Saudi Arabia. The production of petrol has the following impacts:

- Transformation of these countries from self-sufficient to consuming communities.
- Development of the petrochemical industry. Most of its products

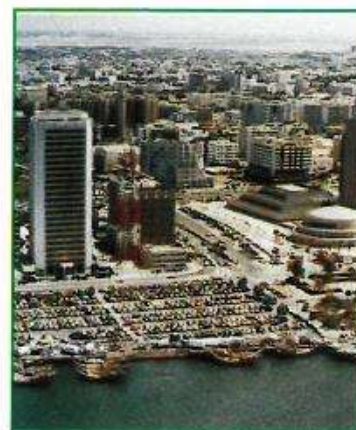


Fig. 13.9: Dubai: modern city in the Gulf area.

Chemistry

~~SPECIMEN~~



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- c-most of the raw materials are exported.
- d-raw materials are produced locally.

III- Refer to Figure 13.7 and Table 13.2 to answer the following questions:

- 1- Identify the important industrial areas in Lebanon.
- 2- Classify the industries as organic or inorganic.
- 3- Identify the main industries that pollute the Litani river.
- 4- Identify where cement plants are located. What is the impact of this industry on the environment?

IV- Relate the raw materials in column A to the corresponding industry in column B.

Column A

- 1- Olive oil
- 2- Polyethylene
- 3- Milk
- 4- Limestone and clay
- 5- Phosphate rocks
- 6- Air
- 7- Sand

Column B

- a- Ceramics
- b- Food processing
- c- Plastics
- d- Detergents
- e- Cement
- f- Soaps
- g- Helium
- h- Glass
- i- Fertilizers

V- Thinking Problems

- 1- a- What are the economic, social, and political advantages of the development of the food processing industry in Lebanon?
b- Where do you propose to locate the new food processing plants? (Refer to Figure 13.7).
- 2- Some of the industries in Lebanon are limited to formulation and packaging, such as detergents and pharmaceutical industries. In your opinion, what is the importance of such industries?